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(12) **United States Patent**
Kim et al.(10) **Patent No.:** **US 12,398,316 B2**(45) **Date of Patent:** **Aug. 26, 2025**(54) **ORGANIC COMPOUND AND ORGANIC ELECTROLUMINESCENT DEVICE COMPRISING SAME**(71) Applicant: **DOOSAN SOLUS CO., LTD.**, Iksan-si (KR)(72) Inventors: **Hong Suk Kim**, Yongin-si (KR);
Young Bae Kim, Yongin-si (KR)(73) Assignee: **DOOSAN SOLUS CO., LTD.**, Iksan-si (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 686 days.

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See application file for complete search history.(56) **References Cited**

U.S. PATENT DOCUMENTS

2008/0036373 A1 * 2/2008 Itoh H10K 85/346 546/88
2010/0237769 A1 9/2010 Park et al.
2013/0299128 A1 11/2013 Bergamini

FOREIGN PATENT DOCUMENTS

CN 106544006 A 3/2017
EP 0 934 847 A2 8/1999
JP 2002123012 A * 4/2002
JP 2003-291629 A 10/2003
JP 2005-082701 A 3/2005
JP 2003277743 A * 1/2008
JP 2008-037848 A 2/2008
JP 2014-055131 A 3/2014
KR 10-2010-0118690 A 11/2010
KR 20150121626 * 10/2012
KR 10-2014-0049861 A 4/2014
KR 10-2015-0106067 A 9/2015
KR 10-2015-0109111 A 10/2015
KR 10-2015-0121626 A 10/2015
KR 2015-0110101 A * 10/2015

(Continued)

OTHER PUBLICATIONS

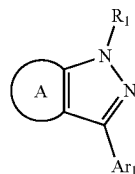
Yu et al. (Tetrahedron Letters (2015), 56 (11), 1432-1436).*

(Continued)

Primary Examiner — Valerie Rodriguez-Garcia(74) *Attorney, Agent, or Firm* — Sughrue Mion, PLLC(57) **ABSTRACT**

A novel compound represented by Chemical Formula 6 below and an organic electroluminescent device including the same are disclosed. The inclusion of the compound in an organic material layer of an organic electroluminescent device, preferably as a light emitting layer, can improve luminous efficiency, driving voltage, lifetime and the like of the organic electroluminescent device:

Chemical Formula 6



(56)

References Cited

FOREIGN PATENT DOCUMENTS

KR	2015-0111441	A	*	10/2015
KR	10-2016-0005944	A		1/2016
KR	10-2016-0045019	A		4/2016
KR	10-2016-0057018		*	5/2016
KR	10-2016-0057018	A		5/2016
KR	10-2017-0005853	A		1/2017
KR	10-2017-0023025	A		3/2017
KR	10-2017-0063394	A		6/2017
WO	2007/065010	A2		6/2007
WO	2010/126270	A1		11/2010
WO	WO 2010/126270		*	11/2010
WO	2014/054912	A1		4/2014
WO	2015175678	A1		11/2015
WO	2018/093107	A1		5/2018
WO	2020/021894	A1		1/2020

OTHER PUBLICATIONS

STN CAS Registry 1350026-65-1—Dec. 7, 2011.*

Communication dated Feb. 25, 2022 from the Korean Intellectual Property Office in Application No. 10-2017-0083287.

Y.T. Tao et al., "Organic Light-Emitting Diodes Based on Variously Substituted Pyrazoloquinolines as Emitting Material", Chem. Mater., 2001, pp. 1207-1212, vol. 13, No. 4.

International Search Report for PCT/KR2018/005079 dated Aug. 24, 2018 PCT/ISA/210.

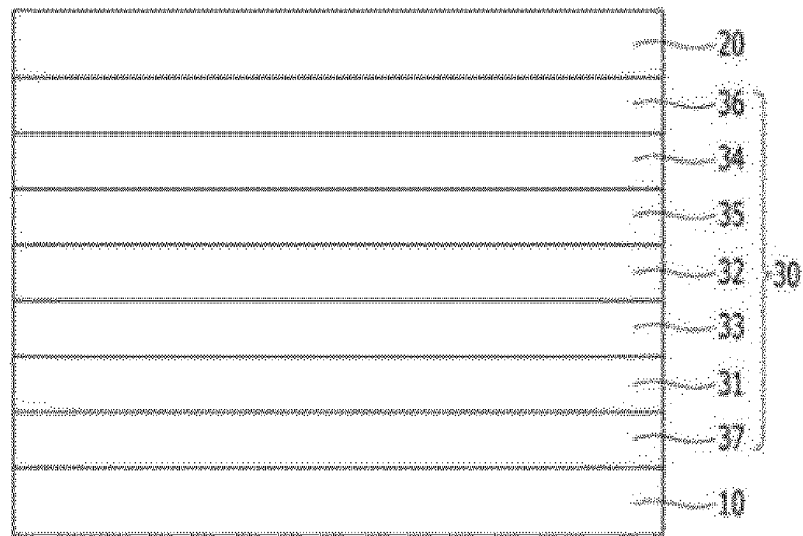
Korean Office Action dated Jan. 16, 2023 in Korean Application No. 10-2017-0083287.

* cited by examiner

【FIG. 1】



【FIG. 2】



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ORGANIC COMPOUND AND ORGANIC ELECTROLUMINESCENT DEVICE COMPRISING SAME

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a National Stage of International Application No. PCT/KR2018/005079 filed May 2, 2018, claiming priority based on Korean Patent Application No. 10-2017-0083287 filed Jun. 30, 2017.

TECHNICAL FIELD

The present invention relates to a novel organic compound capable of being used as a material for an organic electroluminescent device, and an organic electroluminescent device including the same.

BACKGROUND ART

With the observation of organic thin film light emission made by Bemanose in 1950s as a start, studies on organic electroluminescent (EL) devices have been continued leading to blue electroluminescence using a single anthracene crystal in 1965, and in 1987, an organic electroluminescent device having a laminated structure divided into functional layers of a hole layer and a light emitting layer has been proposed by Tang. After that, in order to manufacture organic electroluminescent devices with high efficiency and long lifetime, development has been made in the form of introducing each characteristic organic material layer into the device, which leads to the development of specialized materials used therein.

When a voltage is applied between the two electrodes in an organic electroluminescent device, holes and electrons are injected to an organic material layer from the anode and the cathode, respectively. When the injected holes and electrons meet, excitons are formed, and light emits when these excitons fall back to the ground state. Herein, materials used as the organic material layer may be divided into a light emitting material, a hole injection material, a hole transport material, an electron transport material, an electron injection material and the like depending on the function.

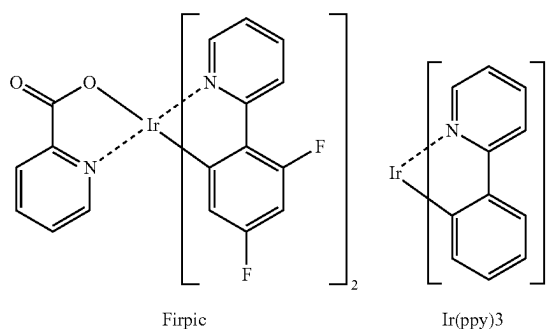
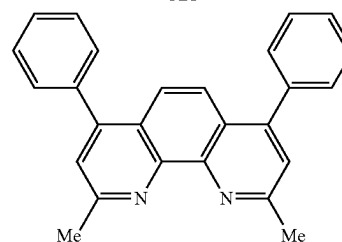
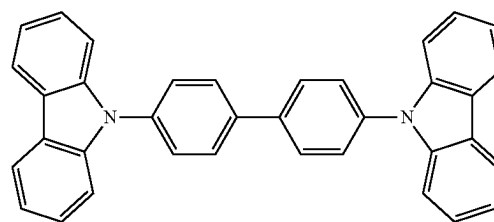
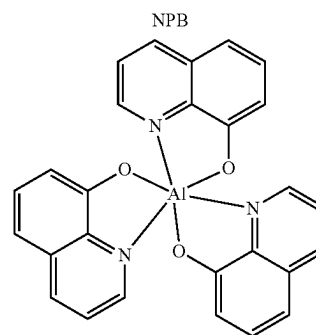
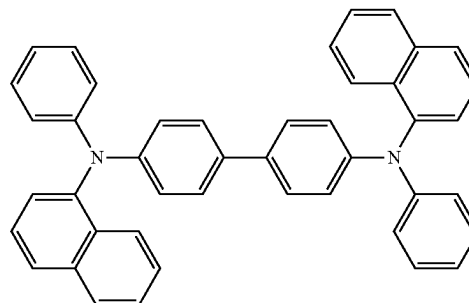
The light emitting material may be divided into, depending on the light emitting color, blue, green and red light emitting materials, and yellow and orange light emitting materials for obtaining better natural colors. In addition, in order to increase color purity and increase luminous efficiency through energy transfer, host/dopant series may be used as the light emitting material.

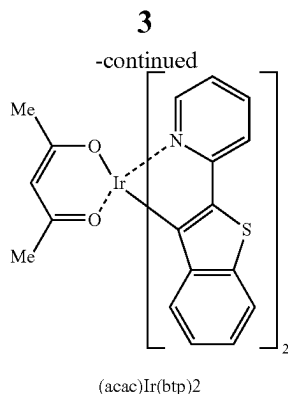
The dopant material may be divided into fluorescent dopants using organic materials and phosphorescent dopants using metal complex compounds including heavy atoms such as Ir or Pt. Herein, development of phosphorescent materials may enhance luminous efficiency up to 4 times compared to fluorescence theoretically, and therefore, studies on phosphorescent host materials have been widely progressed as well as on phosphorescent dopants.

So far, NPB, BCP, Alq₃ and the like have been widely known as materials of a hole injection layer, a hole transport layer, a hole blocking layer and an electron transport layer, and anthracene derivatives have been reported as a material of a light emitting layer. Particularly, among light emitting layer materials, metal complex compounds including Ir such as Firpic, Ir(ppy)₃ or (acac)Ir(btp)₂ having advantages in terms of efficiency enhancement have been used as blue,

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green and red phosphorescent dopant materials, and 4,4-dicarbazolylbiphenyl (CBP) has been used as a phosphorescent host material.





However, although being advantageous in terms of a light emission property, existing organic material layer materials have a low glass transition temperature and thereby have very unfavorable thermal stability, which is not satisfactory in terms of an organic electroluminescent device lifetime. Accordingly, development of organic material layer materials having superior performance has been required.

DISCLOSURE

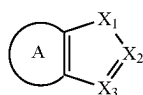
Technical Problem

The present invention is directed to providing a novel organic compound capable of being used in an organic electroluminescent device, and having excellent hole and electron injection and transport abilities, a light emitting ability and the like.

The present invention is also directed to providing an organic electroluminescent device including the novel organic compound, and thereby exhibiting a low driving voltage, high luminous efficiency, and an enhanced lifetime.

Technical Solution

In view of the above, one embodiment of the present invention provides a compound represented by the following Chemical Formula 1:



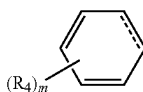
[Chemical Formula 1]

in Chemical Formula 1,

X₁ is selected from the group consisting of S, O, N(R₁) and C(R₂)(R₃);

X₂ and X₃ are each independently N or C(Ar₁), and at least one thereof is C(Ar₁); and

ring A is represented by any one of the following Chemical Formulae 2 to 4;

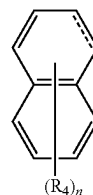


[Chemical Formula 2]

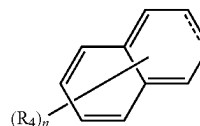
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[Chemical Formula 3]



[Chemical Formula 4]



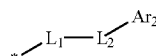
in Chemical Formulae 1 to 4,
a dotted line means a part that is fused;
m is an integer of 0 to 4;
n is an integer of 0 to 6;

R₁ to R₄ are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C₁~C₄₀ alkyl group, a C₂~C₄₀ alkenyl group, a C₂~C₄₀ alkynyl group, a C₃~C₄₀ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C₆~C₆₀ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C₁~C₄₀ alkoxy group, a C₆~C₆₀ aryloxy group, a C₃~C₄₀ alkylsilyl group, a C₆~C₆₀ arylsilyl group, a C₁~C₄₀ alkylboron group, a C₆~C₆₀ arylboron group, a C₆~C₆₀ arylphosphanyl group, a C₆~C₆₀ mono or diarylphosphinyl group and a C₆~C₆₀ arylamine group;

the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkoxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R₁ to R₄ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C₁~C₄₀ alkyl group, a C₂~C₄₀ alkenyl group, a C₂~C₄₀ alkynyl group, a C₆~C₆₀ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C₆~C₆₀ aryloxy group, a C₁~C₄₀ alkoxy group, a C₆~C₆₀ arylamine group, a C₃~C₄₀ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C₁~C₄₀ alkylsilyl group, a C₁~C₄₀ alkylboron group, a C₆~C₆₀ arylboron group, a C₆~C₆₀ arylphosphanyl group, a C₆~C₆₀ mono or diarylphosphinyl group and a C₆~C₆₀ arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other; and

Ar₁ is a substituent represented by the following Chemical Formula 5, and when Ar₁ is present in plural numbers, these are the same as or different from each other;

[Chemical Formula 5]



in Chemical Formula 5,
means a part where a bond is formed;

L₁ and L₂ are each independently selected from the group consisting of a direct bond, a C₆~C₁₈ arylene group and a heteroarylene group having 5 to 18 nuclear atoms;

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Ar₂ is selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C₁~C₄₀ alkyl group, a C₂~C₄₀ alkenyl group, a C₂~C₄₀ alkynyl group, a C₃~C₄₀ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C₆~C₆₀ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C₁~C₄₀ alkoxy group, a C₆~C₆₀ aryloxy group, a C₃~C₄₀ alkylsilyl group, a C₆~C₆₀ arylsilyl group, a C₁~C₄₀ alkylboron group, a C₆~C₆₀ arylboron group, a C₆~C₆₀ arylphosphanyl group, a C₆~C₆₀ mono or diarylphosphinyl group and a C₆~C₆₀ arylamine group; and

the arylene group and the heteroarylene group of L₁ and L₂, and the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkoxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of Ar₂ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C₁~C₄₀ alkyl group, a C₂~C₄₀ alkenyl group, a C₂~C₄₀ alkynyl group, a C₆~C₆₀ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C₆~C₆₀ aryloxy group, a C₁~C₄₀ alkoxy group, a C₆~C₆₀ arylamine group, a C₃~C₄₀ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C₁~C₄₀ alkylsilyl group, a C₁~C₄₀ alkylboron group, a C₆~C₆₀ arylboron group, a C₆~C₆₀ arylphosphanyl group, a C₆~C₆₀ mono or diarylphosphinyl group and a C₆~C₆₀ arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

Another embodiment of the present invention provides an organic electroluminescent device including an anode, a cathode, and one or more organic material layers provided between the anode and the cathode, wherein at least one of the one or more organic material layers includes the compound of Chemical Formula 1.

The “alkyl” in the present invention is a monovalent substituent derived from linear or branched saturated hydrocarbon having 1 to 40 carbon atoms. Examples thereof may include methyl, ethyl, propyl, isobutyl, sec-butyl, pentyl, iso-amyl, hexyl and the like, but are not limited thereto.

The “alkenyl” in the present invention is a monovalent substituent derived from linear or branched unsaturated hydrocarbon having one or more carbon-carbon double bonds and having 2 to 40 carbon atoms. Examples thereof may include vinyl, allyl, isopropenyl, 2-butenyl and the like, but are not limited thereto.

The “alkynyl” in the present invention is a monovalent substituent derived from linear or branched unsaturated hydrocarbon having one or more carbon-carbon triple bonds and having 2 to 40 carbon atoms. Examples thereof may include ethynyl, 2-propynyl and the like, but are not limited thereto.

The “aryl” in the present invention means a monovalent substituent derived from aromatic hydrocarbon having a single ring or two or more rings combined and having 6 to 60 carbon atoms. In addition, a monovalent substituent having two or more rings fused with each other, including only carbon (for example, the number of carbon atoms may

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be from 8 to 60) as a ring-forming atom, and with the whole molecule having non-aromaticity may also be included. Examples of such aryl may include phenyl, naphthyl, phenanthryl, anthryl, fluorenyl and the like, but are not limited thereto. The “heteroaryl” in the present invention means a monovalent substituent derived from monoheterocyclic or polyheterocyclic aromatic hydrocarbon having 5 to 60 nuclear atoms. Herein, one or more carbons, preferably 1 to 3 carbons, in the ring are substituted with a heteroatom selected from among N, O, P, S and Se. In addition, a monovalent group having two or more rings simply attached (pendant) or fused with each other, including a heteroatom selected from among N, O, P, S and Se as a ring-forming atom in addition to carbon, and with the whole molecule having non-aromaticity is interpreted to be included as well. Examples of such heteroaryl may include 6-membered monocyclic rings such as pyridyl, pyrazinyl, pyrimidinyl, pyridazinyl or triazinyl; polycyclic rings such as phenoxathieryl, indolizynyl, indolyl, purinyl, quinolyl, benzothiazole or carbazolyl; 2-furanyl, N-imidazolyl, 2-isoxazolyl, 2-pyridinyl, 2-pyrimidinyl and the like, but are not limited thereto.

The “aryloxy” in the present invention is a monovalent substituent represented by RO—, and R means aryl having 5 to 60 carbon atoms. Examples of such aryloxy may include phenyloxy, naphthyloxy, diphenyloxy and the like, but are not limited thereto.

The “alkyloxy” in the present invention is a monovalent substituent represented by R'O—, and R' means alkyl having 1 to 40 carbon atoms and is interpreted to include a linear, branched or cyclic structure. Examples of such alkyloxy may include methoxy, ethoxy, n-propoxy, 1-propoxy, t-butoxy, n-butoxy, pentoxy and the like, but are not limited thereto.

The “arylamine” in the present invention means amine substituted with aryl having 6 to 60 carbon atoms.

The “cycloalkyl” in the present invention means a monovalent substituent derived from monocyclic or polycyclic non-aromatic hydrocarbon having 3 to 40 carbon atoms. Examples of such cycloalkyl may include cyclopropyl, cyclopentyl, cyclohexyl, norbornyl, adamantane and the like, but are not limited thereto.

The “heterocycloalkyl” in the present invention means a monovalent substituent derived from non-aromatic hydrocarbon having 3 to 40 nuclear atoms, and one or more carbons, preferably 1 to 3 carbons, in the ring are substituted with a heteroatom such as N, O, S or Se. Examples of such heterocycloalkyl may include morpholine, piperazine and the like, but are not limited thereto.

The “alkylsilyl” in the present invention means silyl substituted with alkyl having 1 to 40 carbon atoms, and the “aryl silyl” means silyl substituted with aryl having 5 to 60 carbon atoms.

The “fused ring” in the present invention means a fused aliphatic ring, a fused aromatic ring, a fused heteroaliphatic ring, a fused heteroaromatic ring, or a combined form thereof.

Advantageous Effects

A compound of the present invention has excellent thermal stability, carrier transport ability, light emitting ability and the like, and therefore, is useful as a material of an organic material layer of an organic electroluminescent device.

In addition, an organic electroluminescent device including a compound of the present invention in an organic

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material layer exhibits greatly enhanced properties in terms of light emitting performance, driving voltage, lifetime, efficiency and the like, and can be effectively used in a full color display panel and the like.

DESCRIPTION OF DRAWINGS

FIG. 1 is a sectional diagram illustrating an organic electroluminescent device according to one embodiment of the present invention.

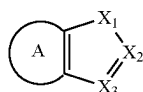
FIG. 2 is a sectional diagram illustrating an organic electroluminescent device according to one embodiment of the present invention.

MODE FOR DISCLOSURE

Hereinafter, the present invention will be described in detail.

1. Novel Organic Compound

A novel compound of the present invention may be represented by the following Chemical Formula 1:



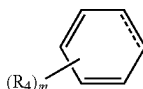
[Chemical Formula 1]

in Chemical Formula 1,

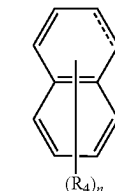
X₁ is selected from the group consisting of S, O, N(R₁) and C(R₂)(R₃);

X₂ and X₃ are each independently N or C(Ar₁), and at least one thereof is C(Ar₁); and

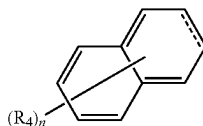
ring A is represented by any one of the following Chemical Formulae 2 to 4;



[Chemical Formula 2]



[Chemical Formula 3]



[Chemical Formula 4]

in Chemical Formulae 1 to 4,

a dotted line means a part that is fused;

m is an integer of 0 to 4;

n is an integer of 0 to 6;

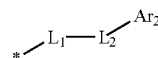
R₁ to R₄ are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C₁~C₄₀ alkyl group, a C₂~C₄₀ alkenyl group, a C₂~C₄₀ alkynyl group, a C₃~C₄₀ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C₆~C₆₀ aryl group, a heteroaryl

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group having 5 to 60 nuclear atoms, a C₁~C₄₀ alkoxy group, a C₆~C₆₀ aryloxy group, a C₃~C₄₀ alkylsilyl group, a C₆~C₆₀ arylsilyl group, a C₁~C₄₀ alkylboron group, a C₆~C₆₀ arylboron group, a C₆~C₆₀ arylphosphanyl group, a C₆~C₆₀ mono or diarylphosphinyl group and a C₆~C₆₀ arylamine group;

the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkoxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R₁ to R₄ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C₁~C₄₀ alkyl group, a C₂~C₄₀ alkenyl group, a C₂~C₄₀ alkynyl group, a C₆~C₆₀ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C₆~C₆₀ aryloxy group, a C₁~C₄₀ alkoxy group, a C₆~C₆₀ arylamine group, a C₃~C₄₀ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C₁~C₄₀ alkylsilyl group, a C₁~C₄₀ alkylboron group, a C₆~C₆₀ arylboron group, a C₆~C₆₀ arylphosphanyl group, a C₆~C₆₀ mono or diarylphosphinyl group and a C₆~C₆₀ arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other; and

Ar₁ is a substituent represented by the following Chemical Formula 5, and when Ar₁ is present in plural numbers, these are the same as or different from each other;



[Chemical Formula 5]

in Chemical Formula 5,

* means a part where a bond is formed;

L₁ and L₂ are each independently selected from the group consisting of a direct bond, a C₆~C₁₈ arylene group and a heteroarylene group having 5 to 18 nuclear atoms;

Ar₂ is selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C₁~C₄₀ alkyl group, a C₂~C₄₀ alkenyl group, a C₂~C₄₀ alkynyl group, a C₃~C₄₀ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C₆~C₆₀ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C₁~C₄₀ alkoxy group, a C₆~C₆₀ aryloxy group, a C₃~C₄₀ alkylsilyl group, a C₆~C₆₀ arylsilyl group, a C₁~C₄₀ alkylboron group, a C₆~C₆₀ arylboron group, a C₆~C₆₀ arylphosphanyl group, a C₆~C₆₀ mono or diarylphosphinyl group and a C₆~C₆₀ arylamine group; and

the arylene group and the heteroarylene group of L₁ and L₂, and the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkoxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of Ar₂ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C₁~C₄₀ alkyl group, a C₂~C₄₀ alkenyl group,

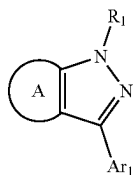
9

a C₂~C₄₀ alkynyl group, a C₆~C₆₀ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C₆~C₆₀ aryloxy group, a C₁~C₄₀ alkyloxy group, a C₆~C₆₀ arylamine group, a C₃~C₄₀ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C₁~C₄₀ alkylsilyl group, a C₁~C₄₀ alkylboron group, a C₆~C₆₀ arylboron group, a C₆~C₆₀ arylphosphanyl group, a C₆~C₆₀ mono or diarylphosphinyl group and a C₆~C₆₀ arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

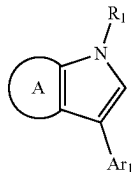
More specifically, the compound represented by Chemical Formula 1 of the present invention has an EWG bonding to a 5-membered aromatic ring or a 5-membered aromatic heteroring such as indole, indazole, indene, benzofuran, benzothiophene or triazolo, and may be used as a host material since, by having a similar energy level with carbazole, the energy level may be adjusted to be higher than a dopant energy level. Particularly, benzofuran and benzothiophene moieties are electron abundant, and when used as an electron transport layer material of an organic electroluminescent device, mobility increases, and as a result, an increase in the luminous efficiency and a decrease in the driving voltage may be expected. In addition, the 5-membered aromatic ring or the 5-membered aromatic heteroring of the present invention has a smaller molecular weight compared to existing compounds, and therefore, may be deposited at a relatively lower deposition temperature when deposited compared to other materials, and favorable processability and enhanced thermal stability may be obtained.

Accordingly, the compound represented by Chemical Formula 1 of the present invention may be used as a material of an organic material layer of an organic electroluminescent device, and may be preferably used as a light emitting layer material (green phosphorescent host material), an electron transport layer/injection layer material, a light emitting auxiliary layer material or an electron transport auxiliary layer material, and more preferably as a light emitting layer material, an electron transport layer material or an electron transport auxiliary layer material. In addition, the organic electroluminescent device including the compound of Chemical Formula 1 may have performance and lifetime properties greatly enhanced, and a full color organic light emitting panel using such an organic electroluminescent device may have its performance maximized.

According to preferred one embodiment of the present invention, the compound may be represented by any one of the following Chemical Formulae 6 to 14:



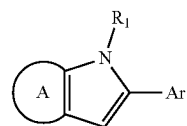
[Chemical Formula 6]



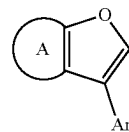
[Chemical Formula 7]

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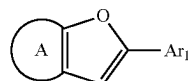
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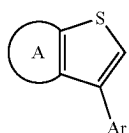
[Chemical Formula 8]



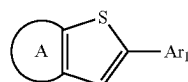
[Chemical Formula 9]



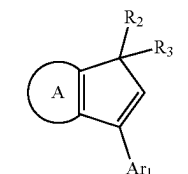
[Chemical Formula 10]



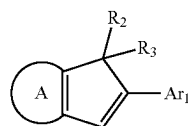
[Chemical Formula 11]



[Chemical Formula 12]



[Chemical Formula 13]



[Chemical Formula 14]

in Chemical Formulae 6 to 14,

ring A, Ar₁ and R₁ to R₃ each have the same definitions as in Chemical Formula 1.

According to preferred one embodiment of the present invention, the compound is preferably represented by any one of Chemical Formulae 6 to 8 in terms of securing low driving voltage and high luminous efficiency, and may be more preferably represented by Chemical Formula 6 or 7, and even more preferably represented by Chemical Formula 8 or 10.

According to preferred one embodiment of the present invention, R₁ to R₄ are each independently selected from the group consisting of a C₁~C₄₀ alkyl group, a C₆~C₆₀ aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and

the alkyl group, the aryl group and the heteroaryl group of R₁ to R₄ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C₁~C₄₀ alkyl group, a C₆~C₆₀ aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these may be the same as or different from each other.

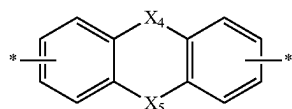
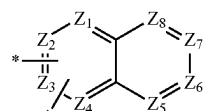
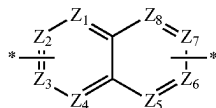
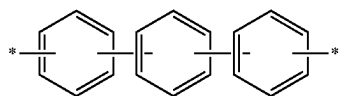
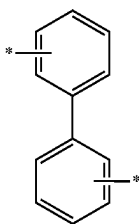
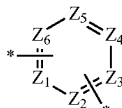
According to preferred one embodiment of the present invention, R₁ to R₄ are each independently selected from the group consisting of a methyl group, an ethyl group, a propyl

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group, a butyl group, a pentyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthalenyl group, a pyridinyl group, a pyrimidinyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a carbazoyl group, a fluorenyl group, a spirofluorenyl group and a dibenzodioxynyl group, and

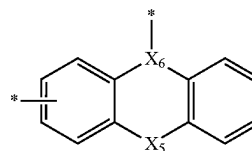
the methyl group, the ethyl group, the propyl group, the butyl group, the pentyl group, the phenyl group, the biphenyl group, the terphenyl group, the naphthalenyl group, the pyridinyl group, the pyrimidinyl group, the triazinyl group, the dibenzofuranyl group, the dibenzothiophenyl group, the carbazoyl group, the fluorenyl group, the spirofluorenyl group and the dibenzodioxynyl group of R_1 to R_4 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C_1 - C_{40} alkyl group, a C_6 - C_{60} alylamine group, a C_6 - C_{60} aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these may be the same as or different from each other.

According to preferred one embodiment of the present invention, L_1 and L_2 are each independently a direct bond, or a linker represented by any one of the following Chemical Formulae A-1 to A-7, but are not limited thereto:



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-continued



A-7

in Chemical Formulae A-1 to A-7,

* means a part where a bond is formed;

Z_1 to Z_8 are each independently N or C(R_5);

any two of Z_1 to Z_6 forming a bond as a linker in Chemical Formula A-1 are C(R_5), and herein, R_5 is not present;

any one of Z_1 to Z_4 and any one of Z_5 to Z_8 forming a bond as a linker in Chemical Formula A-4 are C(R_5), and herein, R_5 is not present;

any two of Z_1 to Z_4 forming a bond as a linker in Chemical Formula A-5 are C(R_5), and herein, R_5 is not present;

X_4 and X_5 are each independently O, S, N(R_6) or C(R_7) (R_8);

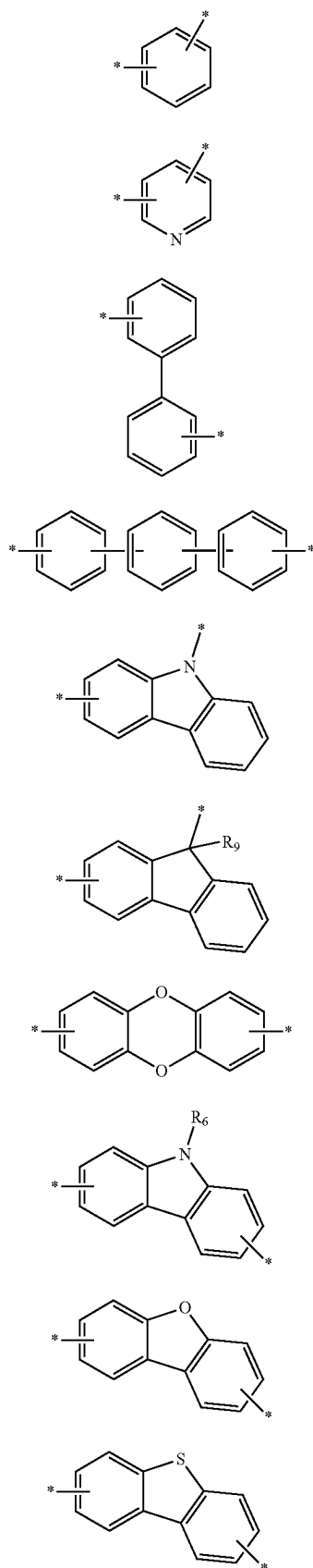
X_6 is N or C(R_9);

R_5 to R_9 are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C_1 - C_{40} alkyl group, a C_2 - C_{40} alkenyl group, a C_2 - C_{40} alkynyl group, a C_3 - C_{40} cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C_6 - C_{60} aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C_1 - C_{40} alkyloxy group, a C_6 - C_{60} aryloxy group, a C_3 - C_{40} alkylsilyl group, a C_6 - C_{60} arylsilyl group, a C_1 - C_{40} alkylboron group, a C_6 - C_{60} arylboron group, a C_6 - C_{60} arylphosphanyl group, a C_6 - C_{60} mono or diarylphosphinyl group and a C_6 - C_{60} arylamine group, or adjacent groups bond to form a fused ring, and when R_5 to R_8 are each present in plural numbers, these are the same as or different from each other; and

the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R_5 to R_9 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C_1 - C_{40} alkyl group, a C_2 - C_{40} alkenyl group, a C_2 - C_{40} alkynyl group, a C_6 - C_{60} aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C_6 - C_{60} aryloxy group, a C_1 - C_{40} alkyloxy group, a C_6 - C_{60} arylamine group, a C_3 - C_{40} cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C_1 - C_{40} alkylsilyl group, a C_1 - C_{40} alkylboron group, a C_6 - C_{60} arylboron group, a C_6 - C_{60} arylphosphanyl group, a C_6 - C_{60} mono or diarylphosphinyl group and a C_6 - C_{60} arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

According to preferred one embodiment of the present invention, L_1 and L_2 are each independently a direct bond, or a linker represented by any one of the following Chemical Formulae B-1 to B-12, but are not limited thereto:

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-continued

B-1

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B-2

10

B-3

15

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B-4

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B-5

30

B-6

35

40

B-7

45

B-8

50

B-9

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B-10

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B-11

B-12

in Chemical Formulae B-1 to B-12,

* means a part where a bond is formed;

R_6 to R_9 are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C_1 - C_{40} alkyl group, a C_2 - C_{40} alkenyl group, a C_2 - C_{40} alkynyl group, a C_3 - C_{40} cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C_6 - C_{60} aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C_1 - C_{40} alkyloxy group, a C_6 - C_{60} aryloxy group, a C_3 - C_{40} alkylsilyl group, a C_6 - C_{60} arylsilyl group, a C_1 - C_{40} alkylboron group, a C_6 - C_{60} arylboron group, a C_6 - C_{60} arylphosphanyl group, a C_6 - C_{60} mono or diarylphosphinyl group and a C_6 - C_{60} arylamine group; and

the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R_6 to R_9 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C_1 - C_{40} alkyl group, a C_2 - C_{40} alkenyl group, a C_2 - C_{40} alkynyl group, a C_6 - C_{60} aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C_6 - C_{60} aryloxy group, a C_1 - C_{40} alkyloxy group, a C_6 - C_{60} arylamine group, a C_3 - C_{40} cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C_1 - C_{40} alkylsilyl group, a C_1 - C_{40} alkylboron group, a C_6 - C_{60} arylboron group, a C_6 - C_{60} arylphosphanyl group, a C_6 - C_{60} mono or diarylphosphinyl group and a C_6 - C_{60} arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

According to preferred one embodiment of the present invention, in Chemical Formulae B-1 to B-12, R_6 to R_9 are each independently selected from the group consisting of hydrogen, a C_1 - C_{40} alkyl group, a C_6 - C_{60} aryl group and a heteroaryl group having 5 to 60 nuclear atoms; and

the alkyl group, the aryl group and the heteroaryl group of R_6 to R_9 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C_1 - C_{40} alkyl group, a C_2 - C_{40} alkenyl group, a C_2 - C_{40} alkynyl group, a C_6 - C_{60} aryl group and a heteroaryl group having 5 to

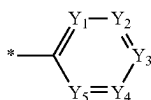
15

60 nuclear atoms, and when substituted with a plurality of the substituents, these are the same as or different from each other.

According to preferred one embodiment of the present invention, L_1 and L_2 are each independently a direct bond, or may be a linker selected from the group consisting of Chemical Formulae B-1 to B-3 and B-7 to B-12.

According to preferred one embodiment of the present invention, Ar_2 is a $C_6\text{--}C_{60}$ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, or an mylamine group, and the aryl group, the heteroaryl group and the mylamine group of Ar_2 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a $C_1\text{--}C_{40}$ alkyl group, a $C_6\text{--}C_{60}$ mylamine group, a $C_6\text{--}C_{60}$ aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these may be the same as or different from each other.

According to preferred one embodiment of the present invention, Ar_2 may be a substituent represented by the following Chemical Formula 15:



[Chemical Formula 15]

in Chemical Formula 15,

* means a part where a bond is formed;

Y_1 to Y_6 are each independently N or C(R_{10});

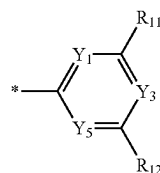
R_{10} is selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a $C_1\text{--}C_{40}$ alkyl group, a $C_2\text{--}C_{40}$ alkenyl group, a $C_2\text{--}C_{40}$ alkynyl group, a $C_3\text{--}C_{40}$ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a $C_6\text{--}C_{60}$ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a $C_1\text{--}C_{40}$ alkoxy group, a $C_6\text{--}C_{60}$ aryloxy group, a $C_3\text{--}C_{40}$ alkylsilyl group, a $C_6\text{--}C_{60}$ arylsilyl group, a $C_1\text{--}C_{40}$ alkylboron group, a $C_6\text{--}C_{60}$ arylboron group, a $C_6\text{--}C_{60}$ arylphosphanyl group, a $C_6\text{--}C_{60}$ mono or diarylphosphinyl group and a $C_6\text{--}C_{60}$ arylamine group, and when R_{10} is present in plural numbers, these are the same as or different from each other; and

the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkoxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R_{10} are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a $C_1\text{--}C_{40}$ alkyl group, a $C_2\text{--}C_{40}$ alkenyl group, a $C_2\text{--}C_{40}$ alkynyl group, a $C_6\text{--}C_{60}$ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a $C_6\text{--}C_{60}$ aryloxy group, a $C_1\text{--}C_{40}$ alkoxy group, a $C_6\text{--}C_{60}$ arylamine group, a $C_3\text{--}C_{40}$ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a $C_1\text{--}C_{40}$ alkylsilyl group, a $C_1\text{--}C_{40}$ alkylboron group, a $C_6\text{--}C_{60}$ arylboron group, a $C_6\text{--}C_{60}$ arylphosphanyl group, a $C_6\text{--}C_{60}$ mono or diarylphosphinyl group and a $C_6\text{--}C_{60}$ arylsilyl

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group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

According to preferred one embodiment of the present invention, the substituent represented by Chemical Formula 15 may be a substituent represented by the following Chemical Formula 16:



[Chemical Formula 16]

in Chemical Formula 16,

* means a part where a bond is formed;

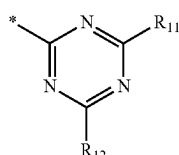
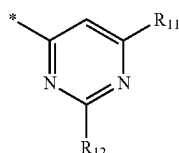
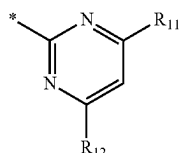
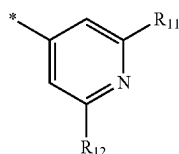
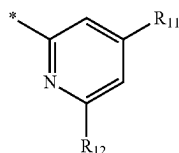
R_{11} and R_{12} are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a $C_1\text{--}C_{40}$ alkyl group, a $C_2\text{--}C_{40}$ alkenyl group, a $C_2\text{--}C_{40}$ alkynyl group, a $C_3\text{--}C_{40}$ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a $C_6\text{--}C_{60}$ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a $C_1\text{--}C_{40}$ alkoxy group, a $C_6\text{--}C_{60}$ aryloxy group, a $C_3\text{--}C_{40}$ alkylsilyl group, a $C_6\text{--}C_{60}$ arylsilyl group, a $C_1\text{--}C_{40}$ alkylboron group, a $C_6\text{--}C_{60}$ arylboron group, a $C_6\text{--}C_{60}$ arylphosphanyl group, a $C_6\text{--}C_{60}$ mono or diarylphosphinyl group and a $C_6\text{--}C_{60}$ arylamine group, or bond to adjacent groups to form a fused ring;

the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkoxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R_{11} and R_{12} are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a $C_1\text{--}C_{40}$ alkyl group, a $C_2\text{--}C_{40}$ alkenyl group, a $C_2\text{--}C_{40}$ alkynyl group, a $C_6\text{--}C_{60}$ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a $C_6\text{--}C_{60}$ aryloxy group, a $C_1\text{--}C_{40}$ alkoxy group, a $C_6\text{--}C_{60}$ arylamine group, a $C_3\text{--}C_{40}$ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a $C_1\text{--}C_{40}$ alkylsilyl group, a $C_1\text{--}C_{40}$ alkylboron group, a $C_6\text{--}C_{60}$ arylboron group, a $C_6\text{--}C_{60}$ arylphosphanyl group, a $C_6\text{--}C_{60}$ mono or diarylphosphinyl group and a $C_6\text{--}C_{60}$ arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other; and

Y_1 , Y_3 and Y_5 each have the same definitions as in Chemical Formula 15.

According to preferred one embodiment of the present invention, the substituent represented by Chemical Formula 15 may be a substituent represented by any one of the following Chemical Formulae C-1 to C-5:

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in Chemical Formulae C-1 to C-5,

* means a part where a bond is formed,

R₁₁ and R₁₂ are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C₁~C₄₀ alkyl group, a C₂~C₄₀ alkenyl group, a C₂~C₄₀ alkynyl group, a C₃~C₄₀ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C₆~C₆₀ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C₁~C₄₀ alkoxy group, a C₆~C₆₀ aryloxy group, a C₃~C₄₀ alkylsilyl group, a C₆~C₆₀ arylsilyl group, a C₁~C₄₀ alkylboron group, a C₆~C₆₀ arylboron group, a C₆~C₆₀ arylphosphanyl group, a C₆~C₆₀ mono or diarylphosphinyl group and a C₆~C₆₀ arylamine group, or bond to adjacent groups to form a fused ring; and the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkoxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R₁₁ and R₁₂ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C₁~C₄₀ alkyl group, a C₂~C₄₀ alkenyl group, a C₂~C₄₀ alkynyl group, a C₆~C₆₀ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C₆~C₆₀ aryloxy group, a C₁~C₄₀ alkoxy group, a C₆~C₆₀ arylamine group, a C₃~C₄₀ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a

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C₁~C₄₀ alkylsilyl group, a C₁~C₄₀ alkylboron group, a C₆~C₆₀ arylboron group, a C₆~C₆₀ arylphosphanyl group, a C₆~C₆₀ mono or diarylphosphinyl group and a C₆~C₆₀ arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

According to preferred one embodiment of the present invention, R₁₁ and R₁₂ are each independently selected from the group consisting of a C₁~C₄₀ alkyl group, a C₆~C₆₀ aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and

the alkyl group, the aryl group and the heteroaryl group of R₁₁ and R₁₂ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C₁~C₄₀ alkyl group, a C₆~C₆₀ aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these are the same as or different from each other.

According to preferred one embodiment of the present invention, R₁₁ and R₁₂ are each independently selected from the group consisting of a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a dibenzofuranyl group, a carbazolyl group, a fluorenyl group and a dibenzothiophenyl group, and

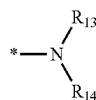
the phenyl group, the biphenyl group, the pyridinyl group, the pyrimidinyl group, the dibenzofuranyl group, the carbazolyl group, the fluorenyl group and the dibenzothiophenyl group of R₁₁ and R₁₂ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C₁~C₄₀ alkyl group, a C₆~C₆₀ aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these are the same as or different from each other.

According to preferred one embodiment of the present invention, R₁₁ and R₁₂ are each independently selected from the group consisting of a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a dibenzofuranyl group, a carbazolyl group, a fluorenyl group and a dibenzothiophenyl group, and

the phenyl group, the biphenyl group, the pyridinyl group, the pyrimidinyl group, the dibenzofuranyl group, the carbazolyl group, the fluorenyl group and the dibenzothiophenyl group of R₁₁ and R₁₂ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a methyl group, an ethyl group, a propyl group, a butyl group, a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a dibenzofuranyl group, a carbazolyl group, a fluorenyl group and a dibenzothiophenyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

According to preferred one embodiment of the present invention, Ar₂ may be a substituent represented by the following Chemical Formula 17:

[Chemical Formula 17]



in Chemical Formula 17,

* means a part where a bond is formed;

R₁₃ and R₁₄ are each independently selected from the group consisting of hydrogen, deuterium, halogen, a

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cyano group, a nitro group, a C₁~C₄₀ alkyl group, a C₂~C₄₀ alkenyl group, a C₂~C₄₀ alkynyl group, a C₃~C₄₀ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C₆~C₆₀ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C₁~C₄₀ alkyloxy group, a C₆~C₆₀ aryloxy group, a C₃~C₄₀ alkylsilyl group, a C₆~C₆₀ arylsilyl group, a C₁~C₄₀ alkylboron group, a C₆~C₆₀ arylboron group, a C₆~C₆₀ arylphosphanyl group, a C₆~C₆₀ mono or diarylphosphinyl group and a C₆~C₆₀ arylamine group, or bond to adjacent groups to form a fused ring; and the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R₁₃ and R₁₄ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C₁~C₄₀ alkyl group, a C₂~C₄₀ alkenyl group, a C₂~C₄₀ alkynyl group, a C₆~C₆₀ aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C₆~C₆₀ aryloxy group, a C₁~C₄₀ alkyloxy group, a C₆~C₆₀ arylamine group, a C₃~C₄₀ cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C₁~C₄₀ alkylsilyl group, a C₁~C₄₀ alkylboron group, a C₆~C₆₀ arylboron group, a C₆~C₆₀ arylphosphanyl group, a C₆~C₆₀ mono or diarylphosphinyl group and a C₆~C₆₀ arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

According to preferred one embodiment of the present invention, R₁₃ and R₁₄ are each independently selected from the group consisting of a C₁~C₄₀ alkyl group, a C₆~C₆₀ aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and

the alkyl group, the aryl group and the heteroaryl group of R₁₃ and R₁₄ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C₁~C₄₀ alkyl group, a C₆~C₆₀ aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these are the same as or different from each other.

According to preferred one embodiment of the present invention, R₁₃ and R₁₄ are each independently selected from the group consisting of a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a dibenzofuranyl group, a carbazolyl group, a fluorenyl group and a dibenzothienophenyl group, and

the phenyl group, the biphenyl group, the pyridinyl group, the pyrimidinyl group, the dibenzofuranyl group, the carbazolyl group, the fluorenyl group and the dibenzothienophenyl group of R₁₃ and R₁₄ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C₁~C₄₀ alkyl group, a C₆~C₆₀ aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these are the same as or different from each other.

According to preferred one embodiment of the present invention, R₁₃ and R₁₄ are each independently selected from the group consisting of a phenyl group, a biphenyl group, a

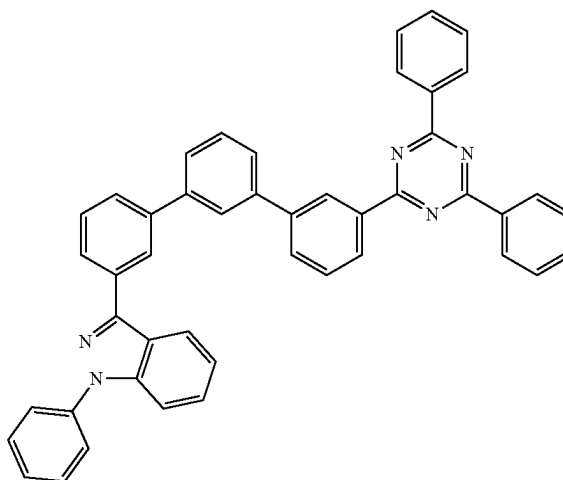
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pyridinyl group, a pyrimidinyl group, a dibenzofuranyl group, a carbazolyl group, a fluorenyl group and a dibenzothienophenyl group, and

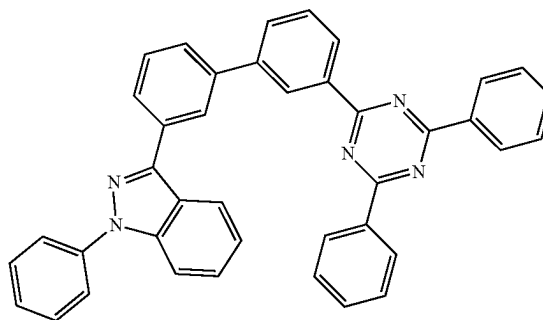
the phenyl group, the biphenyl group, the pyridinyl group, the pyrimidinyl group, the dibenzofuranyl group, the carbazolyl group, the fluorenyl group and the dibenzothienophenyl group of R₁₃ and R₁₄ are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a methyl group, an ethyl group, a propyl group, a butyl group, a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a dibenzofuranyl group, a carbazolyl group, a fluorenyl group and a dibenzothienophenyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

The compound represented by Chemical Formula 1 of the present invention may be represented by the following compounds, but is not limited thereto:

R1

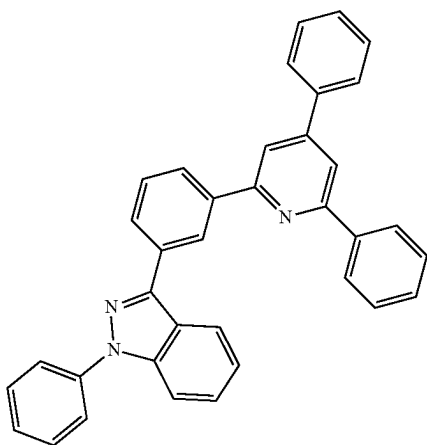
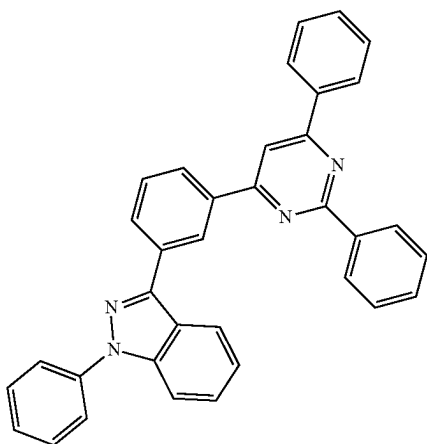
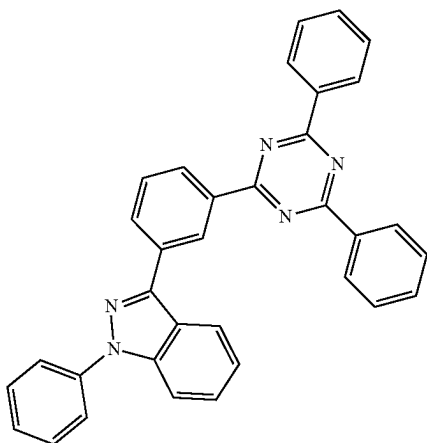


R2



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**22**

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R3

R6

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R4

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R5

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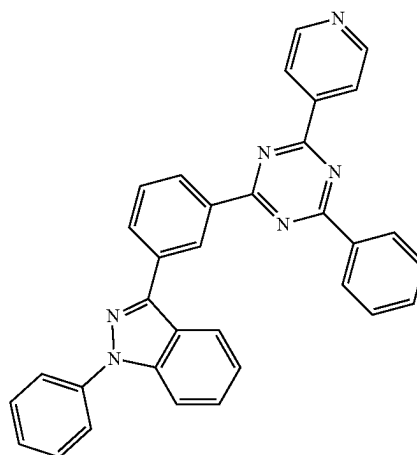
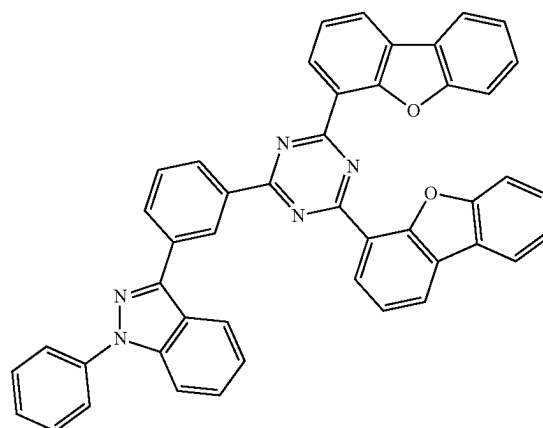
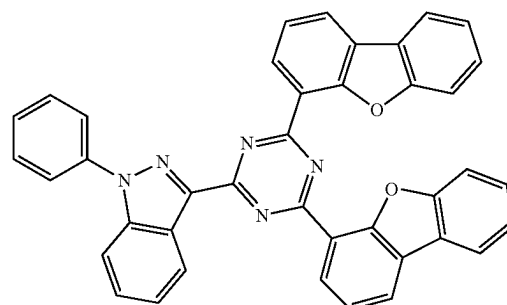
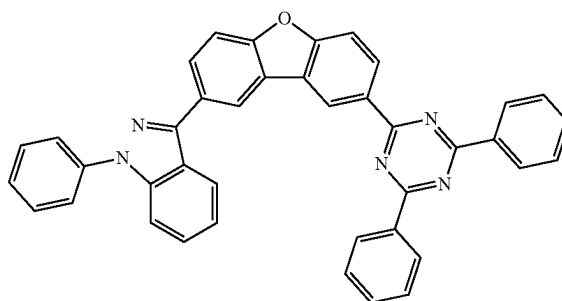
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R7

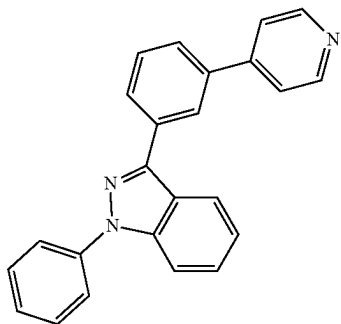
R8

R9



23

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**24**

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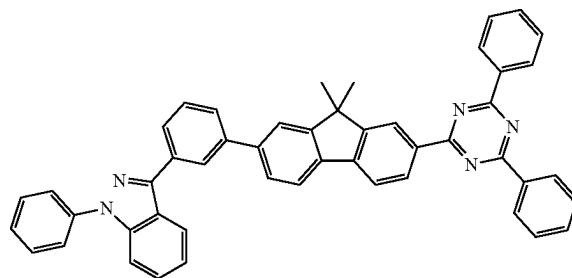
R10

R13

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R14

R11

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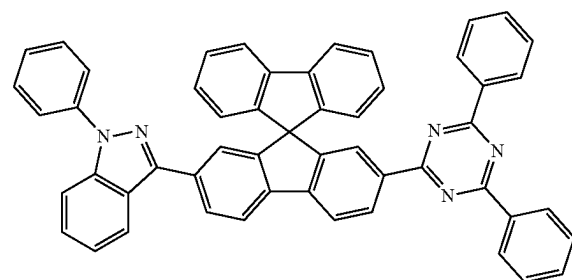
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R15

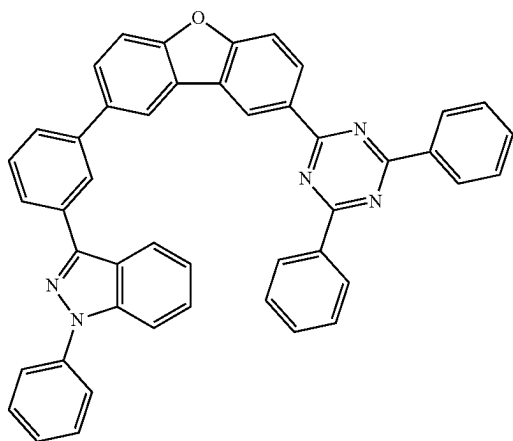
R12

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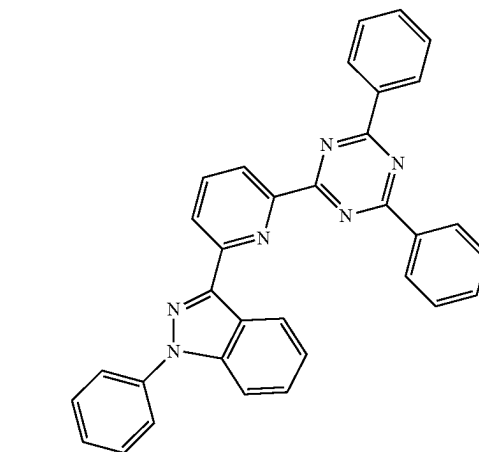
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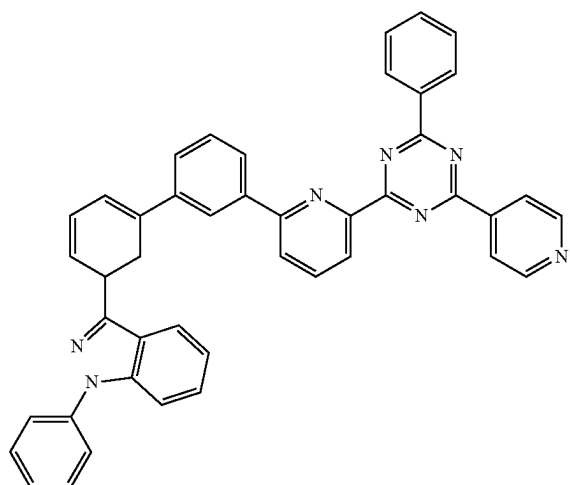
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R16



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R18

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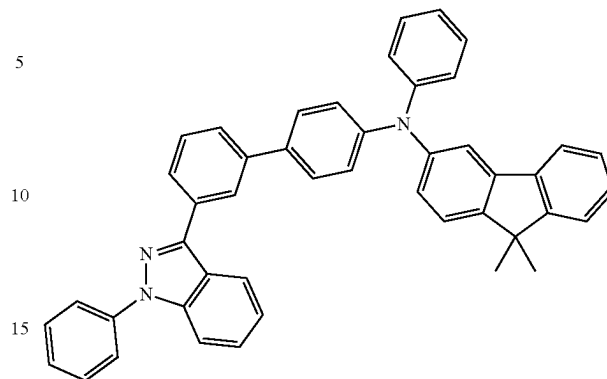
R19

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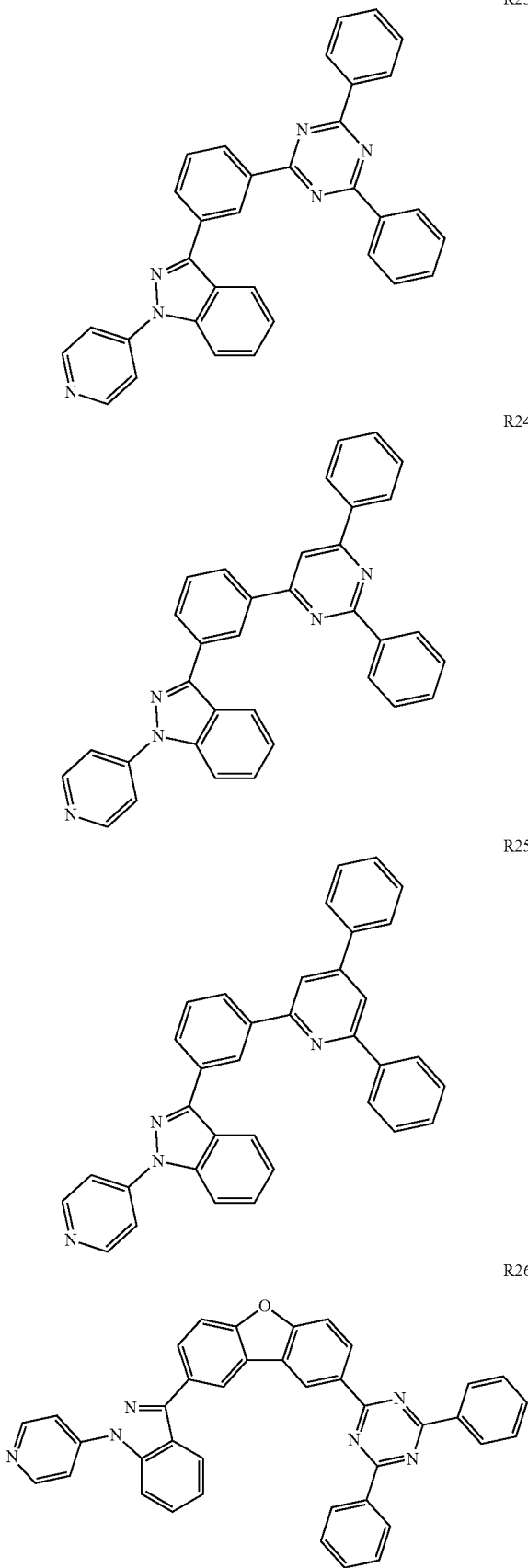
R20

R21

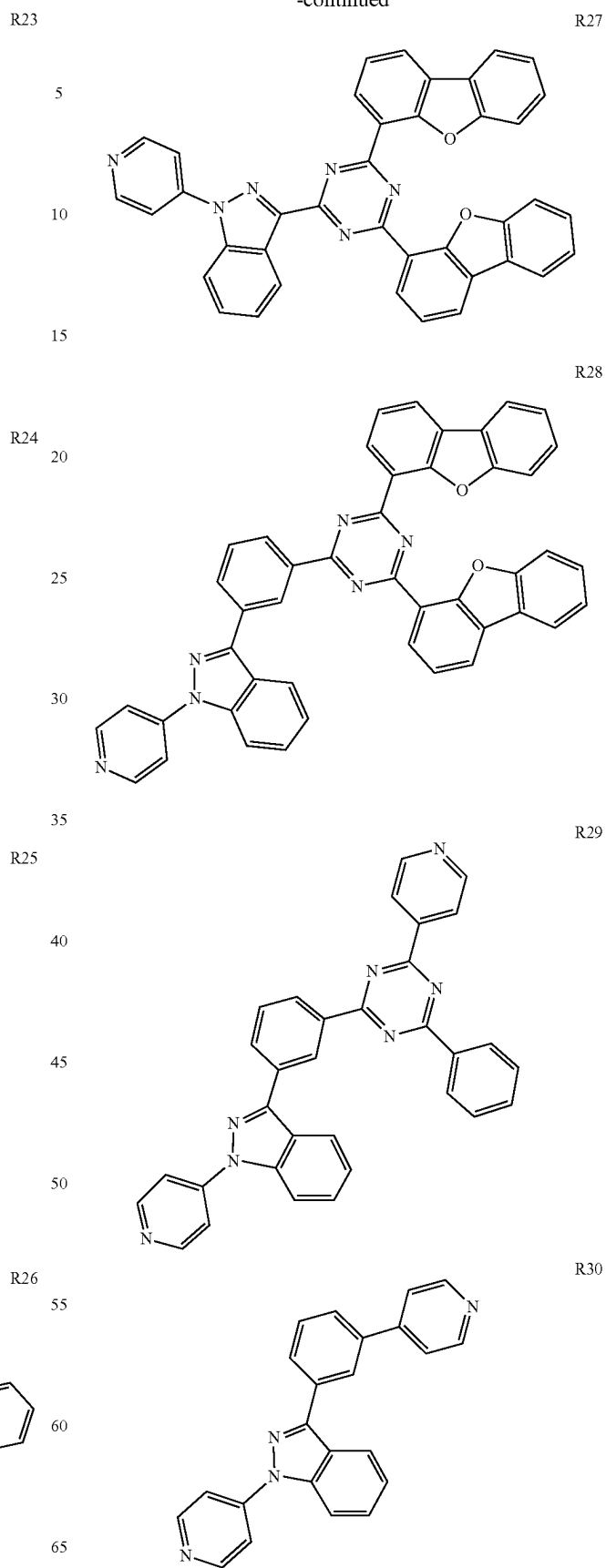
R22

27

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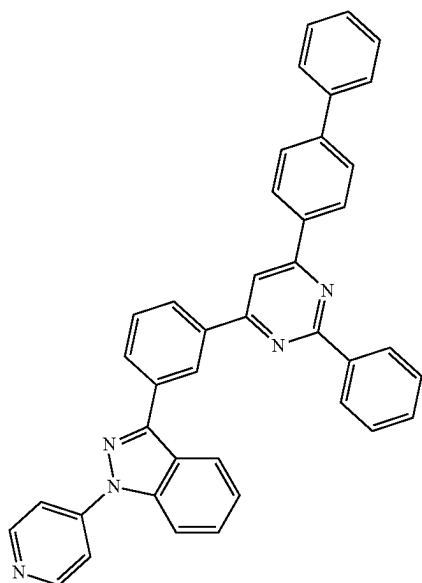
**28**

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R31

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R32 25

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R33

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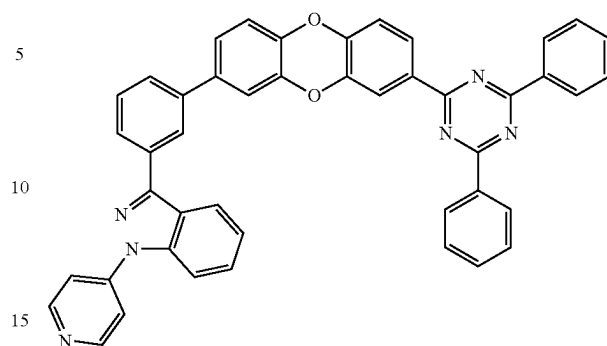
R34 55

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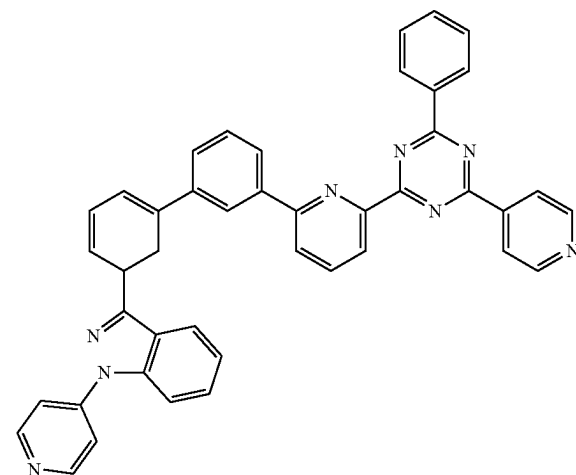
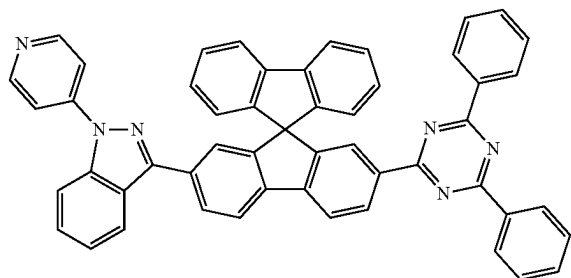
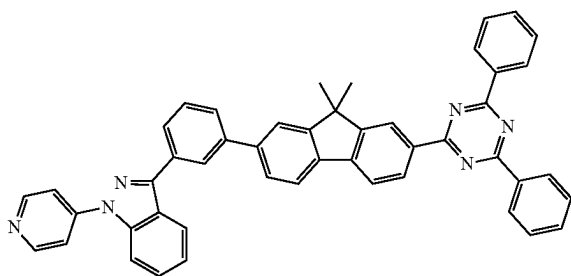
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R35

R36

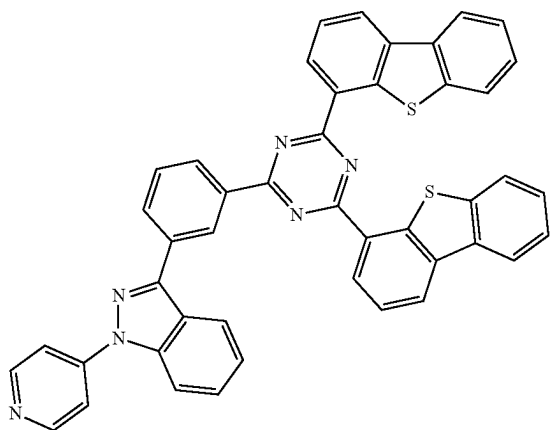
R37



31

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R38



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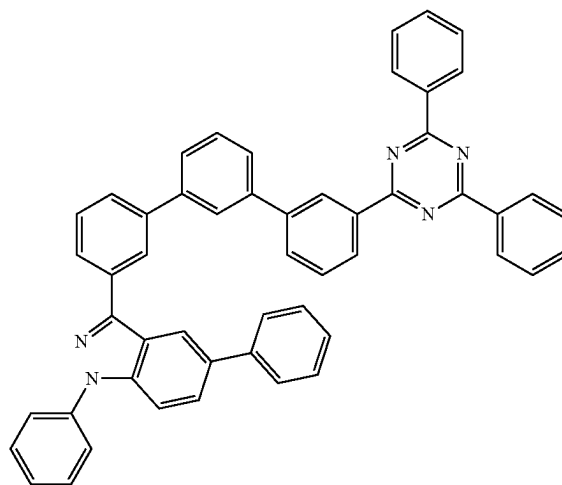
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R41



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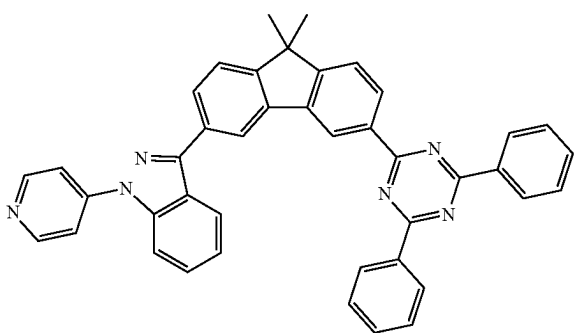
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R39

R40

R42

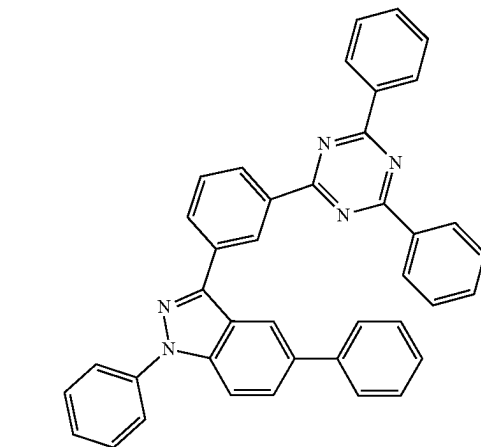
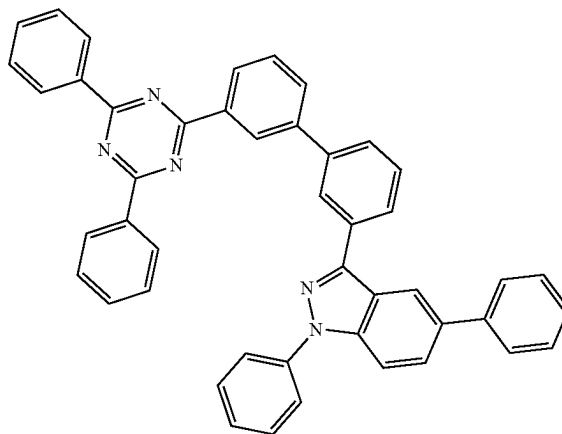
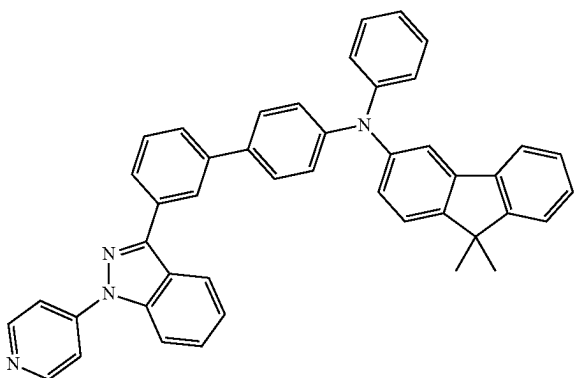
R43



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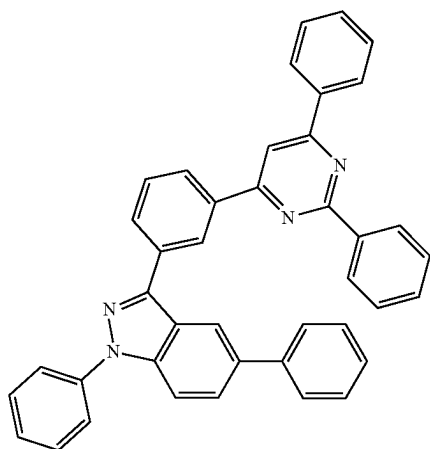
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R44

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R45

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R46

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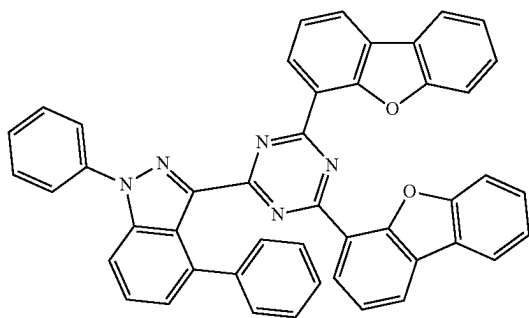
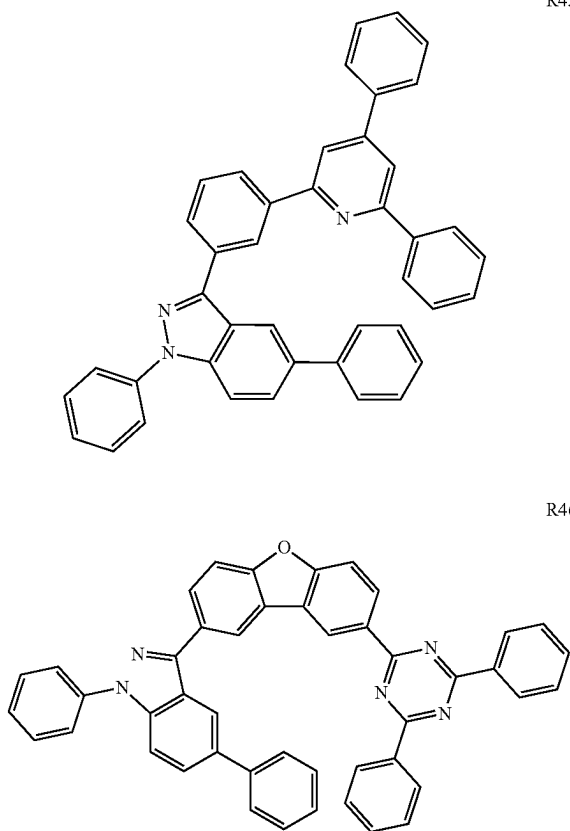
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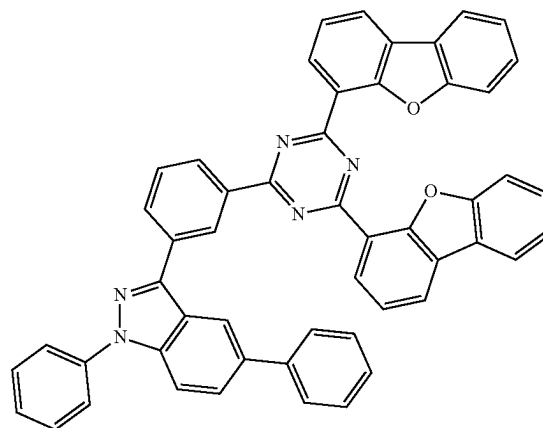
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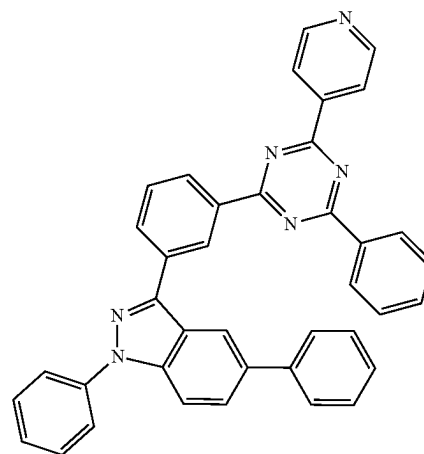
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**34**

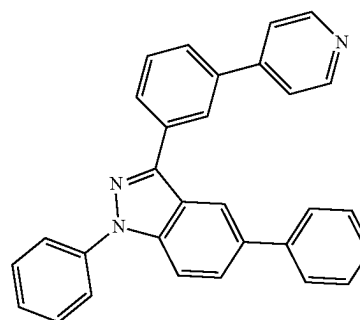
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R48



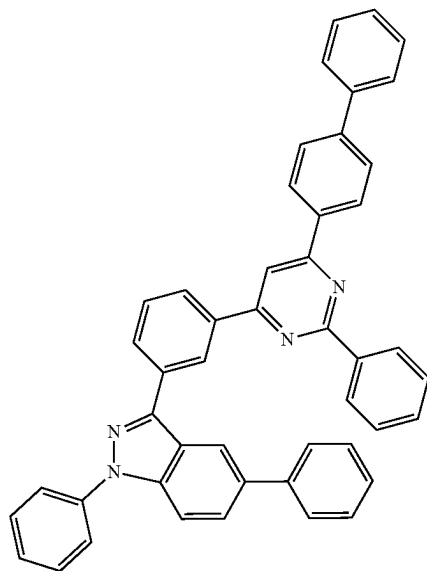
R49



R50

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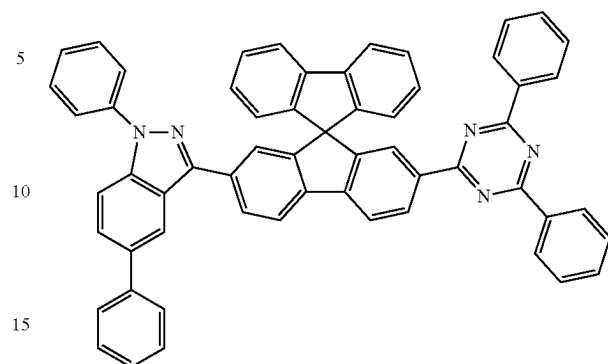
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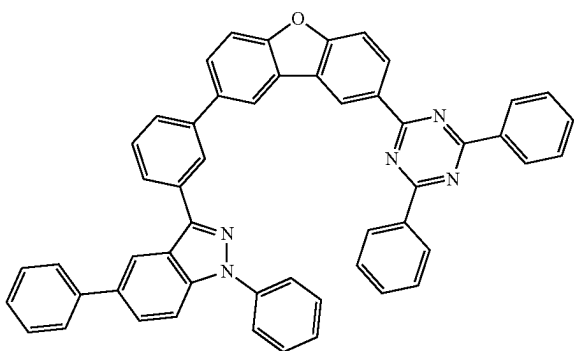
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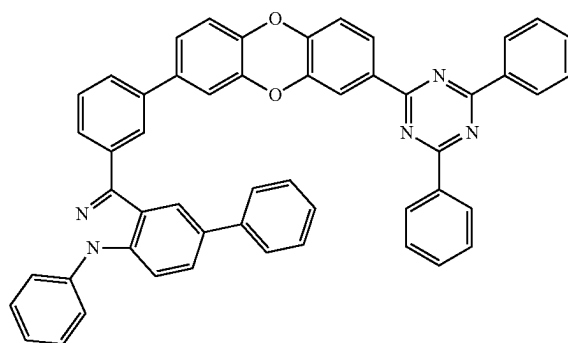
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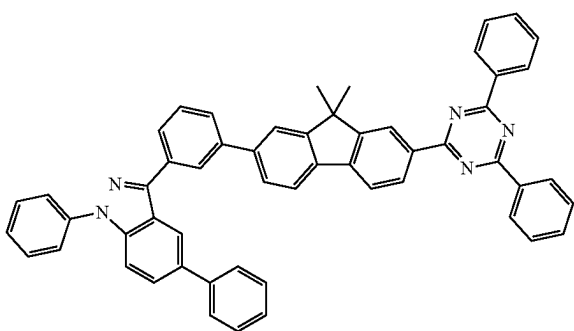
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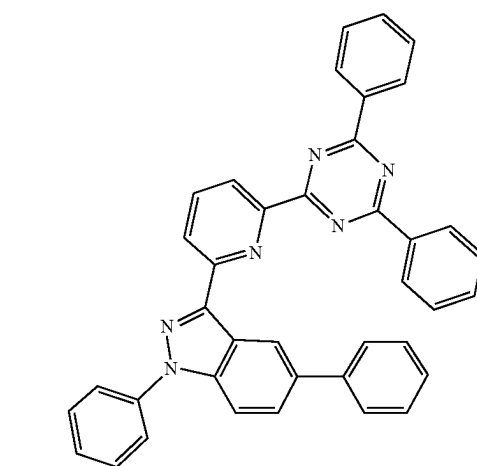
R52



R55



R53

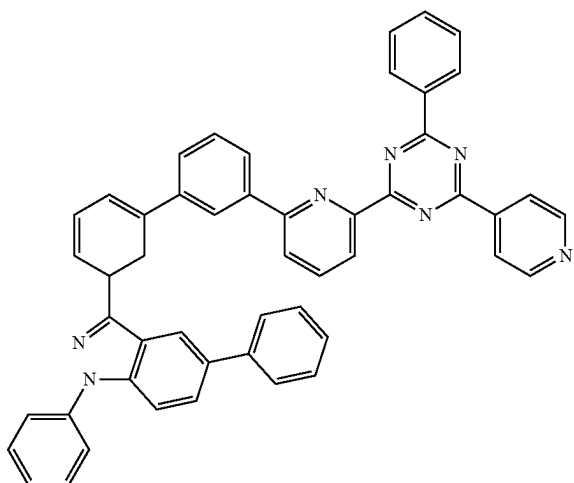


R56

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R57



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R58

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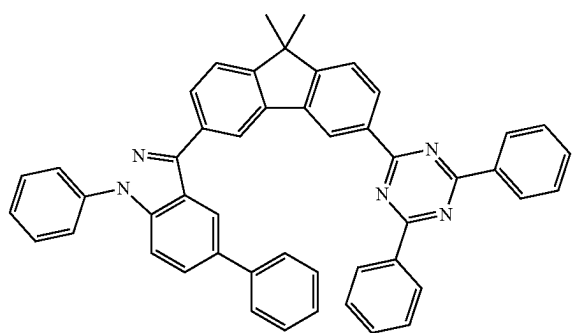
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R59

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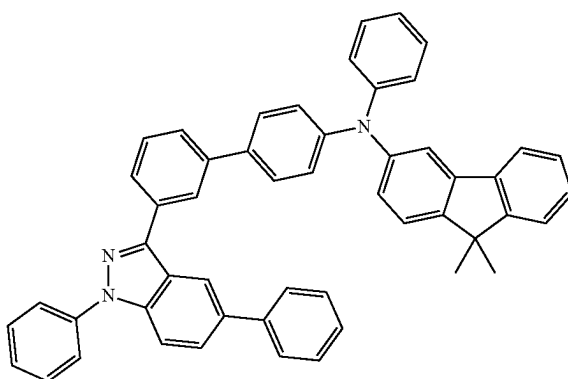
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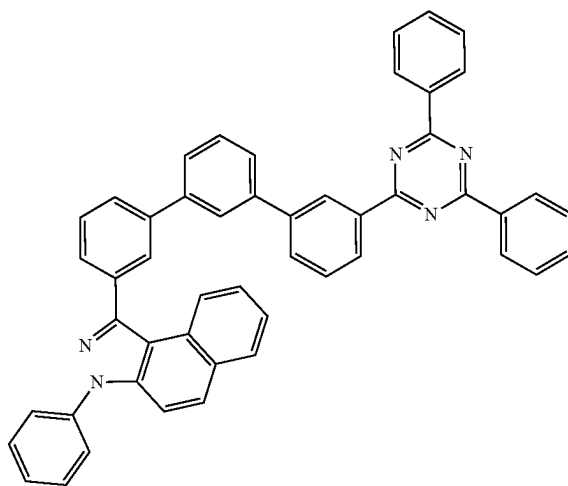
**38**

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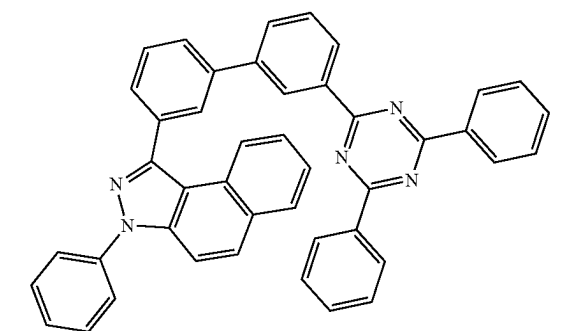
R60



R61

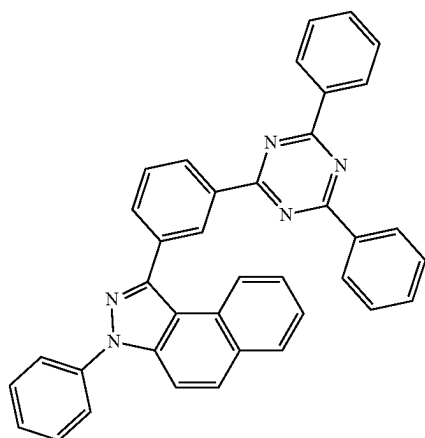


R62



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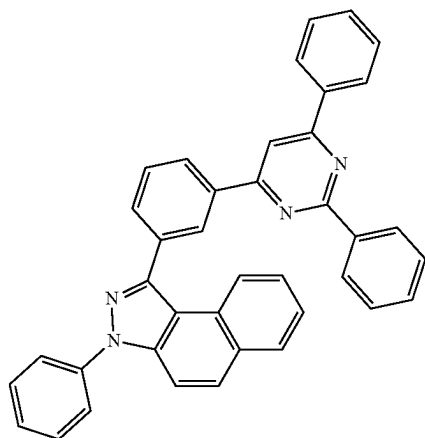


R63

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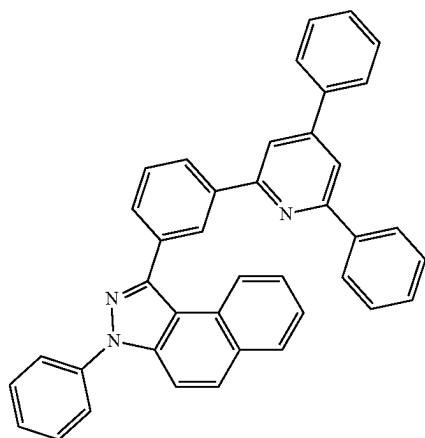
R64

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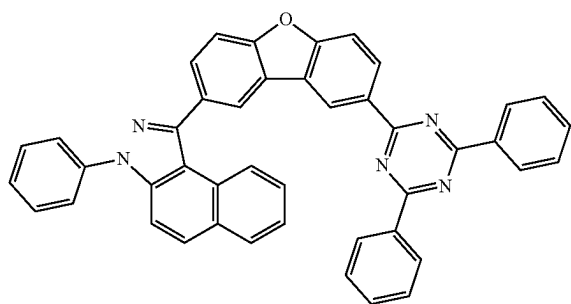


R65

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R66

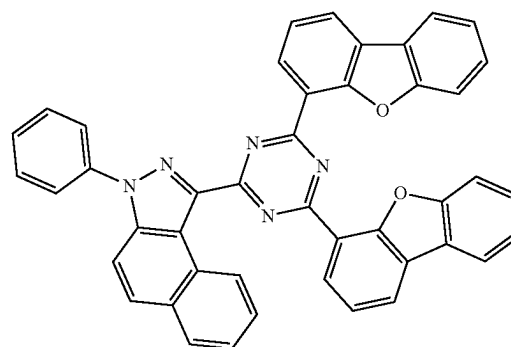
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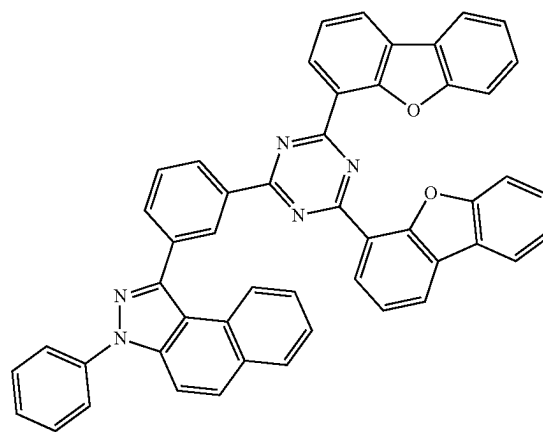
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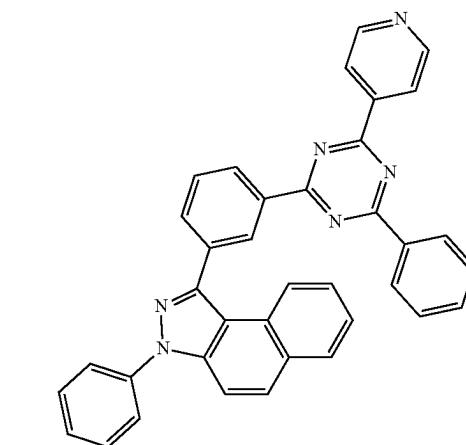
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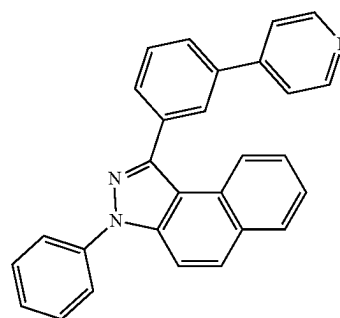
R67



R68



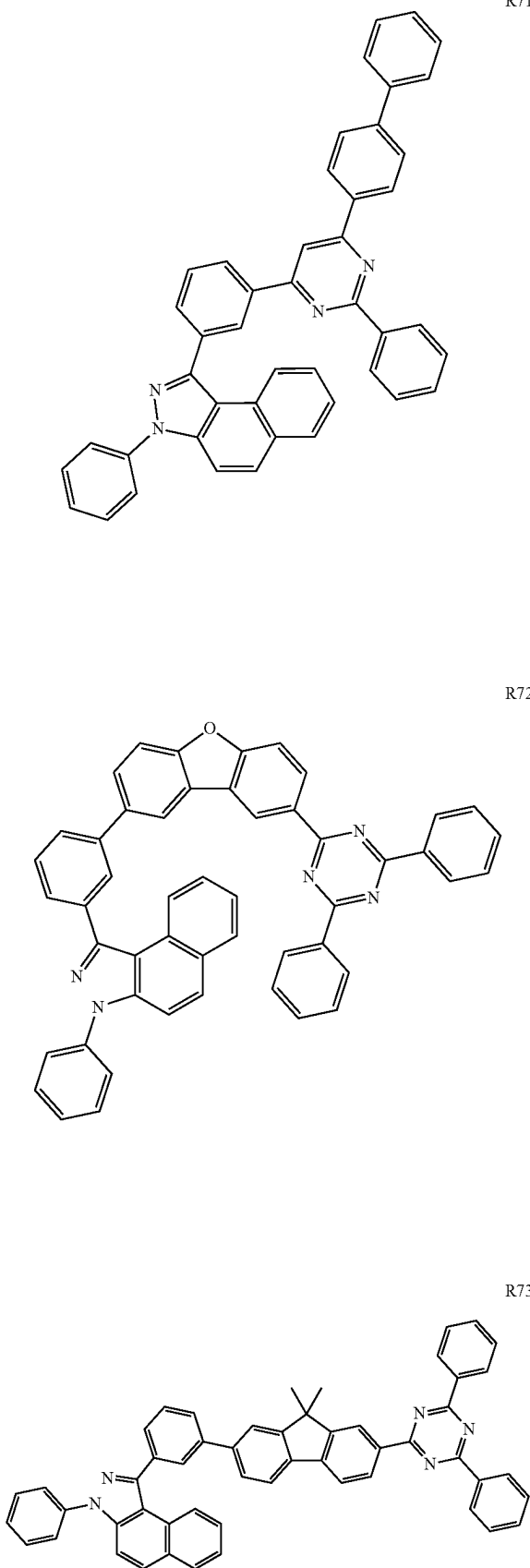
R69



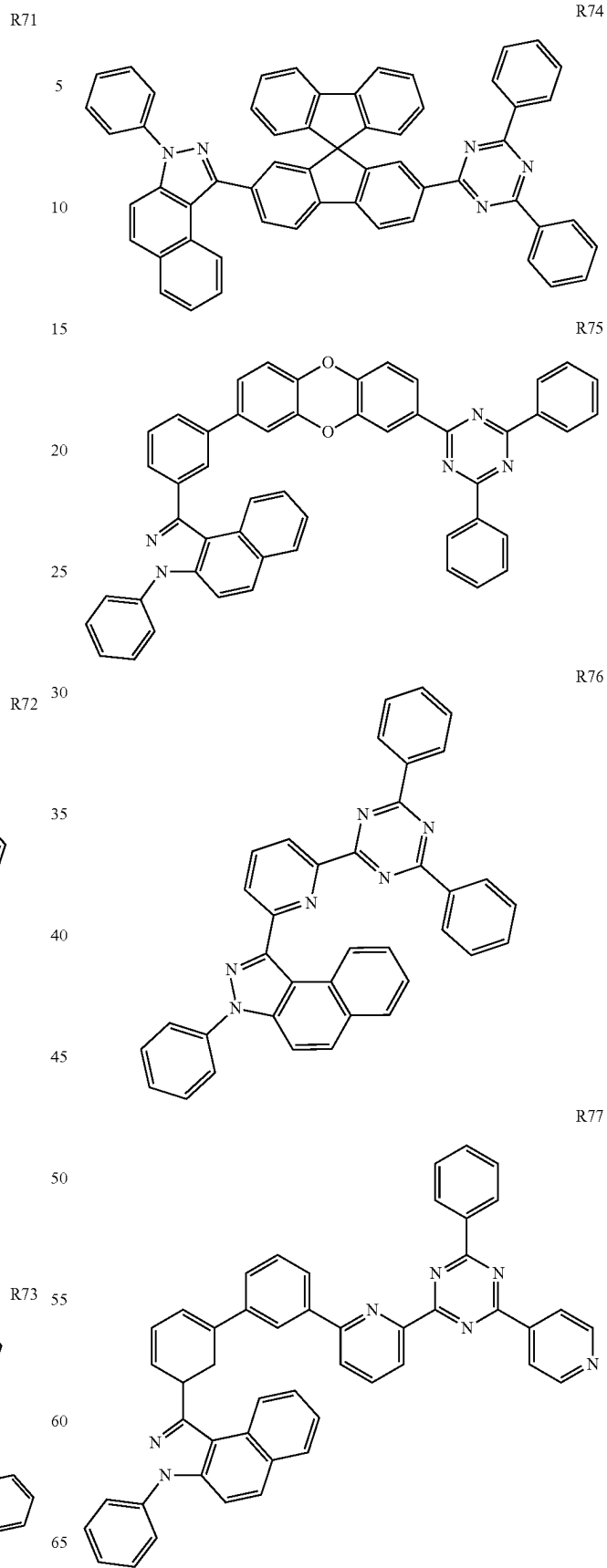
R70

41

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**42**

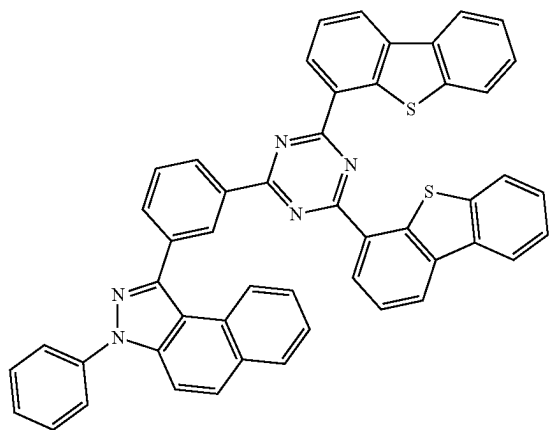
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43

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R78



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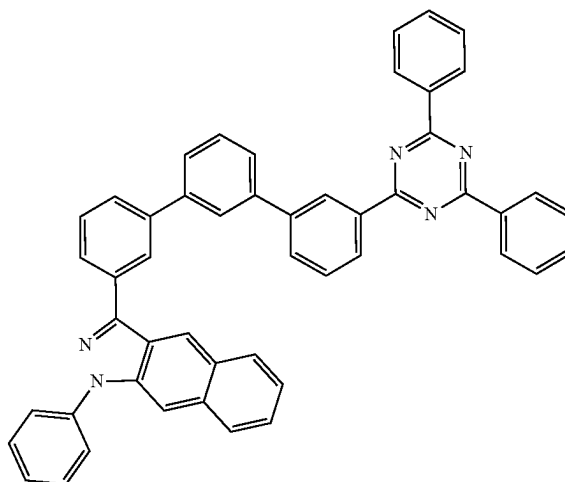
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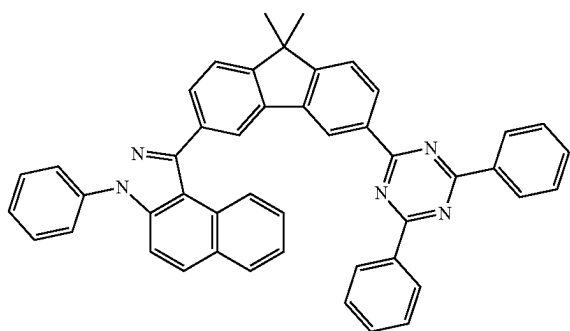
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R101



R79

R102



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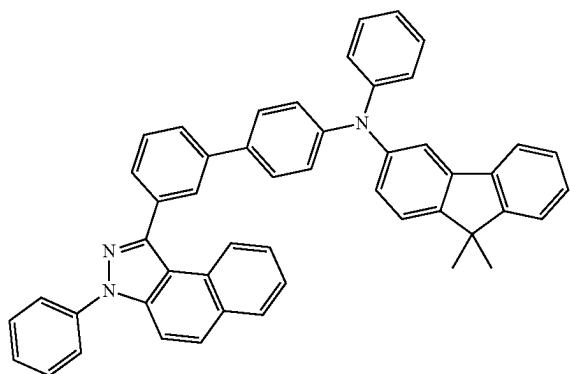
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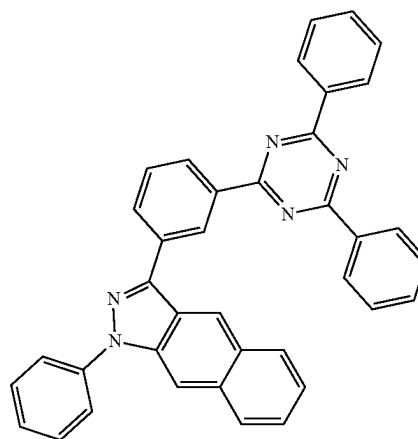
R80



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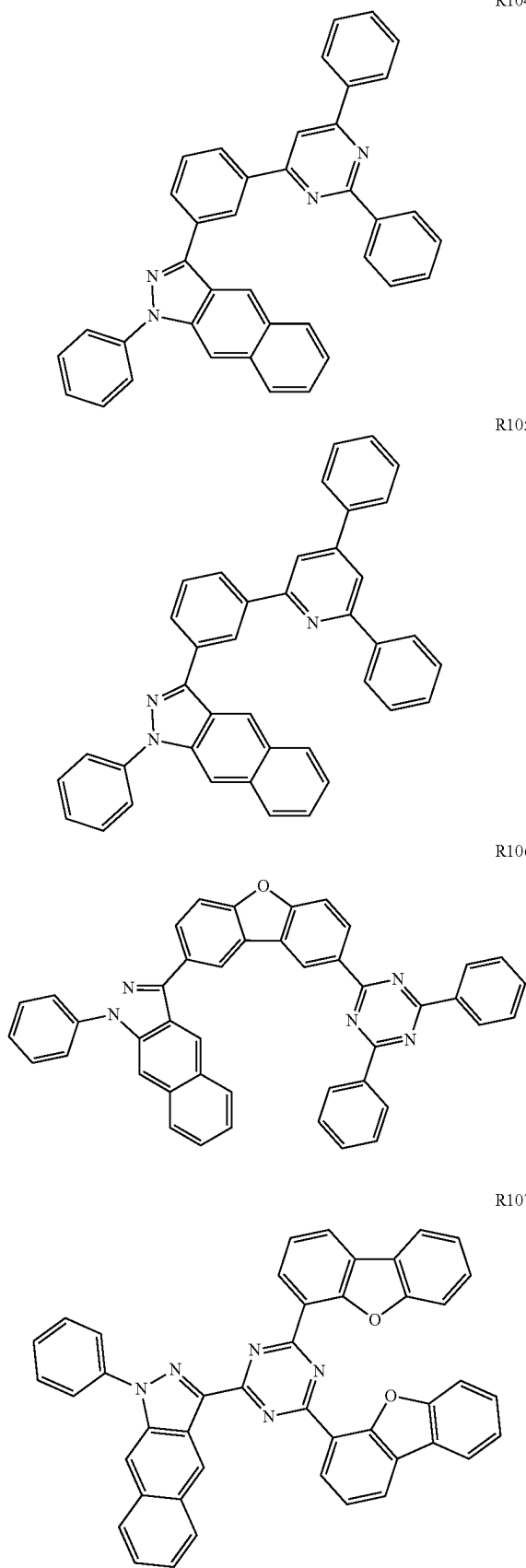
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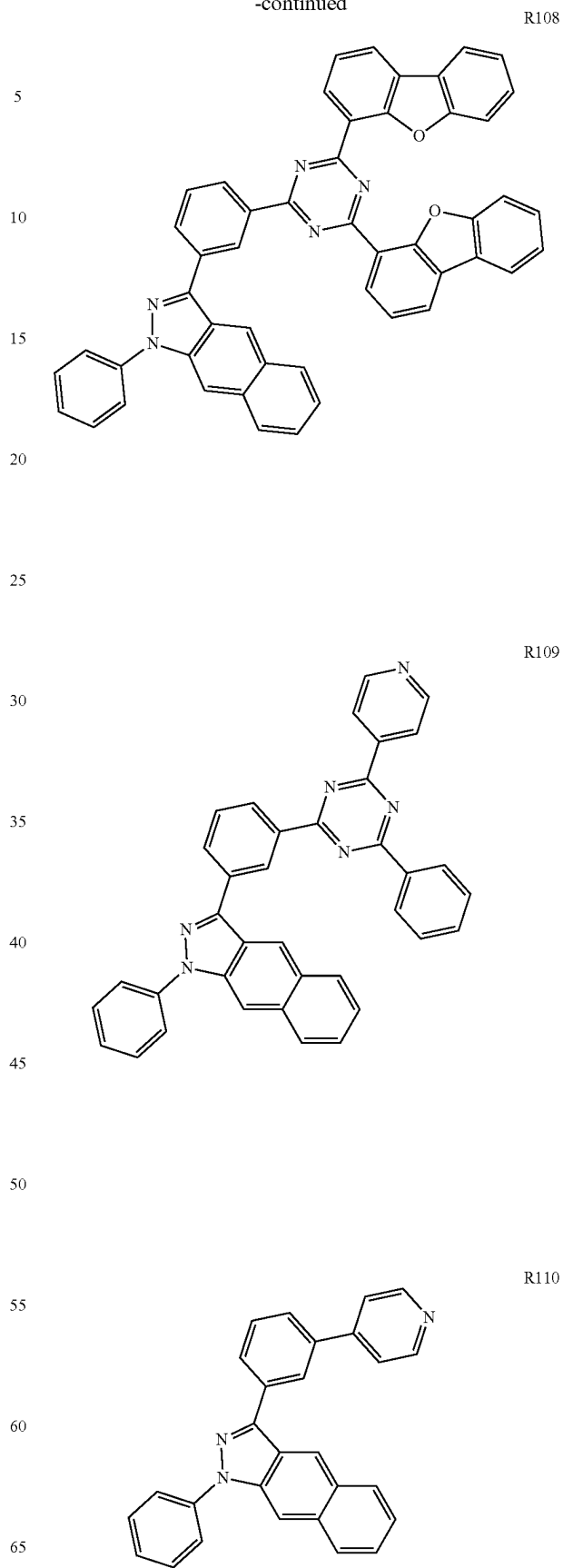
R103

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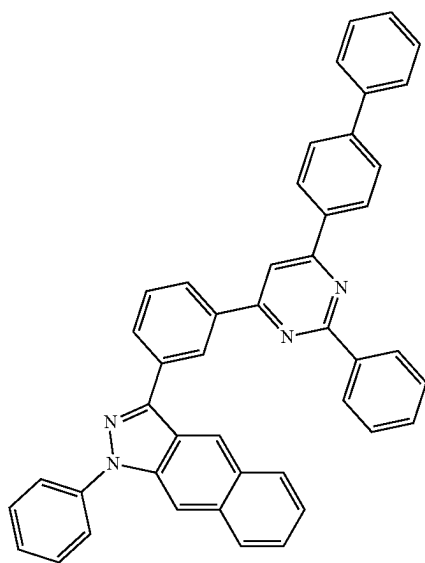
**46**

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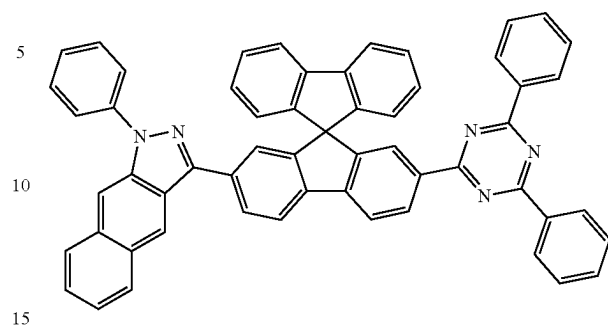
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R111

48

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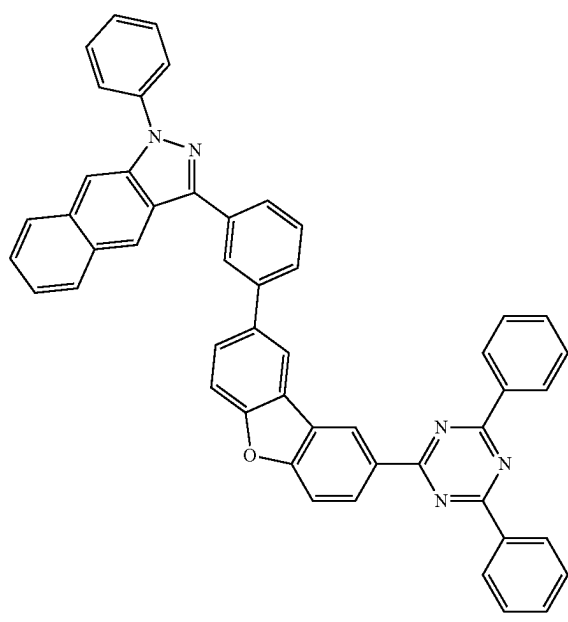


R114

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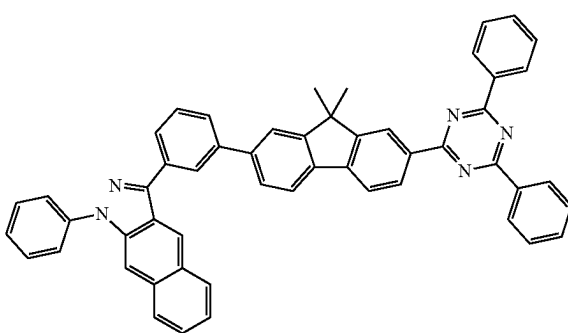
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R112



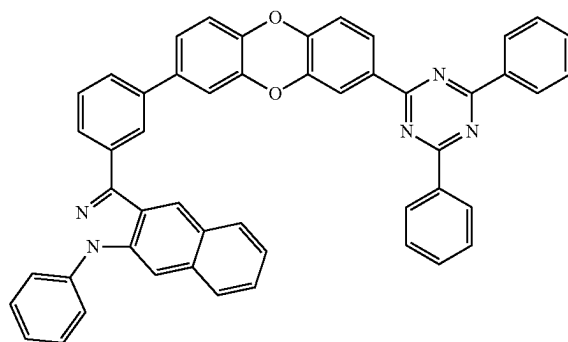
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R113

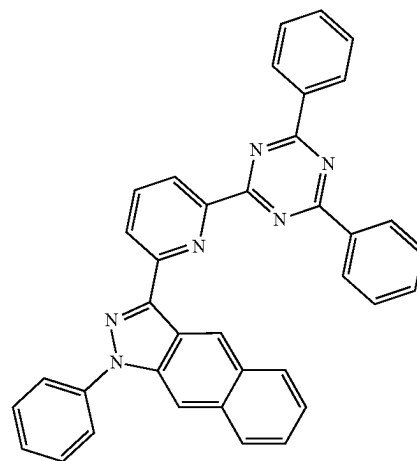


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R115



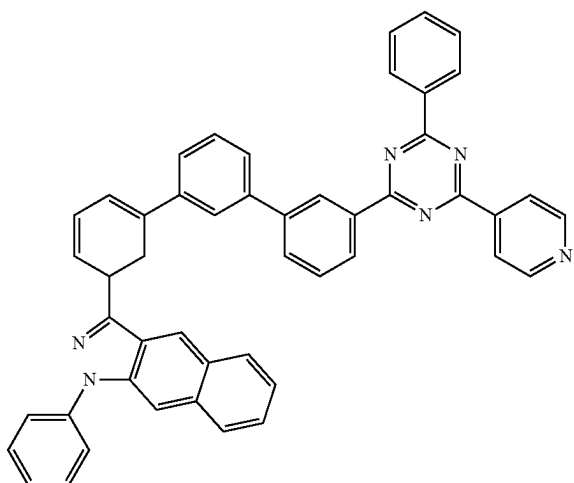
R116



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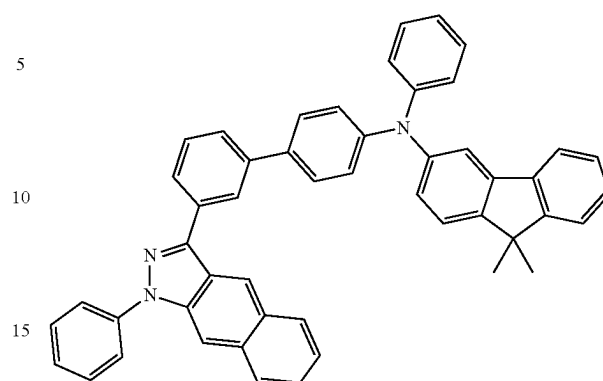
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R117

**50**

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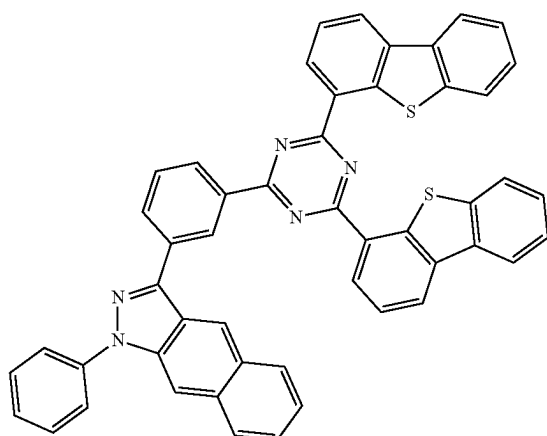
R120



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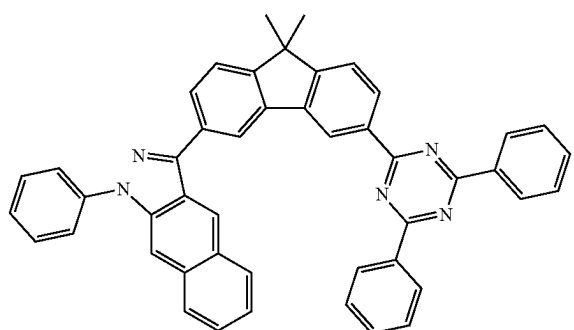
R118



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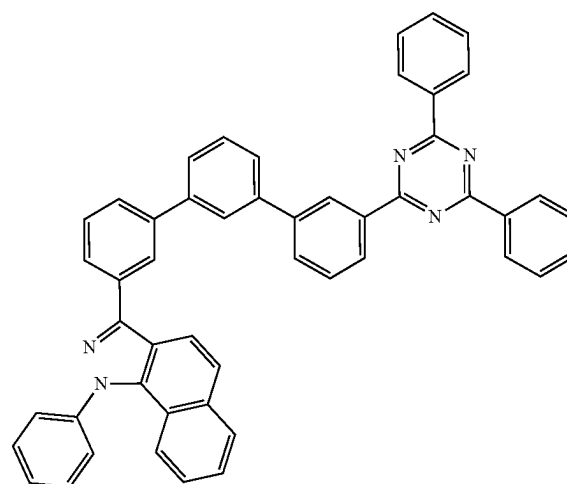
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R119

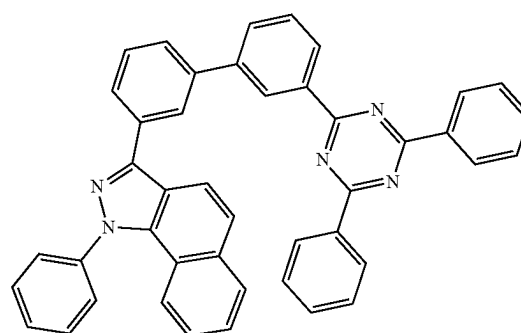


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R121

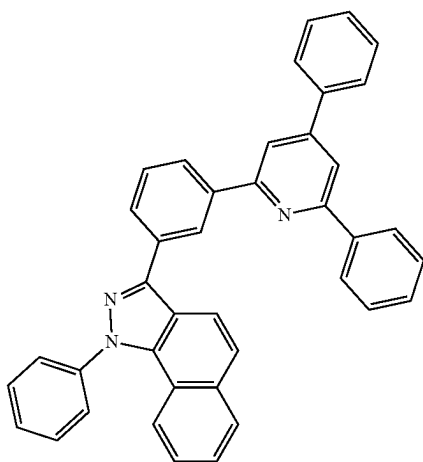
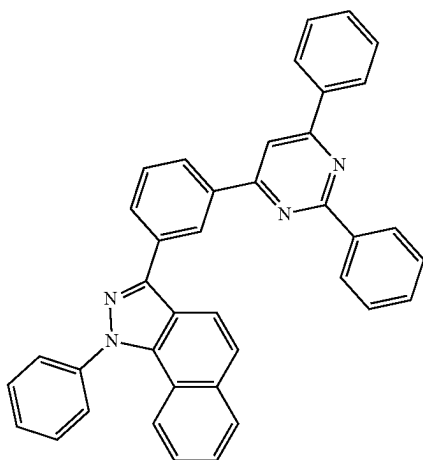
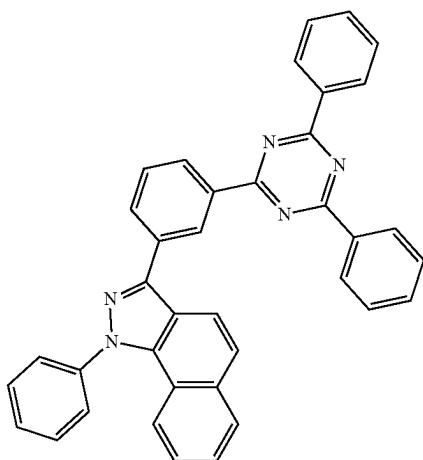


R122



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**52**

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R123

R126

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R124

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R125

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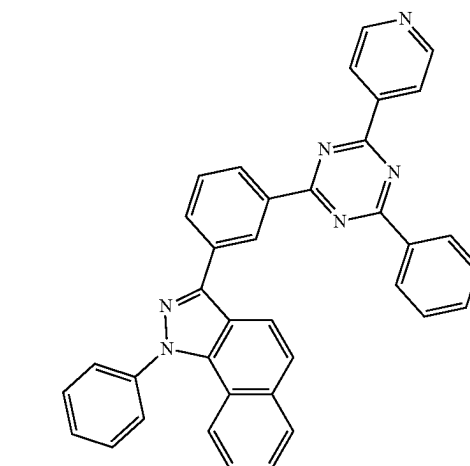
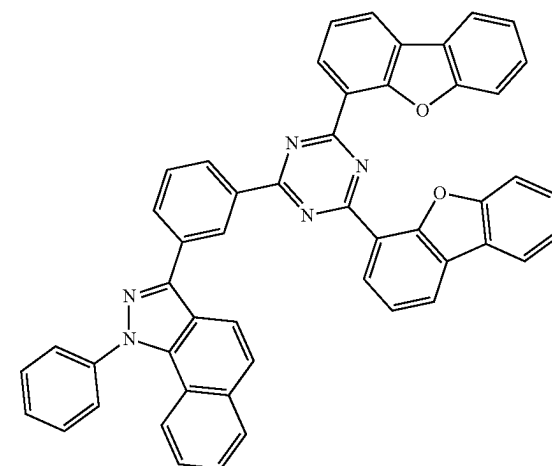
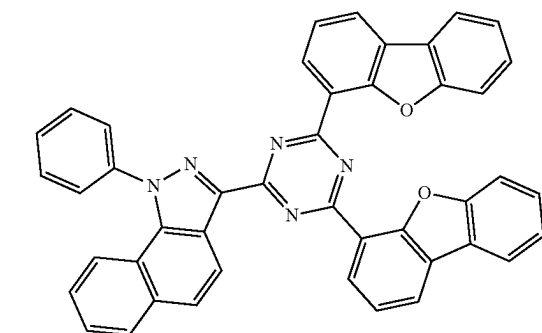
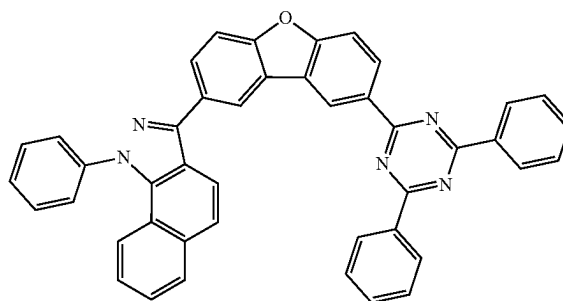
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R127

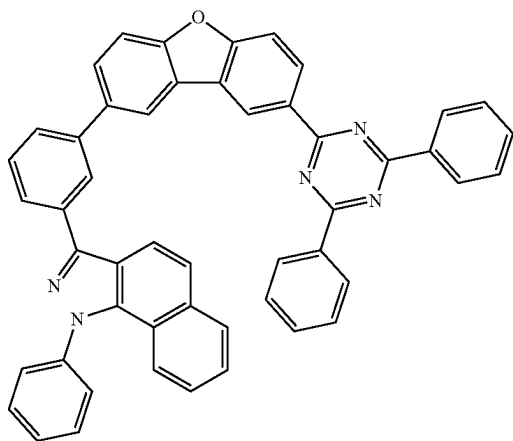
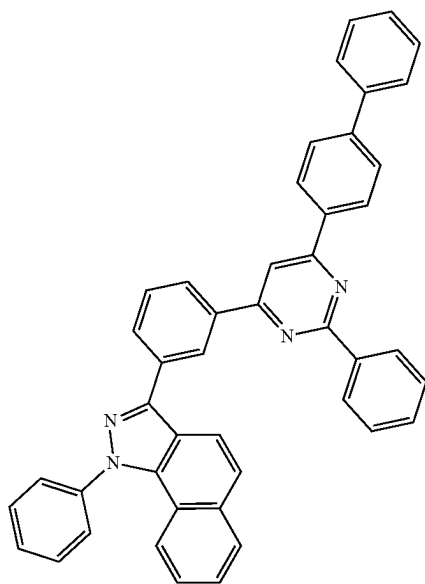
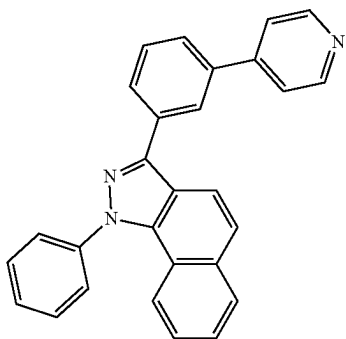
R128

R129



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**54**

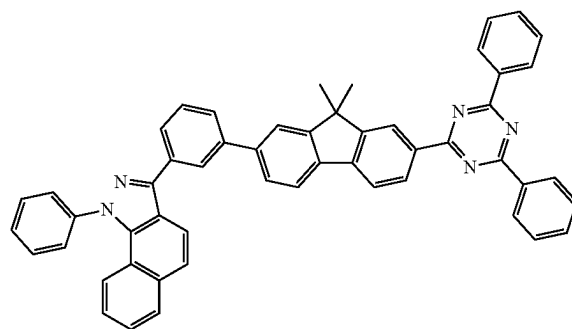
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R130

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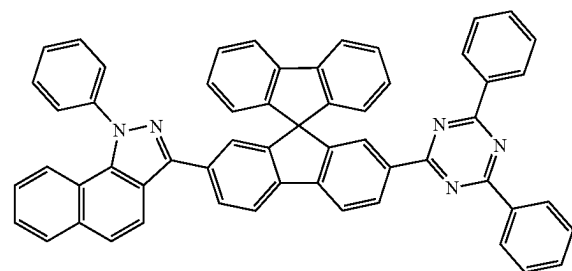
R133

R131

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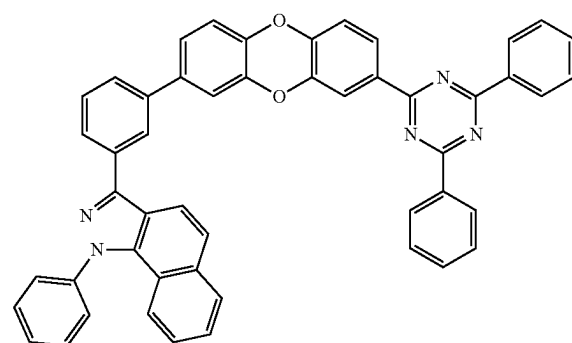


R134

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R135

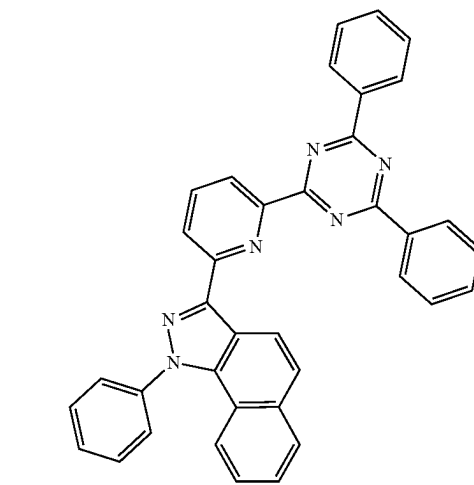
R132

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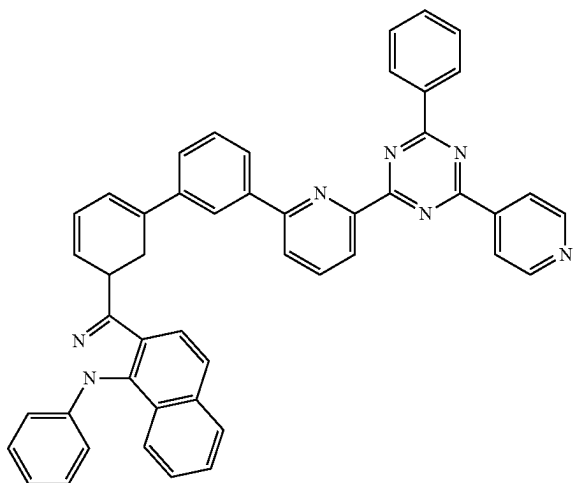


R136

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R137



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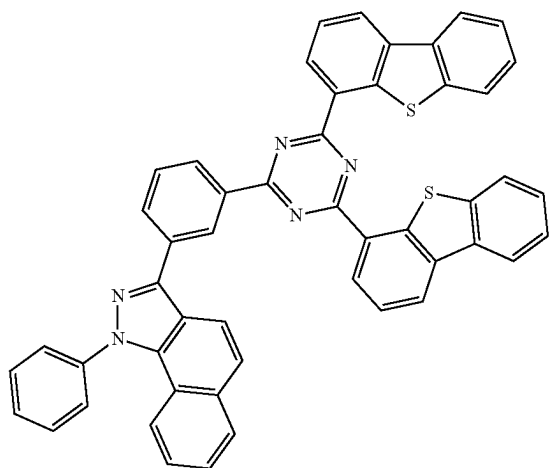
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R138



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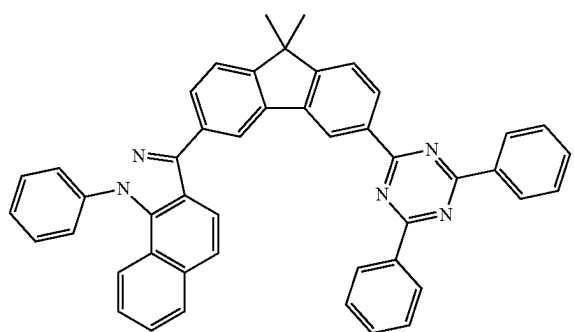
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R139



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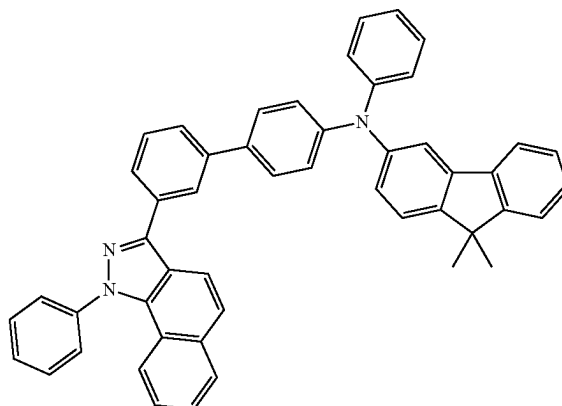
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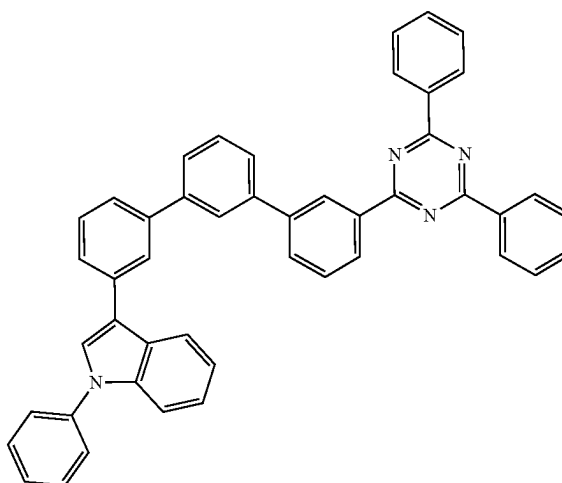
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R140



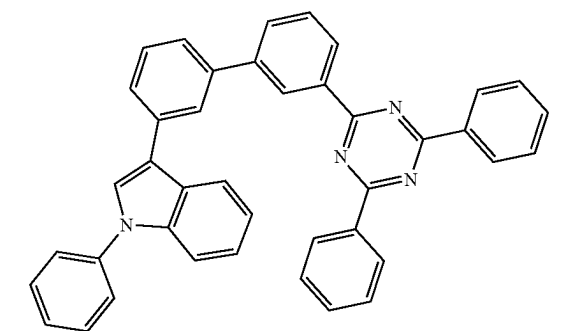
R141



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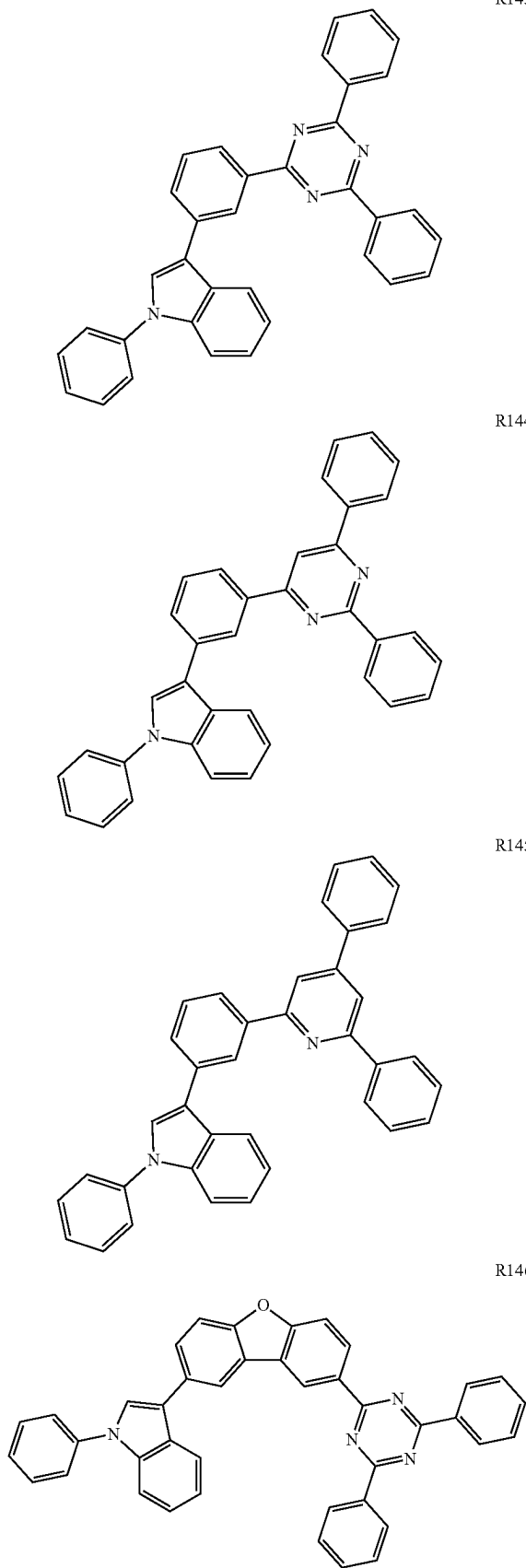
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R142

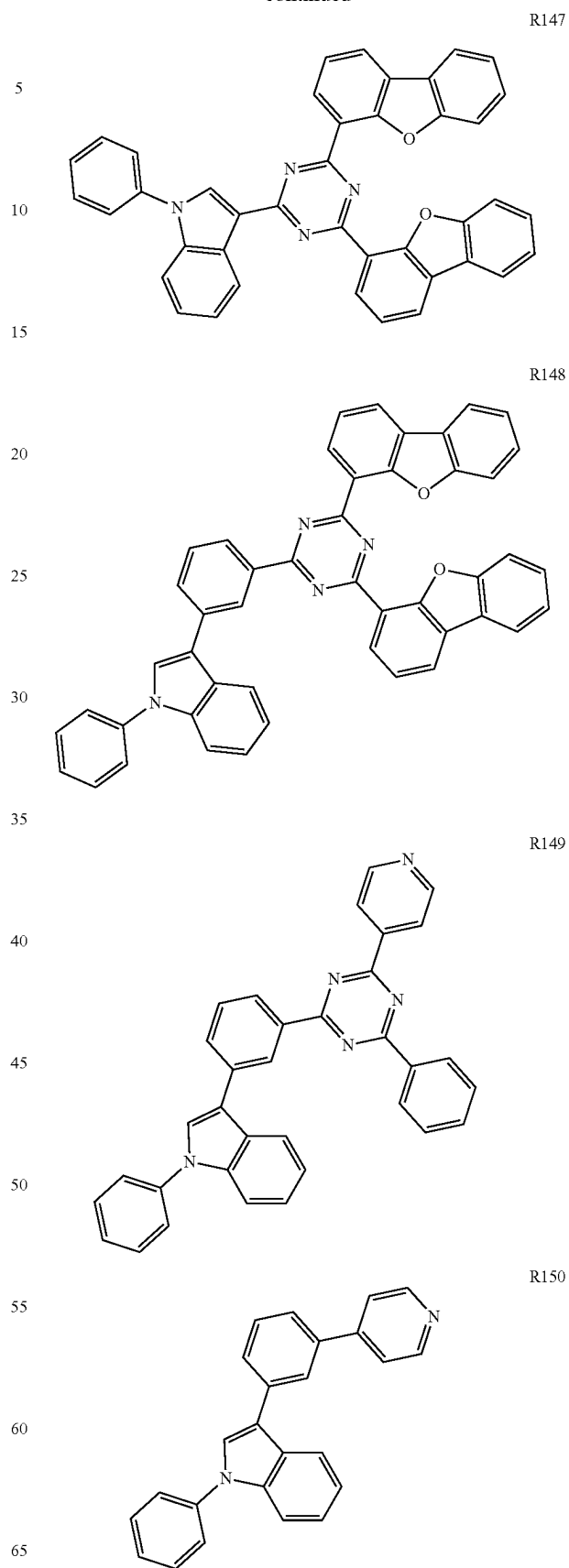


57

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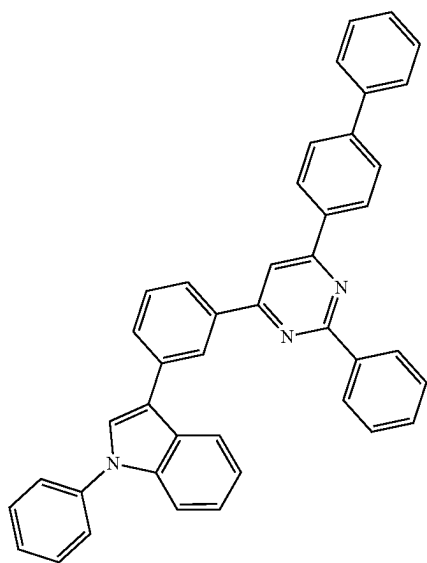
**58**

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59

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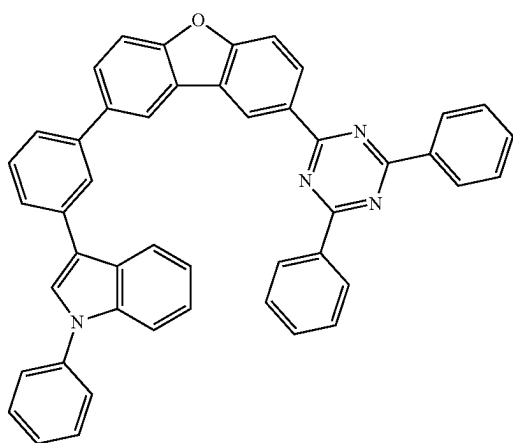
R151

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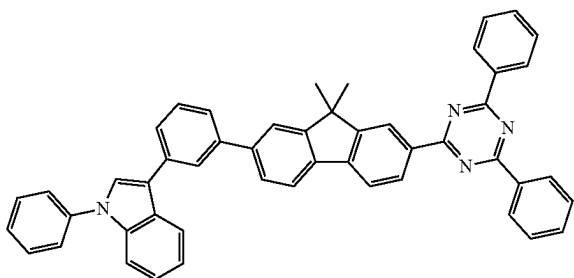
R152

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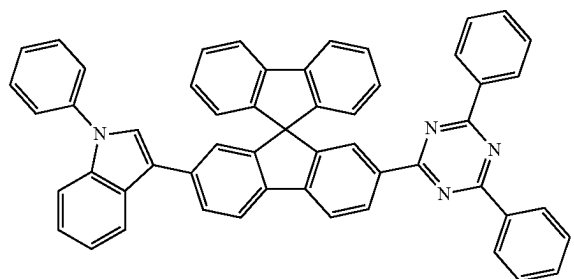
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R153

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R154

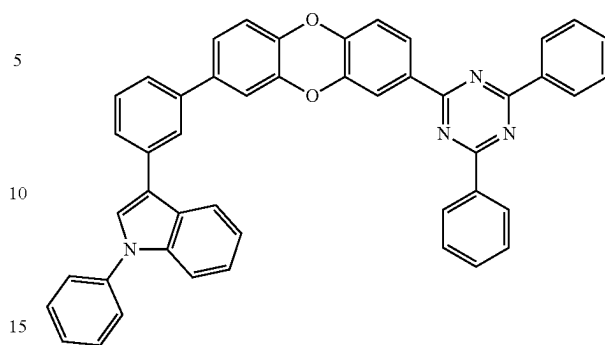
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R155

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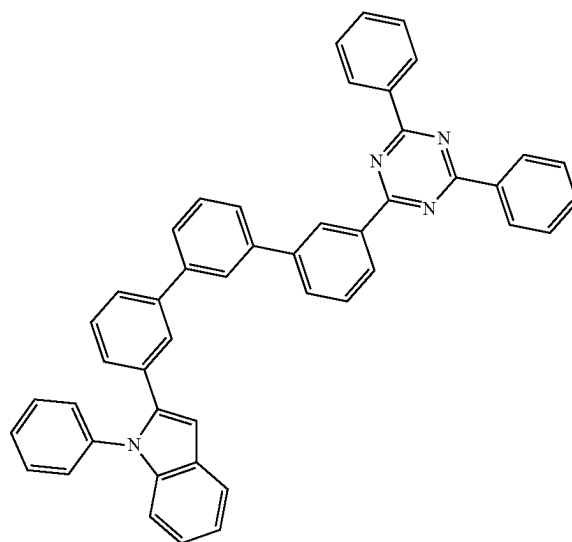
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R156

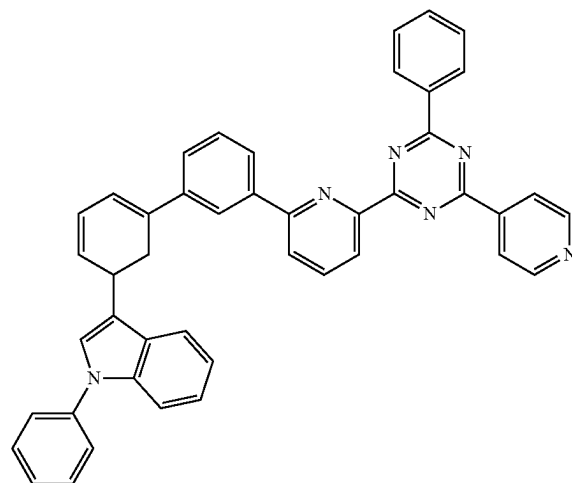
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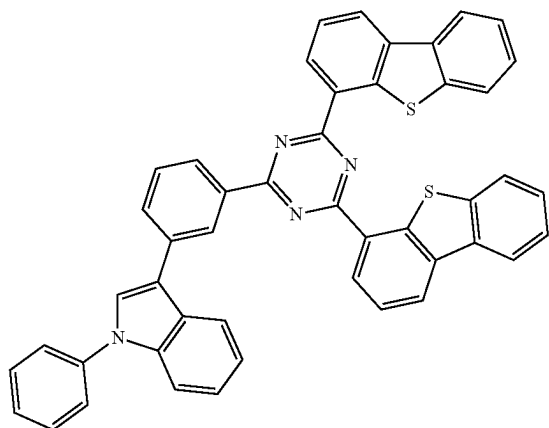


R157

61

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R158



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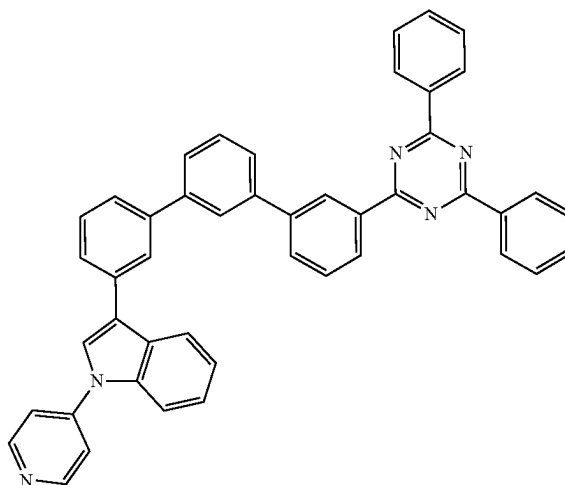
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62

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R161



R159

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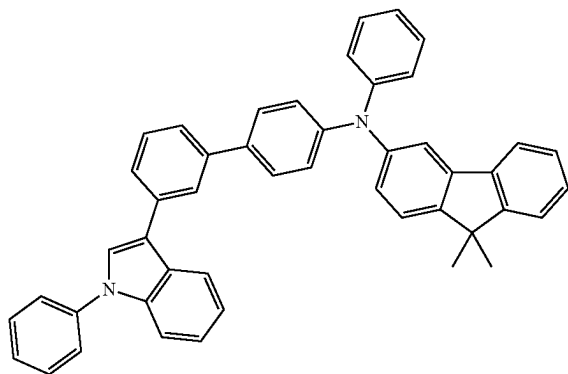
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R160

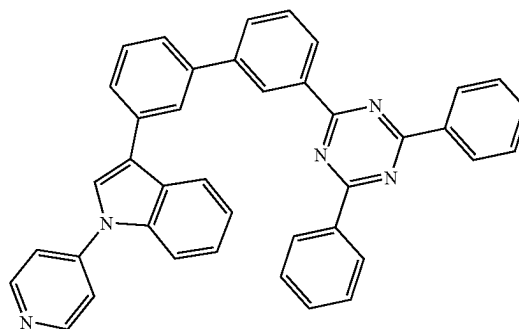


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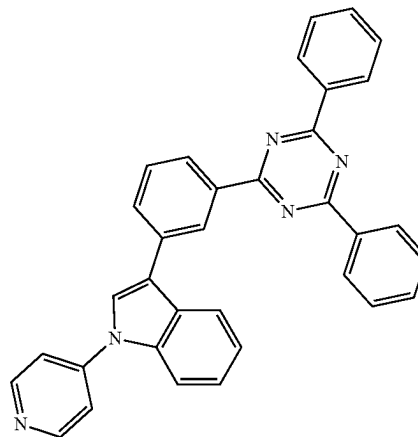
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R162

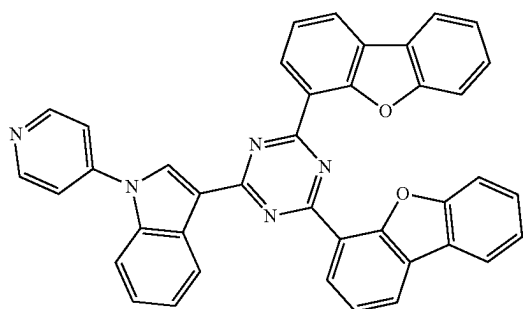
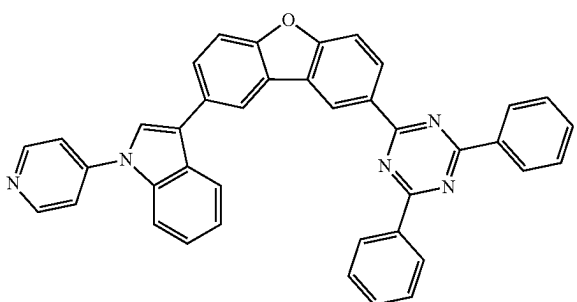
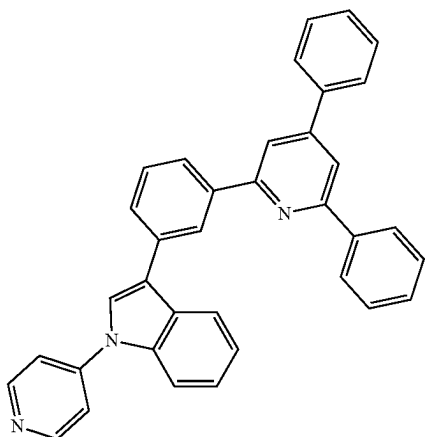
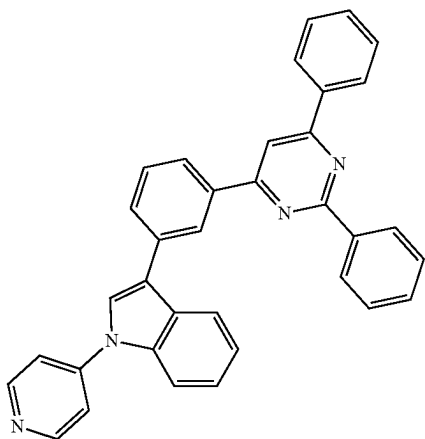


R163

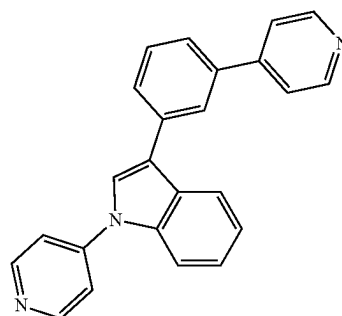
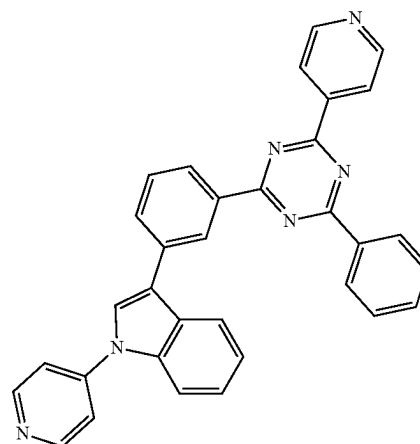
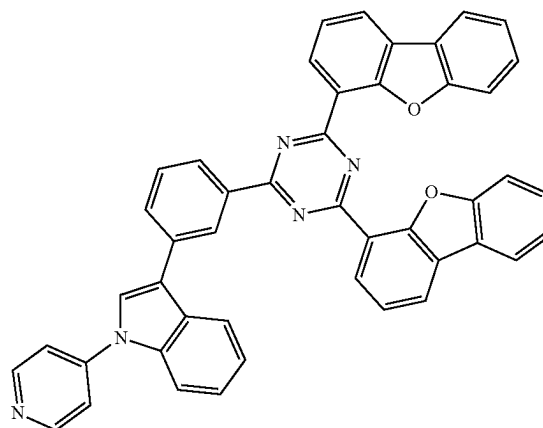


63

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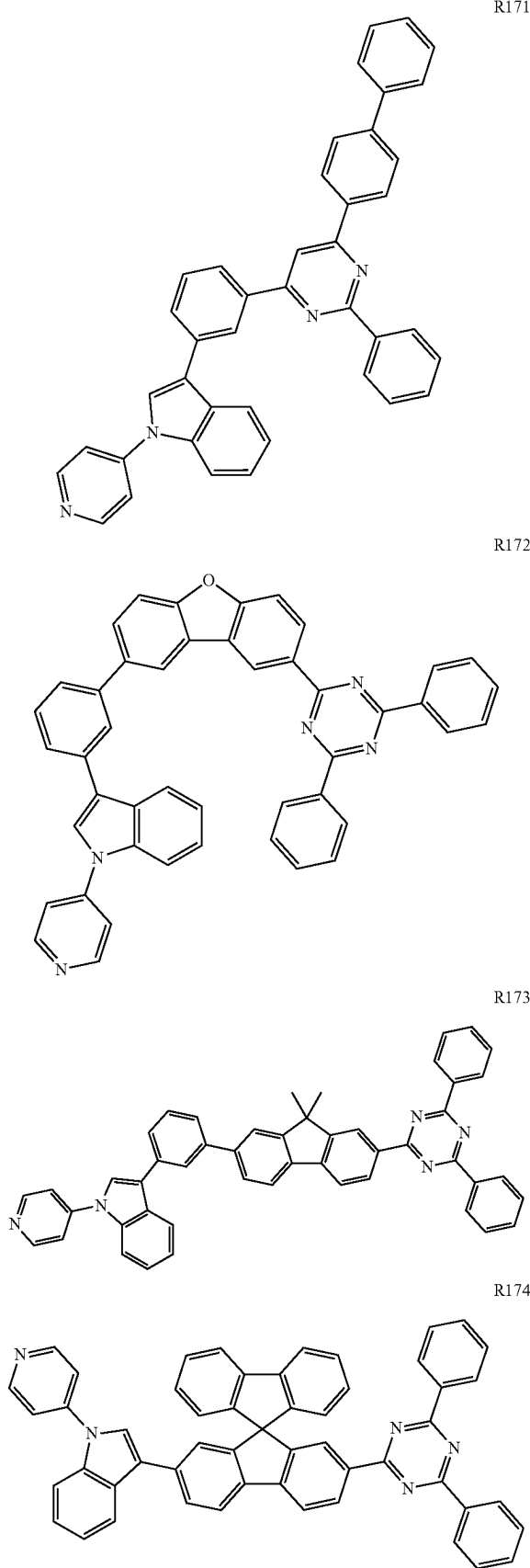
**64**

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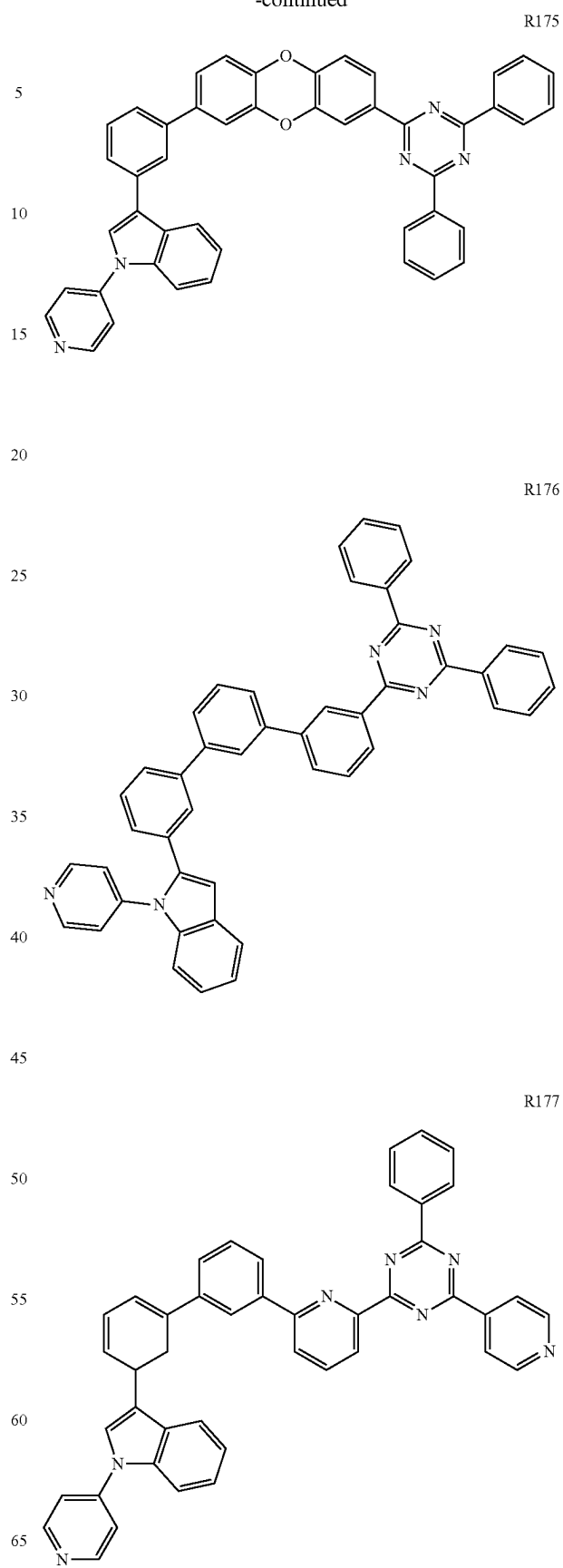


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**66**

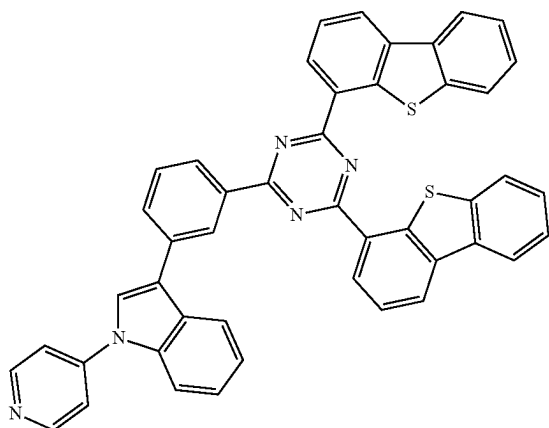
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67

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R178



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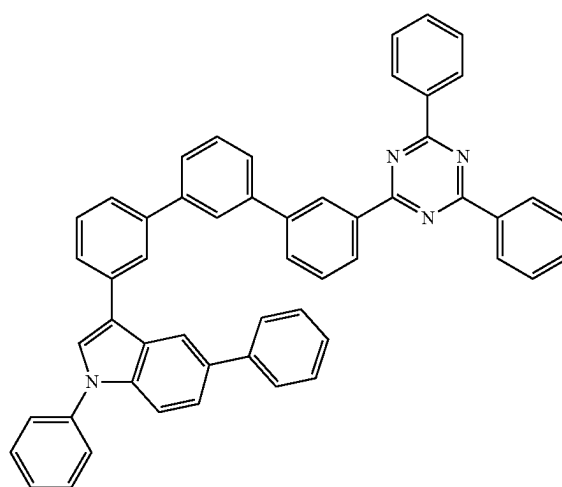
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68

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R181



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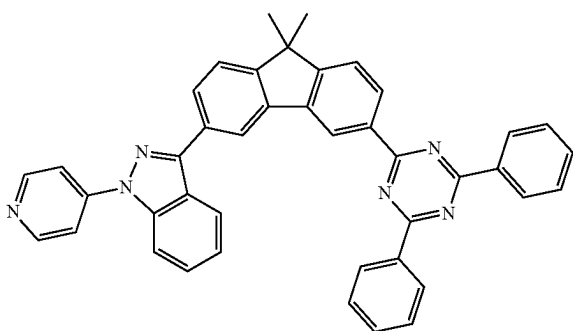
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R179

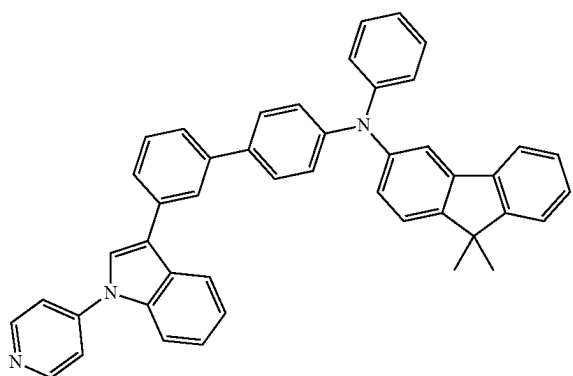


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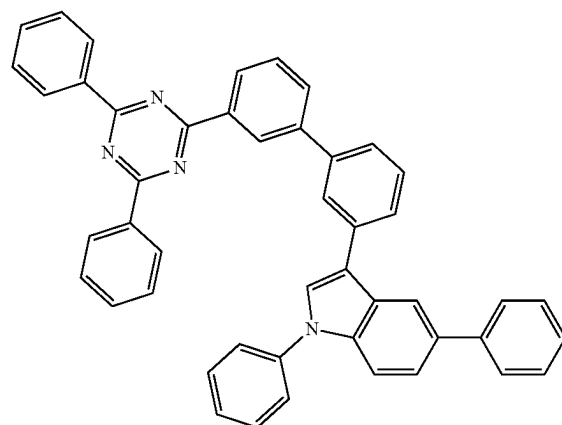
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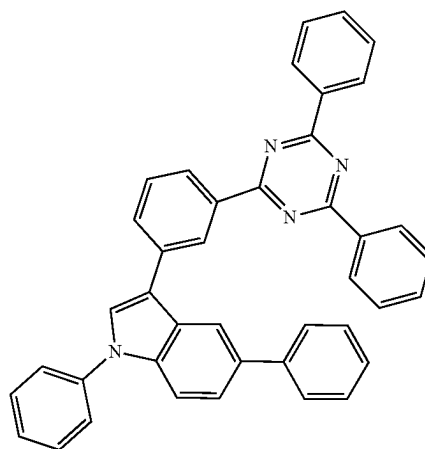
R180



R182

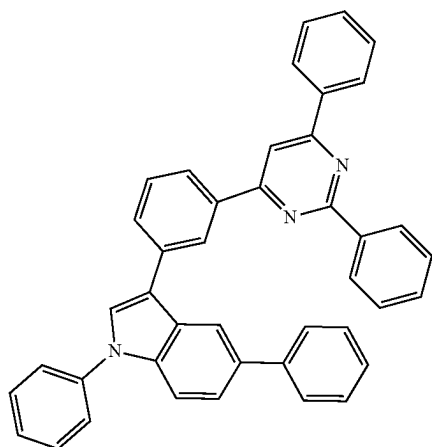


R183



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R184

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R185

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R186

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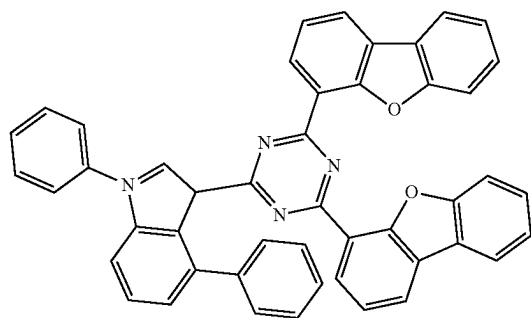
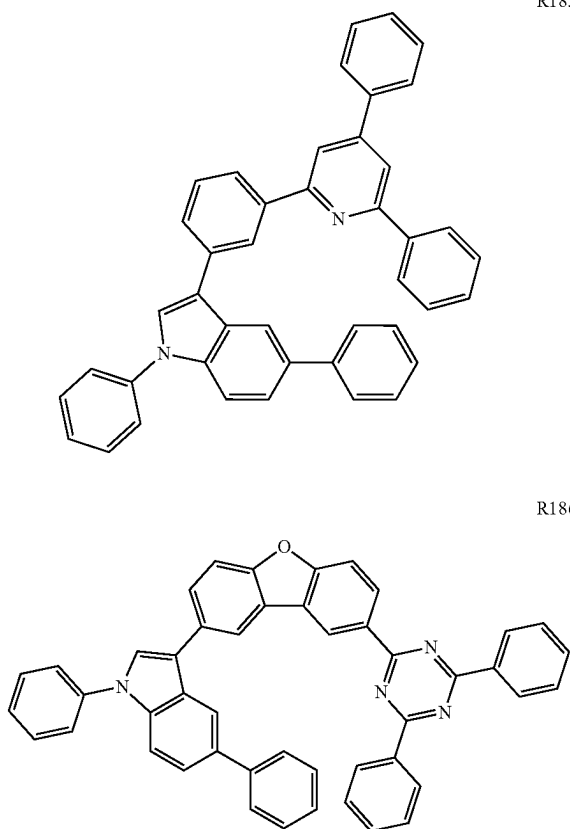
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R187

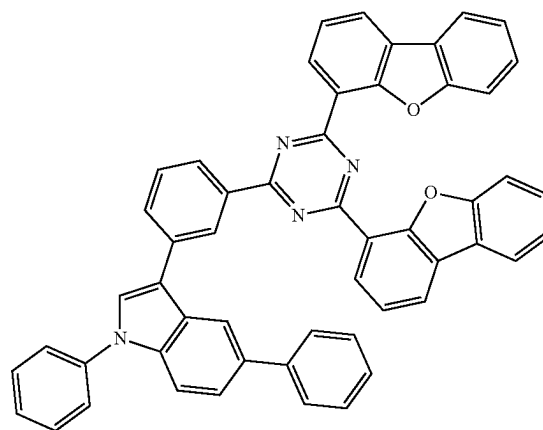
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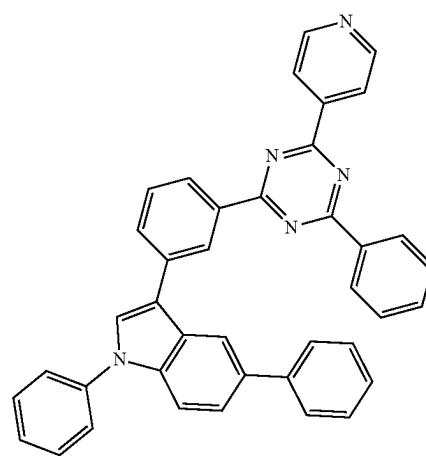
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**70**

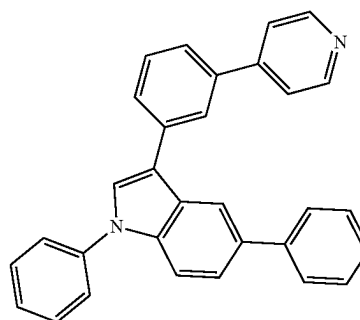
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R188



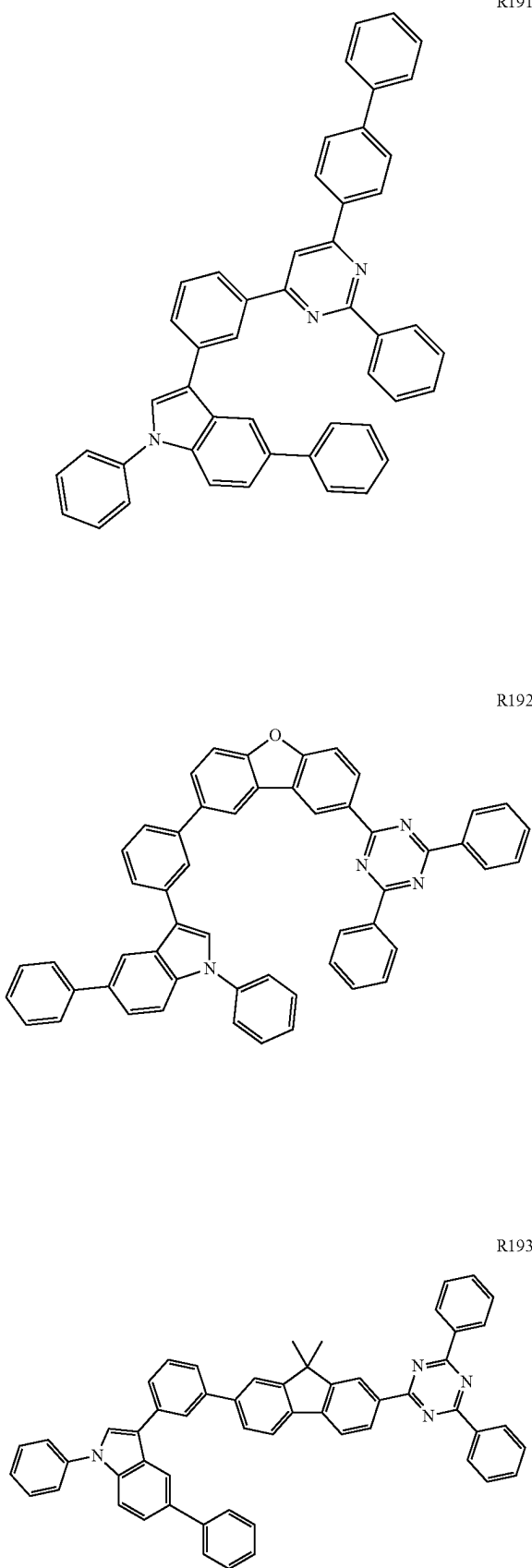
R189



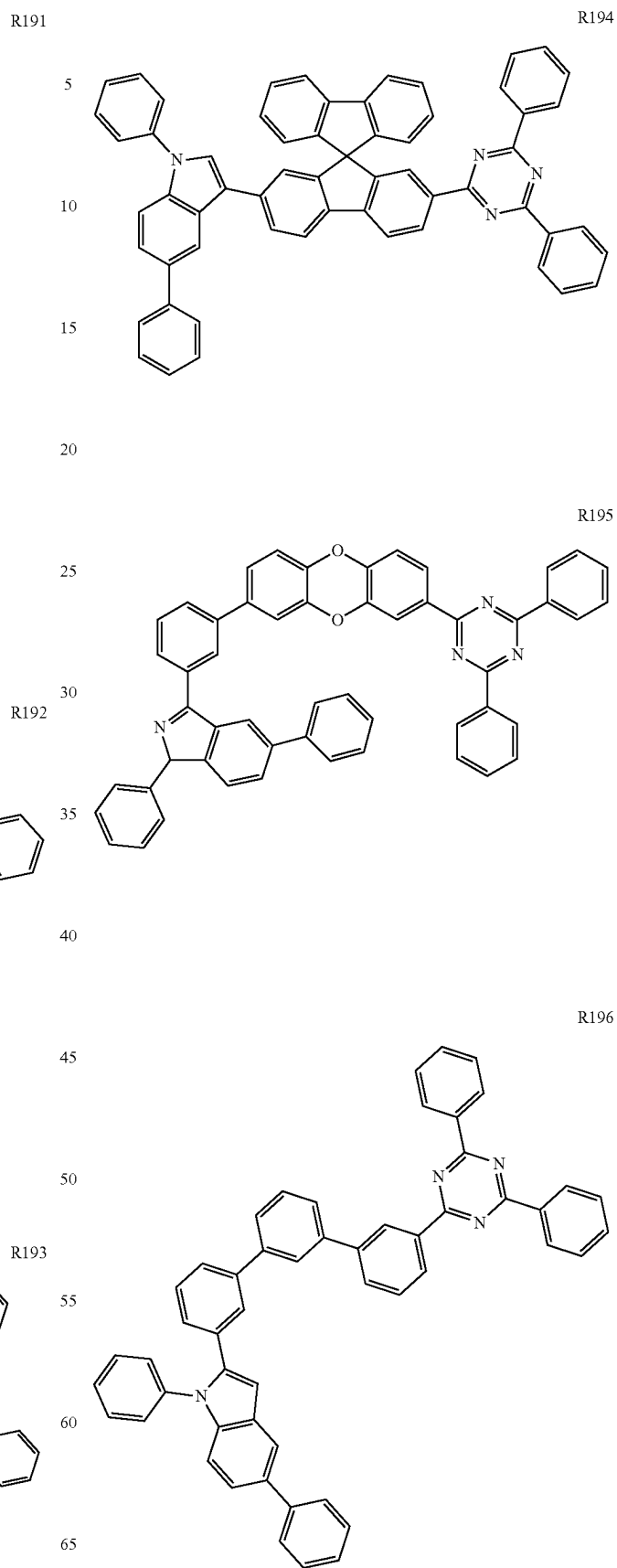
R190

71

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**72**

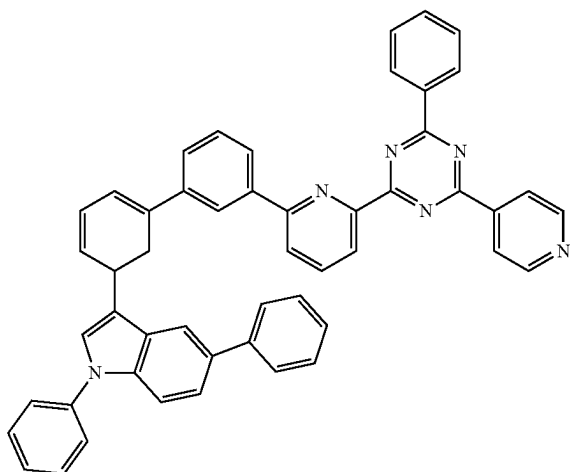
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73

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R197



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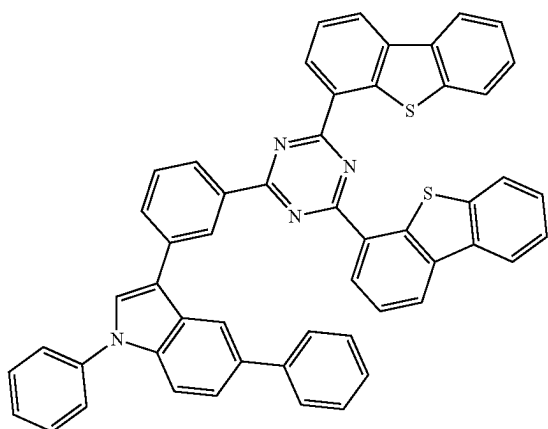
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R198



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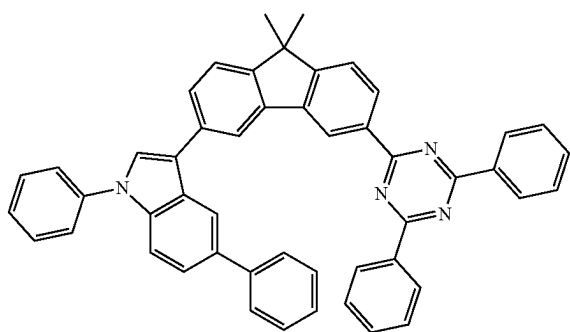
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R199



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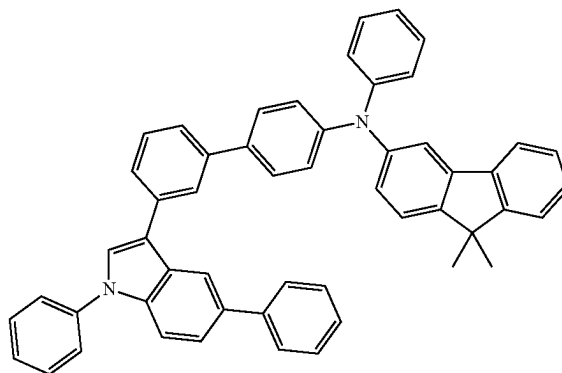
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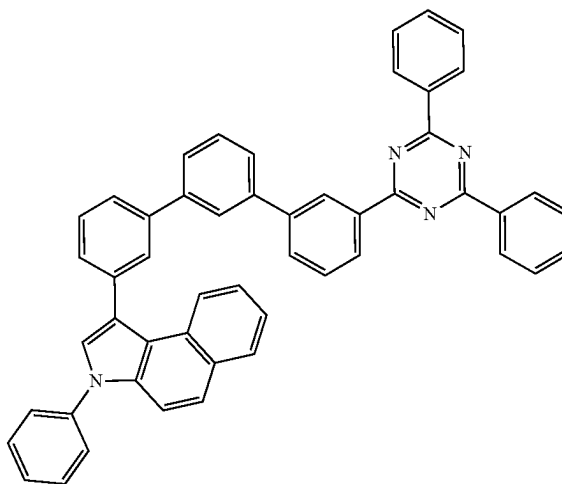
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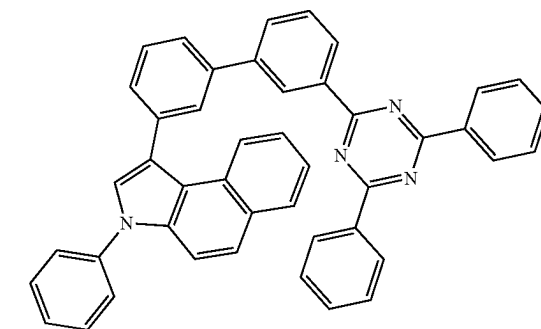
R200



R201

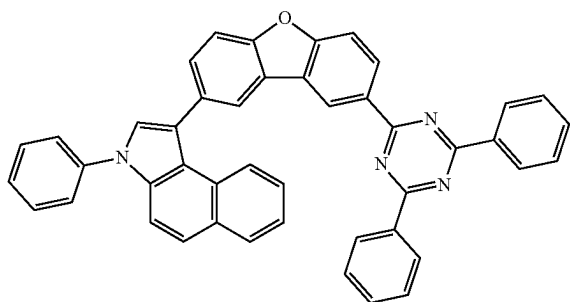
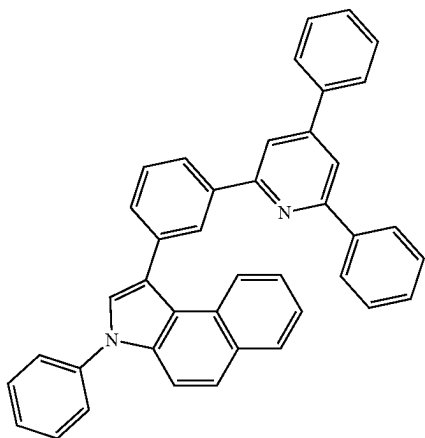
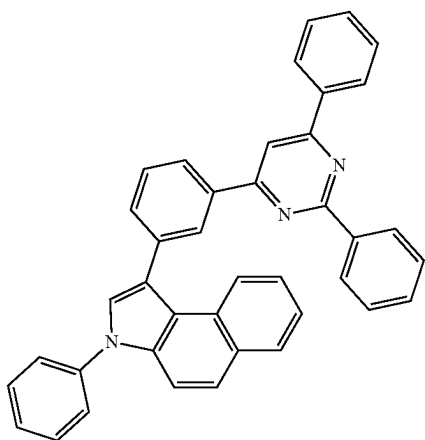
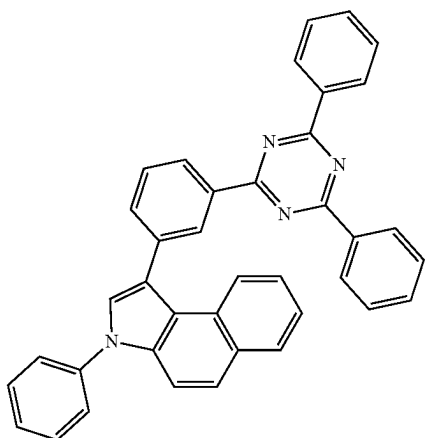


R202

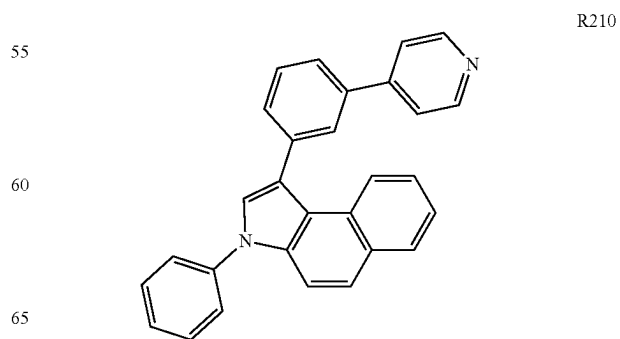
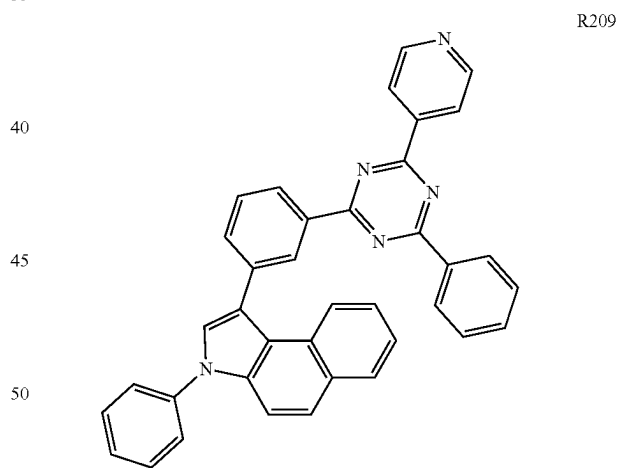
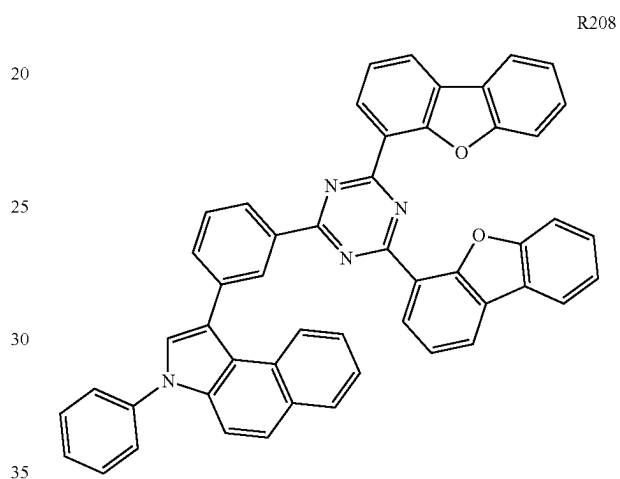
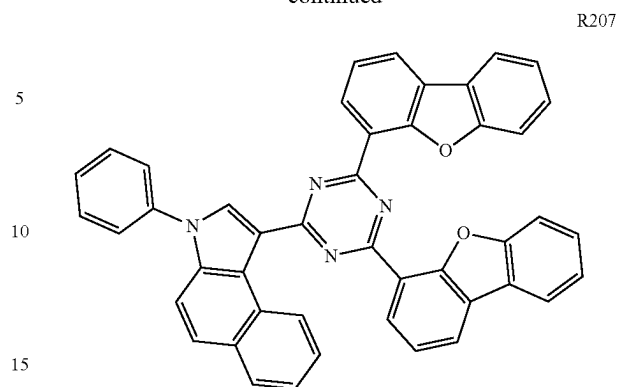


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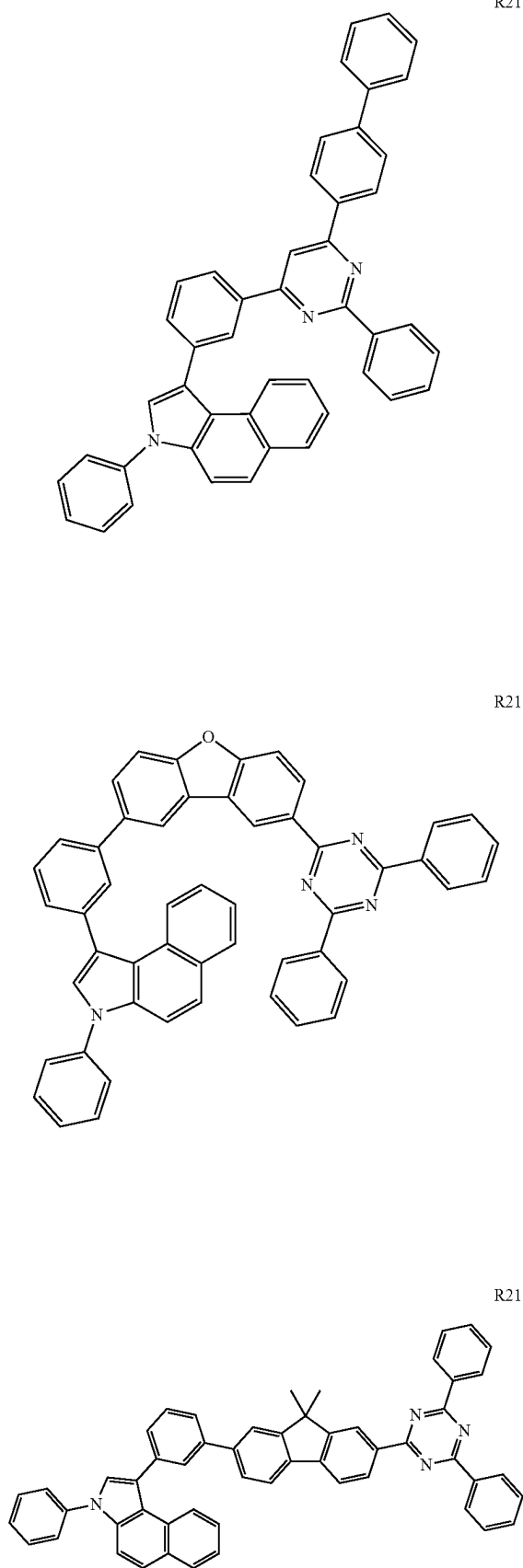
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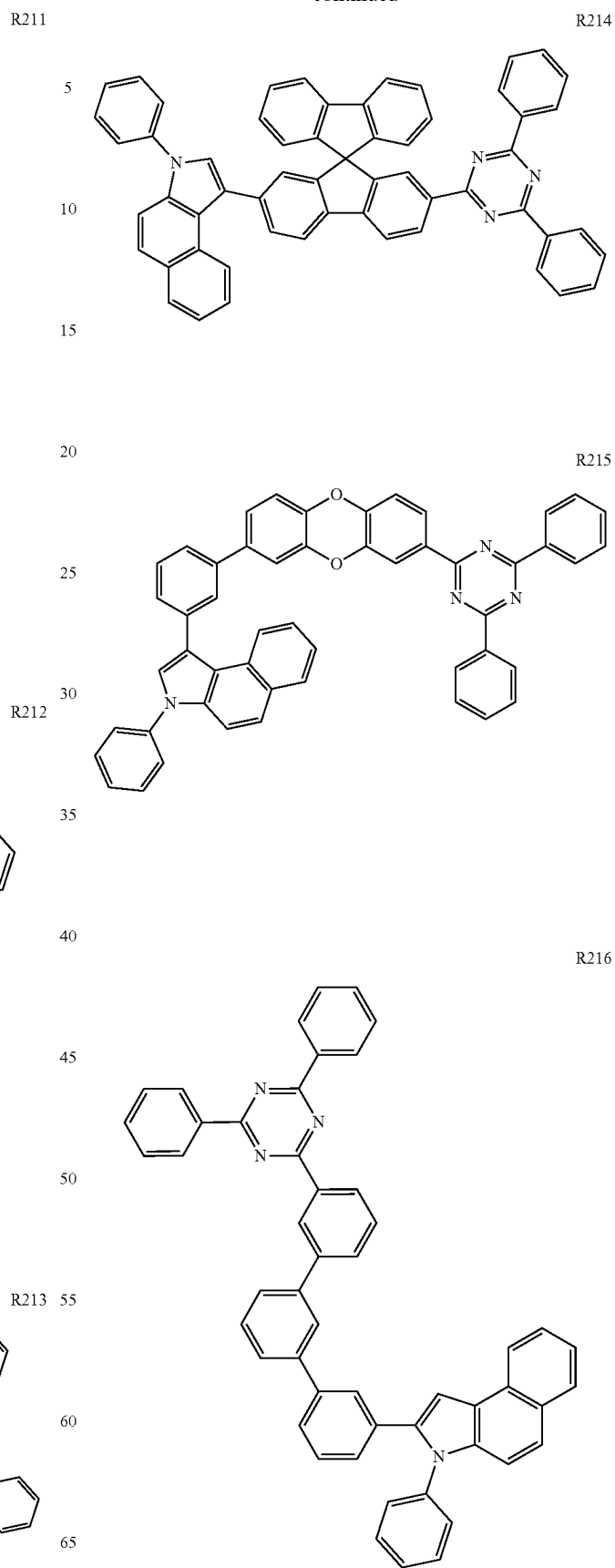


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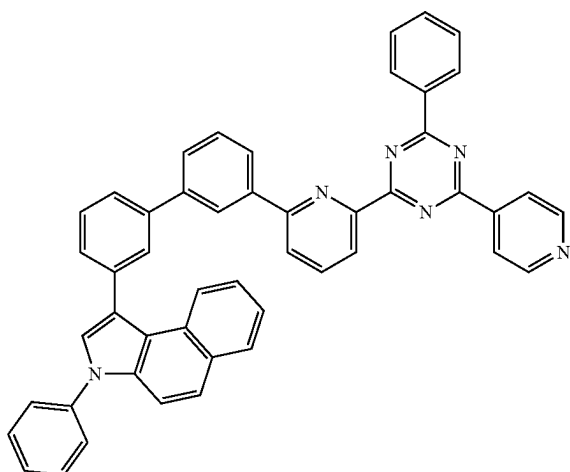
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R217



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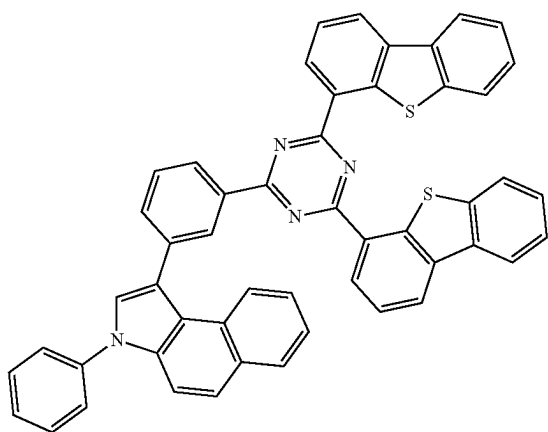
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R218



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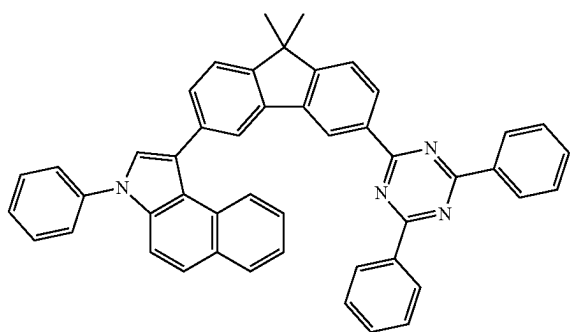
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R219



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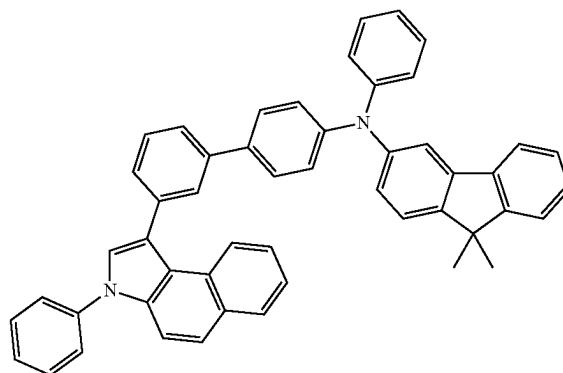
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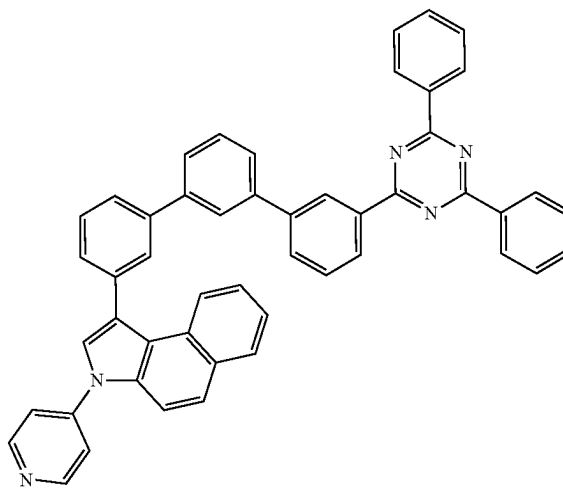
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R220



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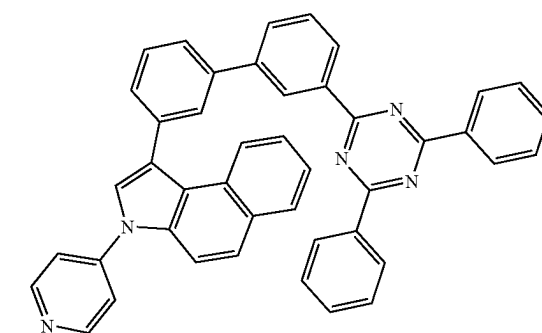
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R222



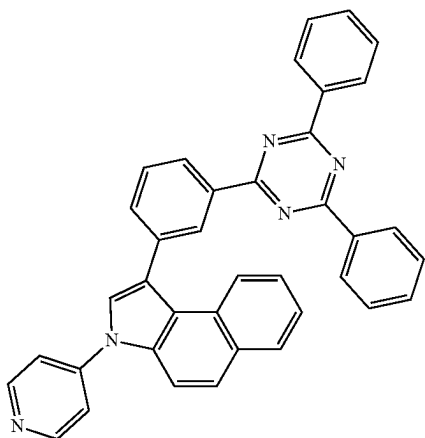
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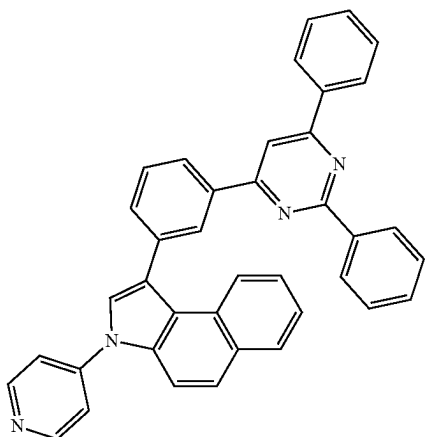


R223

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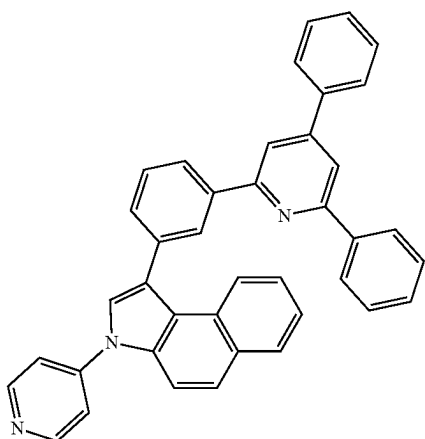
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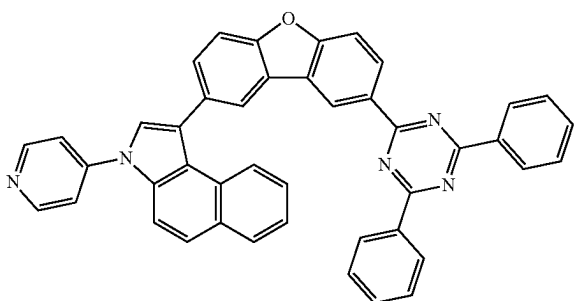


R225

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R226

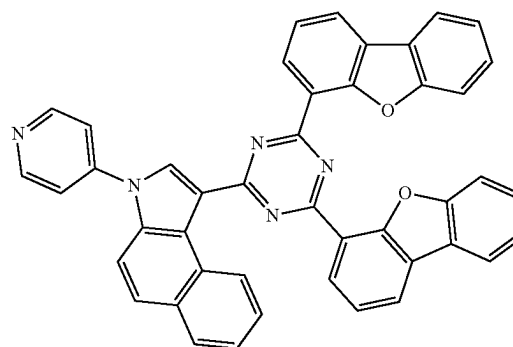
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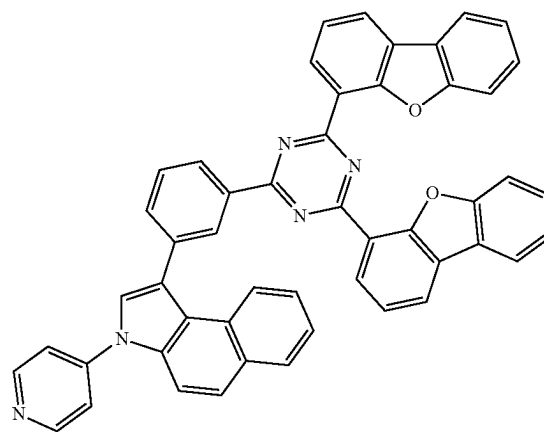
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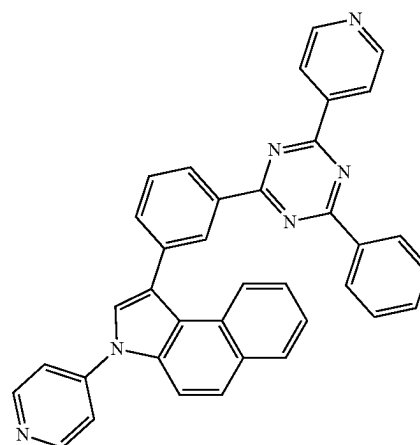


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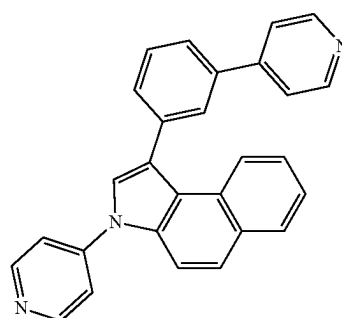
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R229

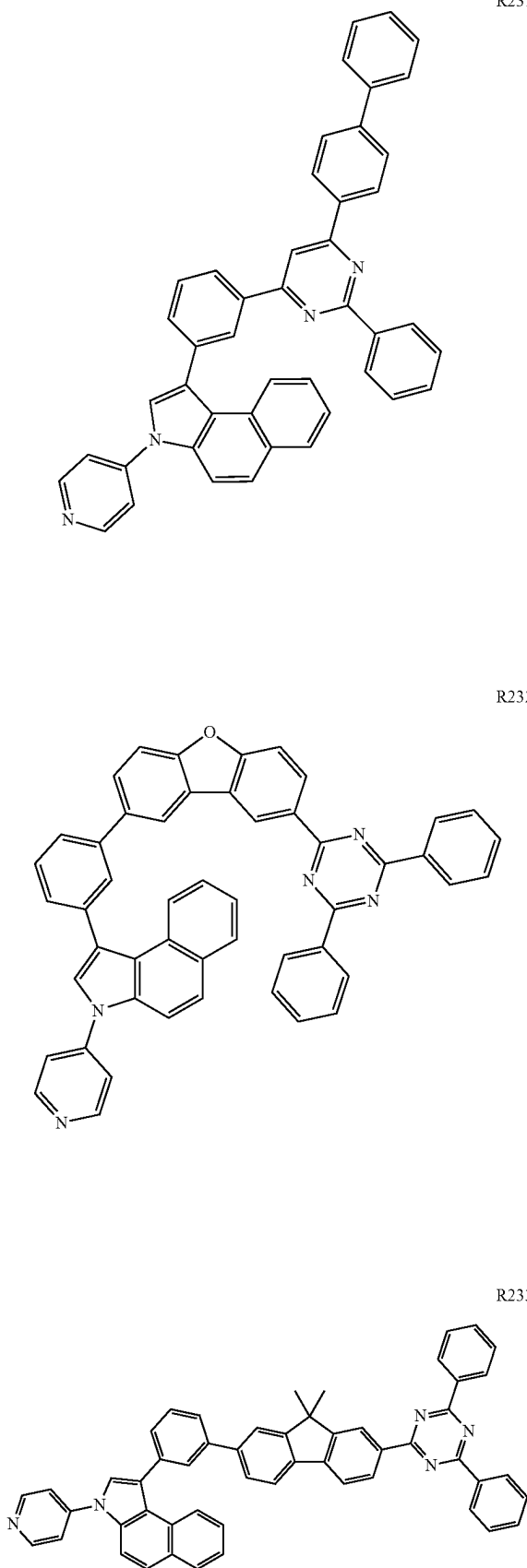
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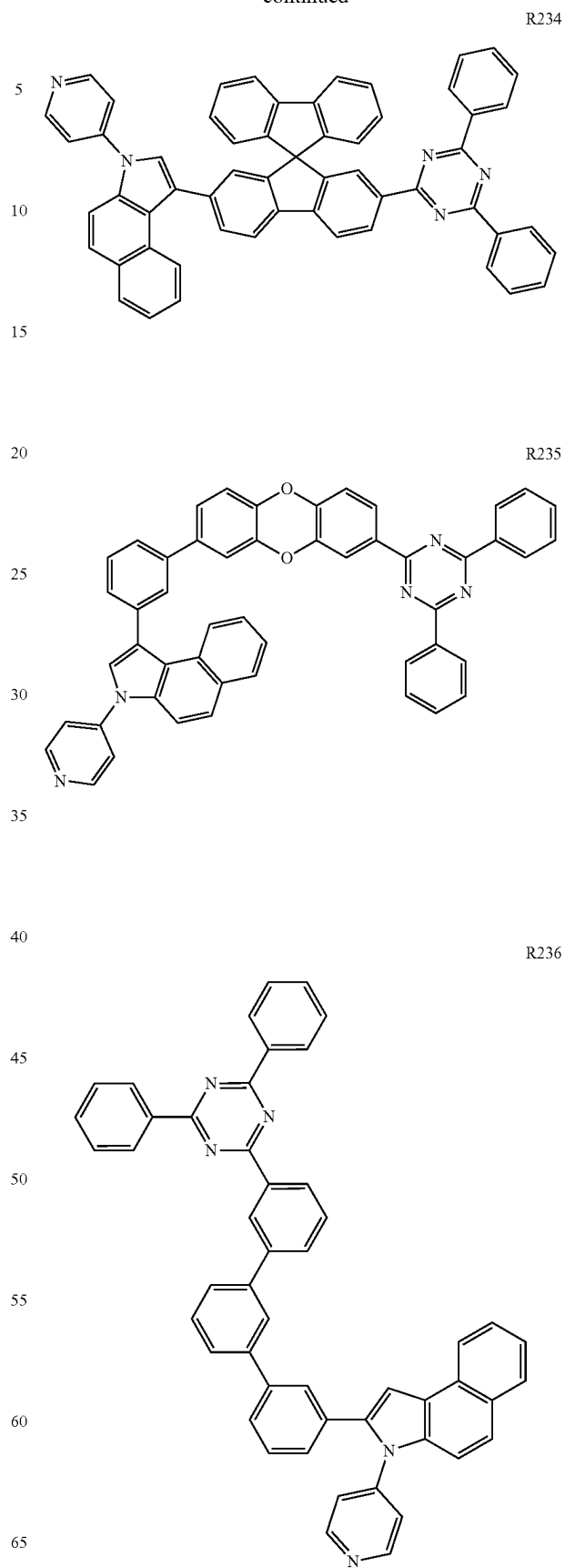
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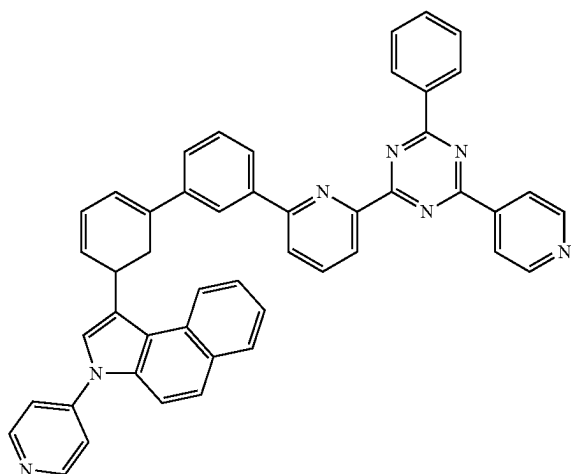
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85

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R237



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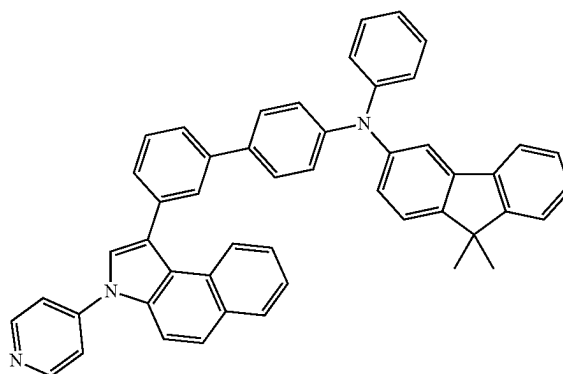
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R240



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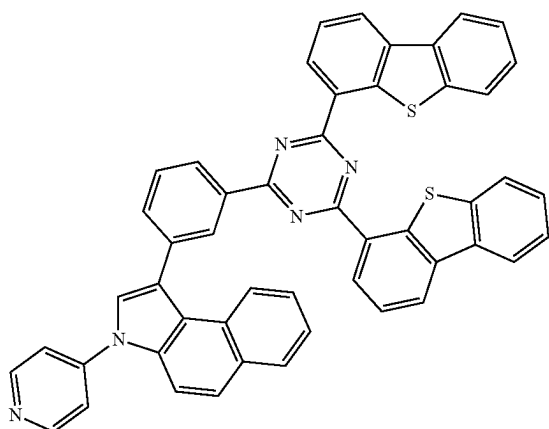
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R238



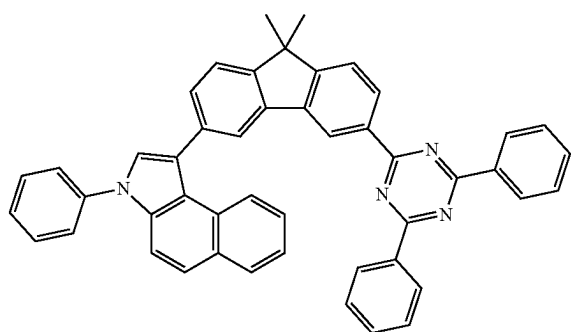
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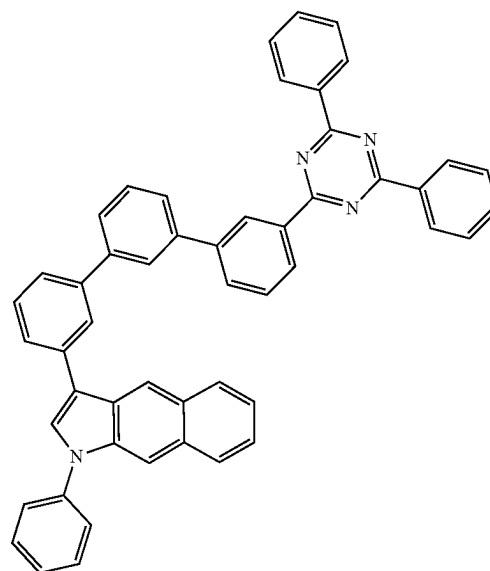
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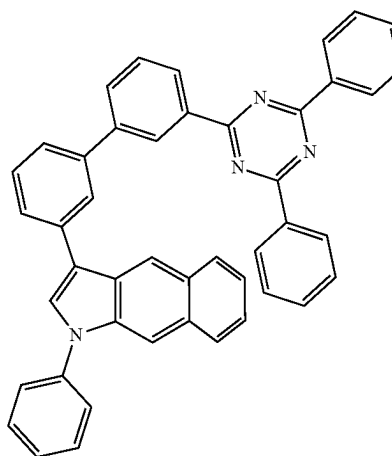
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R241

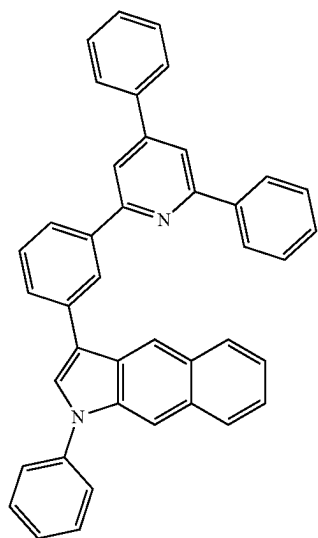
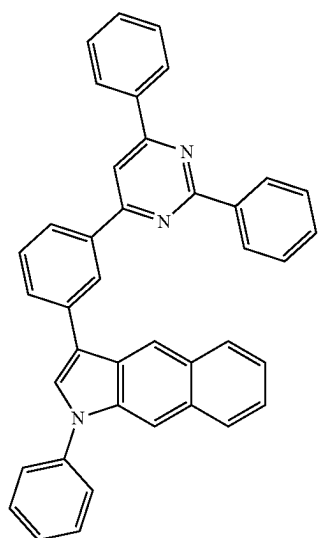
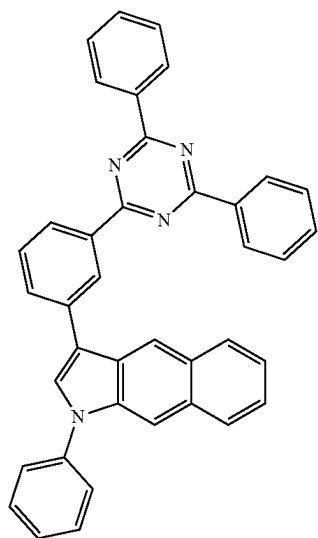


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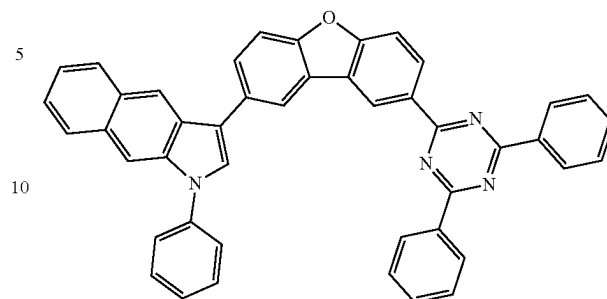
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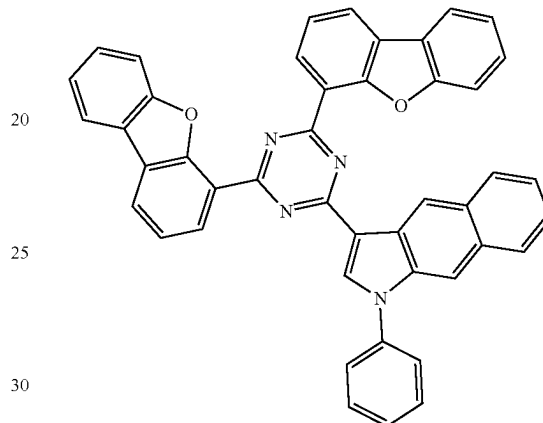
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R246



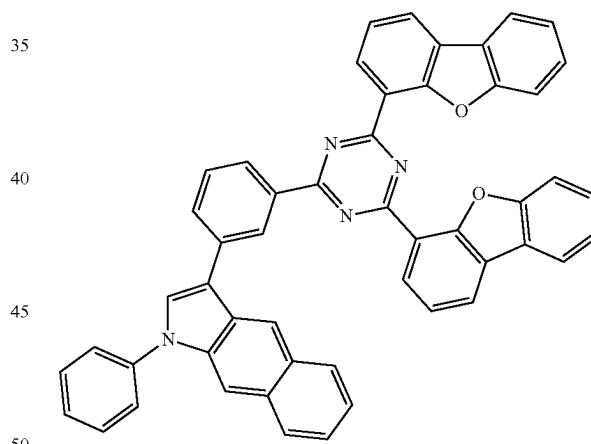
R247

R244

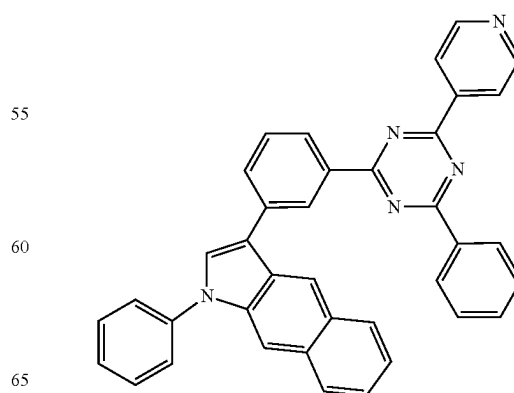


R248

R245

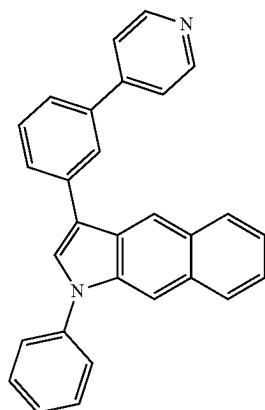


R249



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**90**

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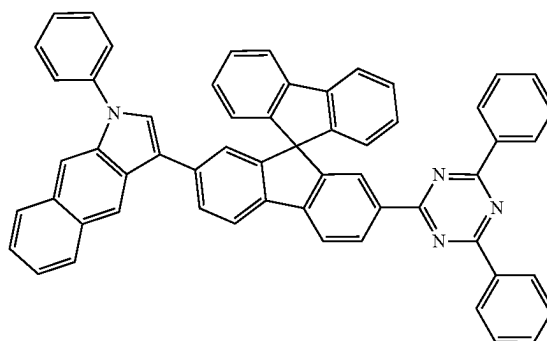
R250

R254

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R25 20

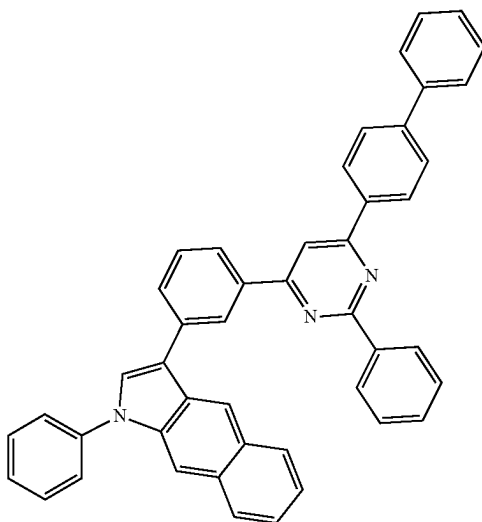
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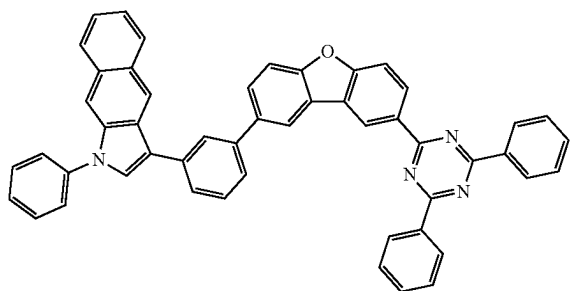
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R252

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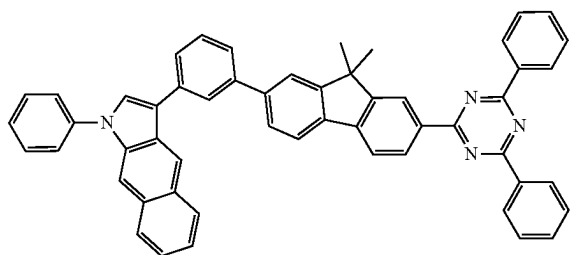


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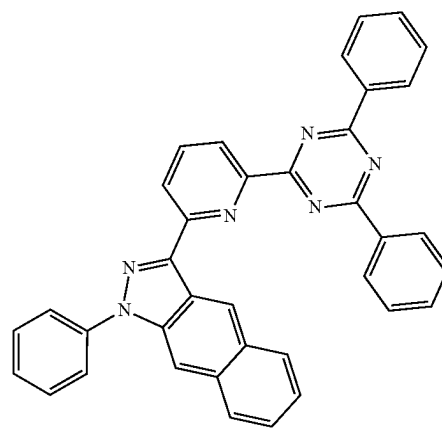
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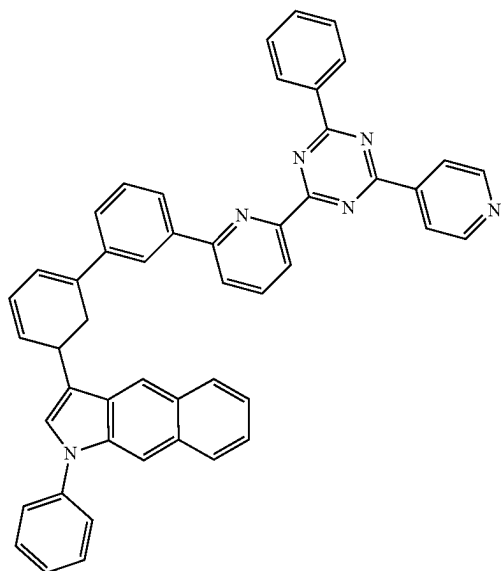


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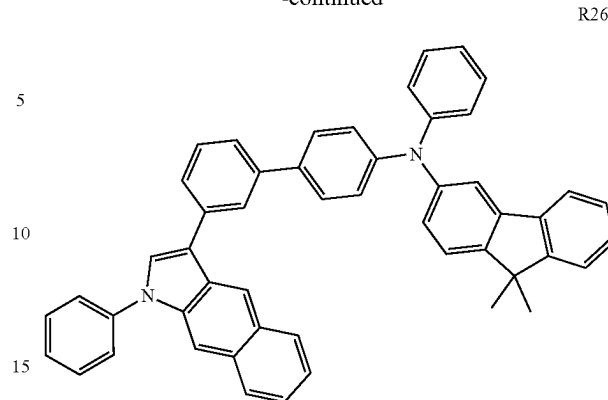
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R257

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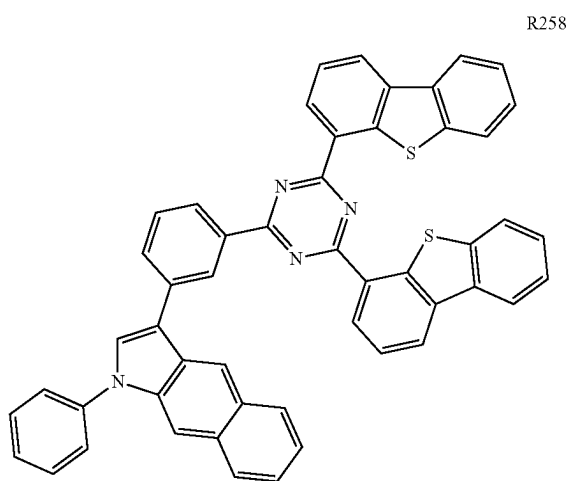


R260

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R261



R258

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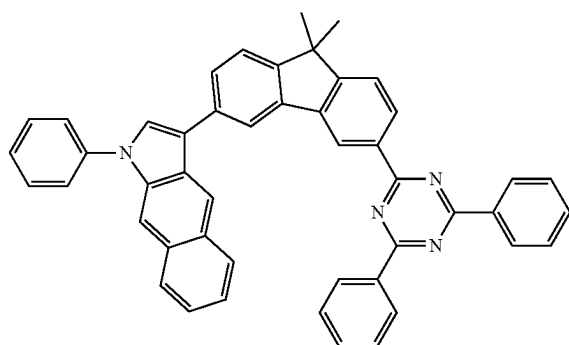
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R259



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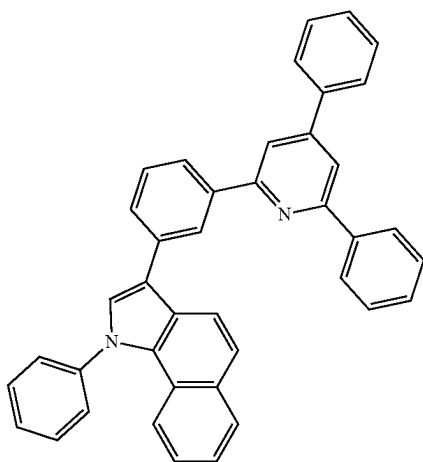
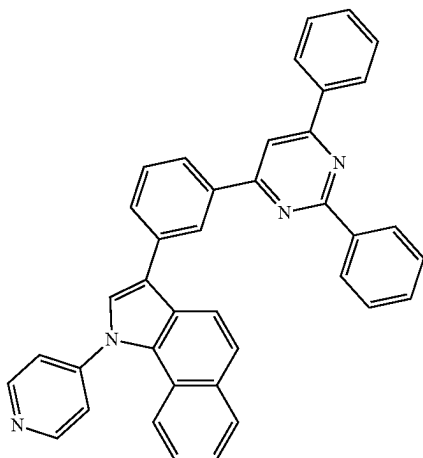
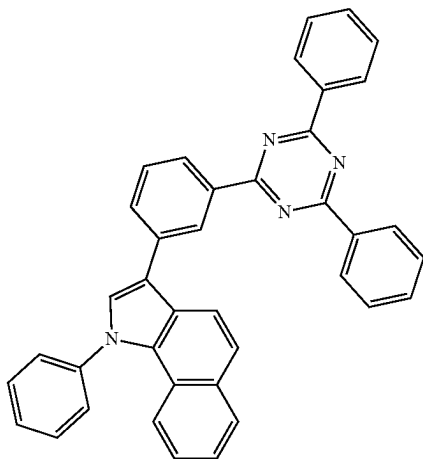
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R262

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**94**

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R263

R266

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R264

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R265

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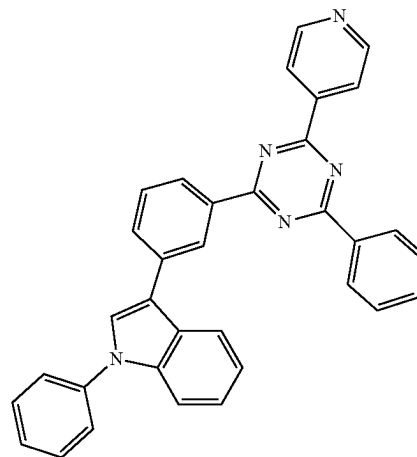
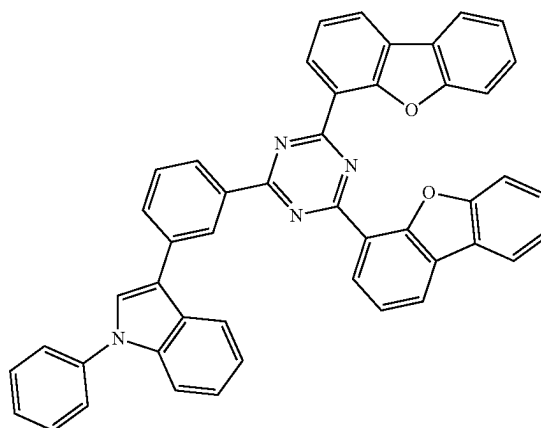
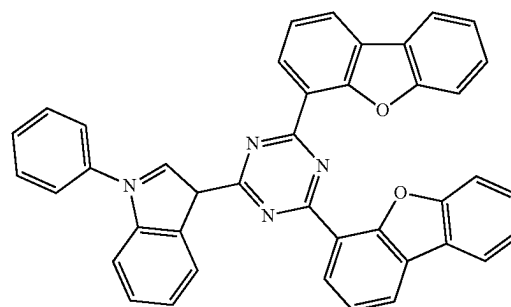
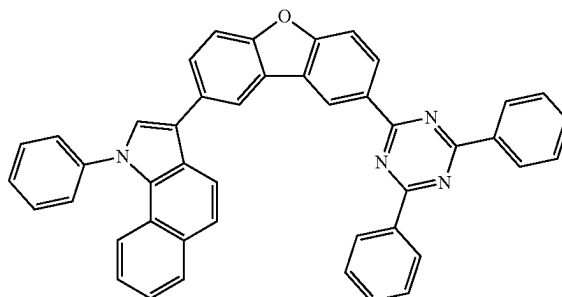
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R267

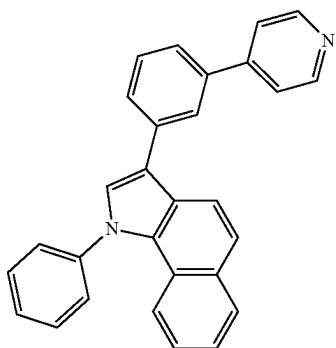
R268

R269



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**96**

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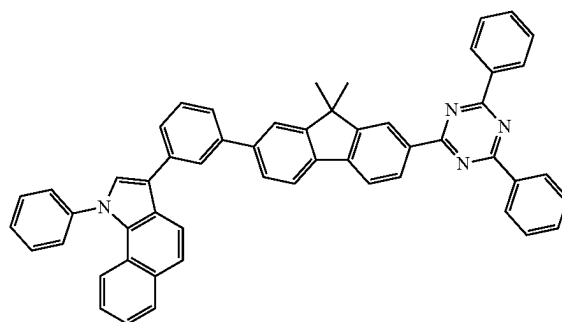
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R273



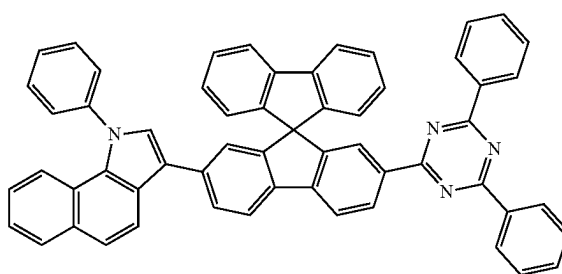
R274

R271

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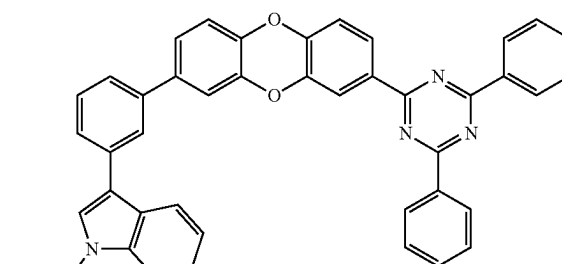


R275

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R276

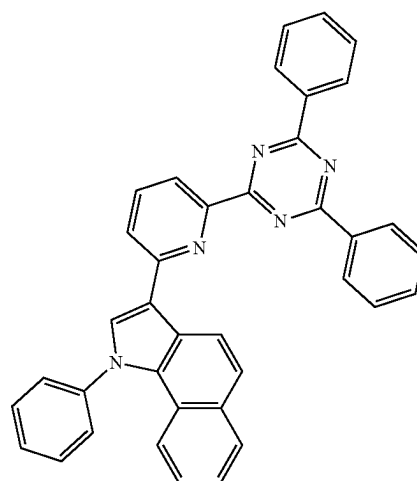
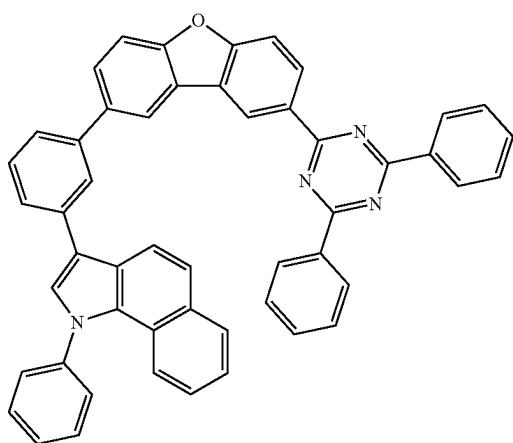
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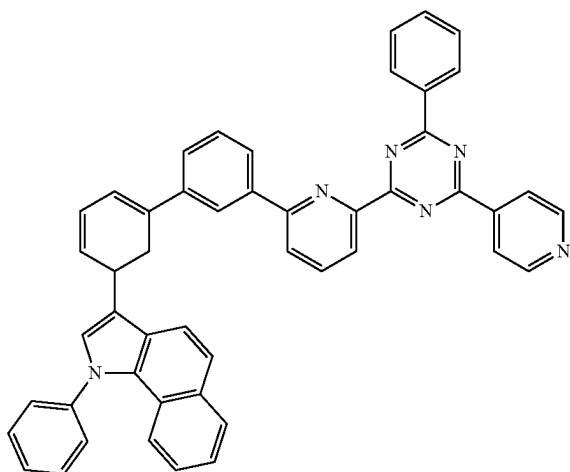
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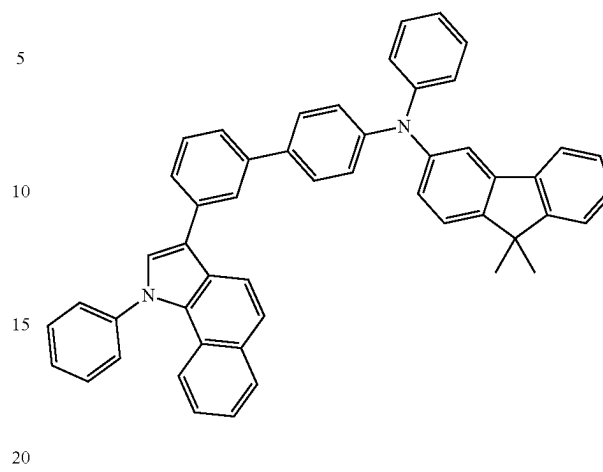
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R277

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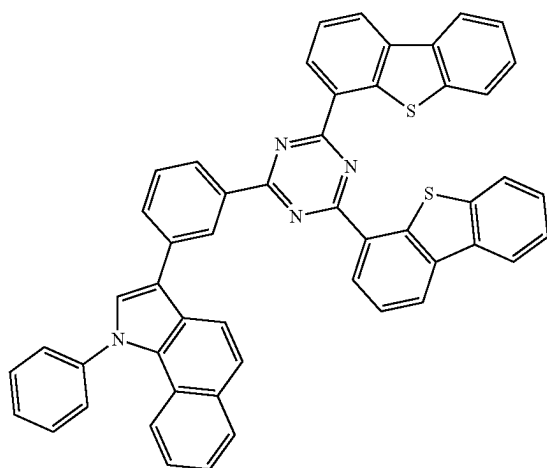
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R280



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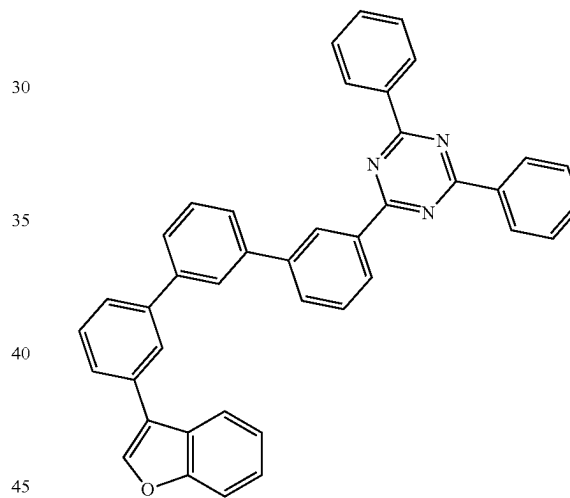
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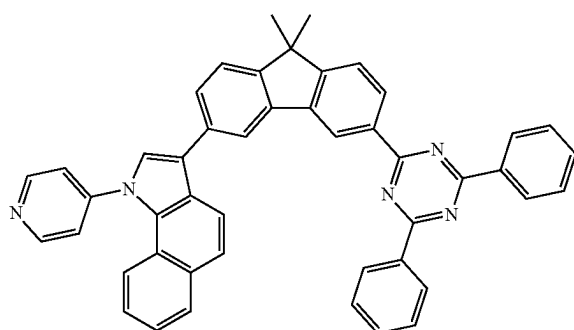
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R281



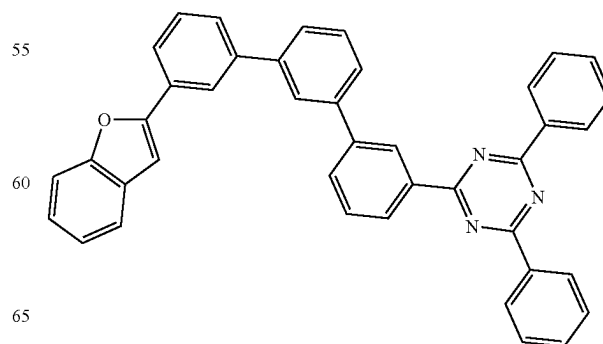
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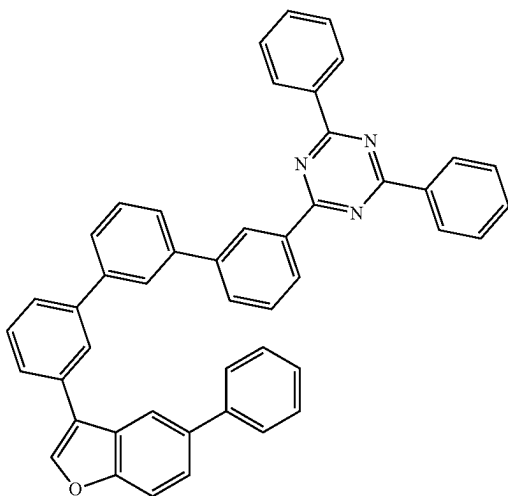
R282



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R283

**100**

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R286

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R284

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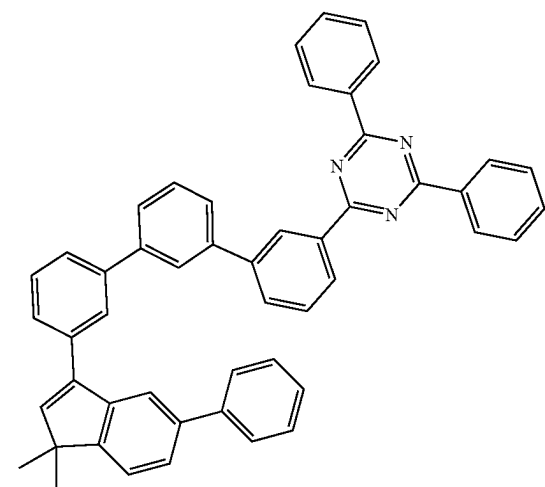
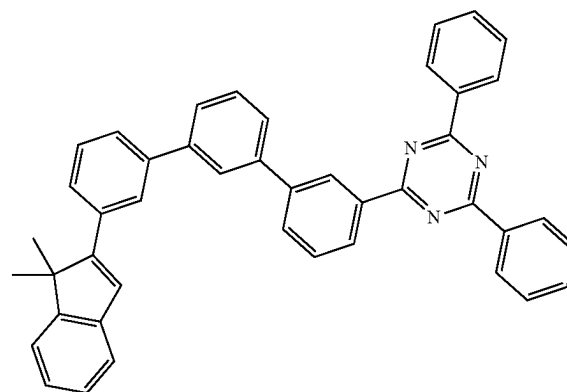
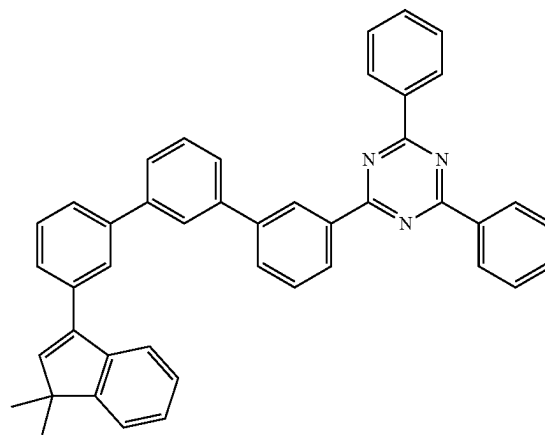
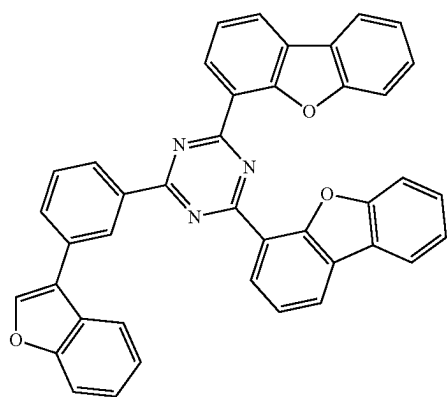
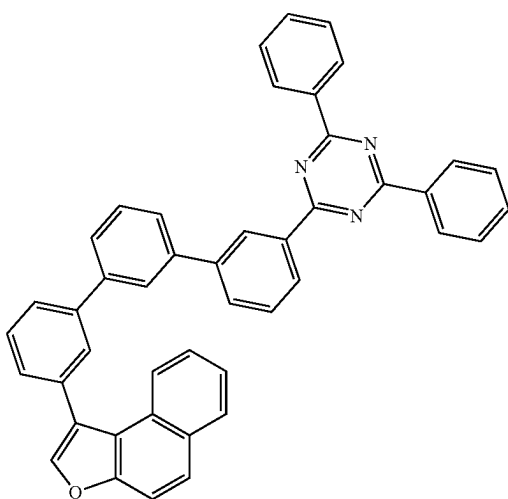
R285

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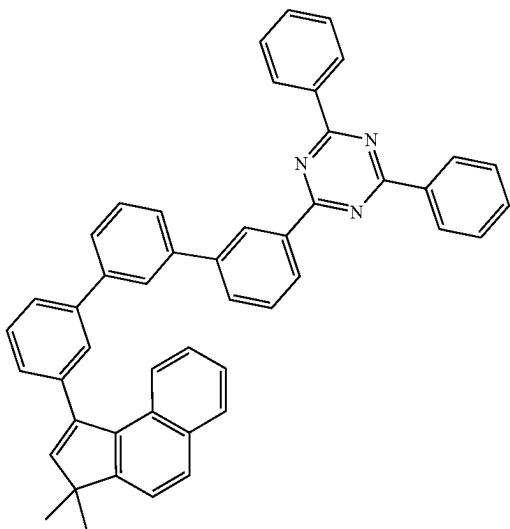
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101

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R289



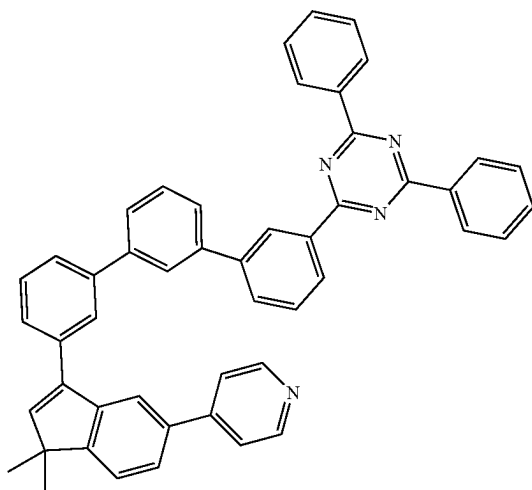
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R290



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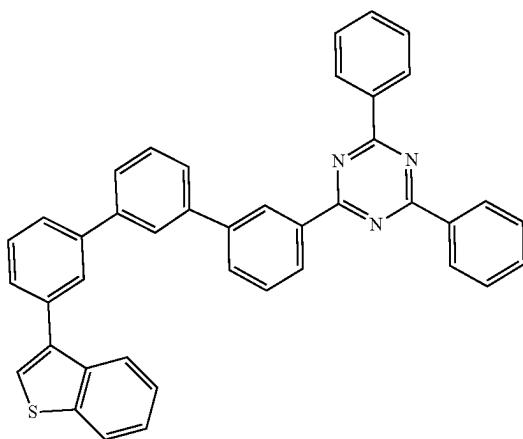
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R291



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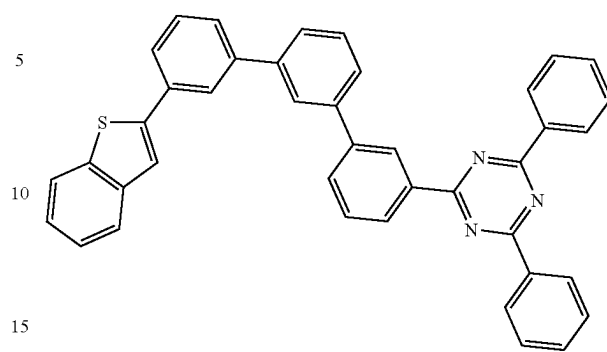
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102

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R292



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R293

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R294

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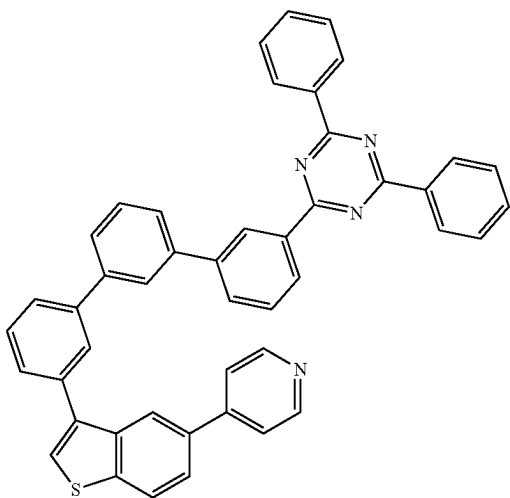
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103

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R295



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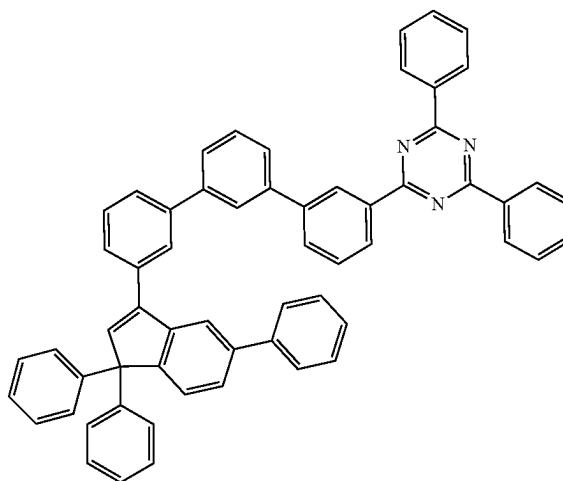
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104

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R298



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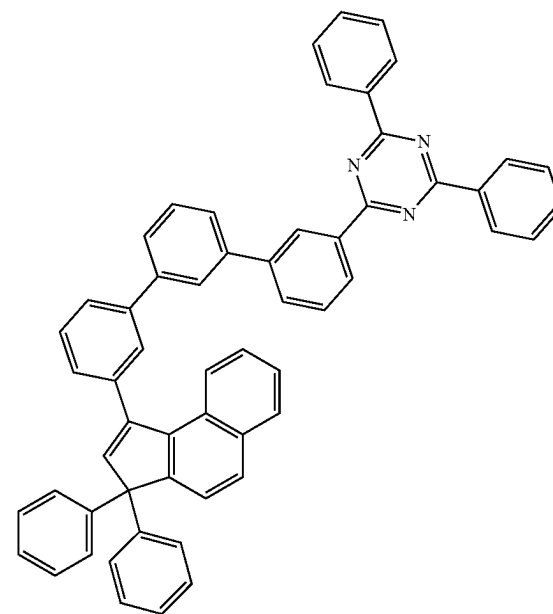
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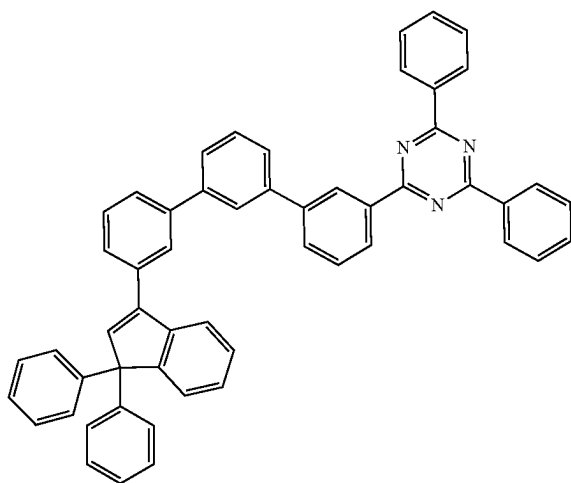
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R299



R296



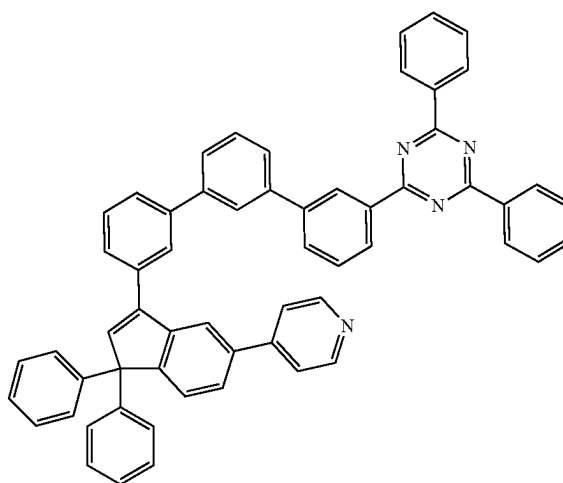
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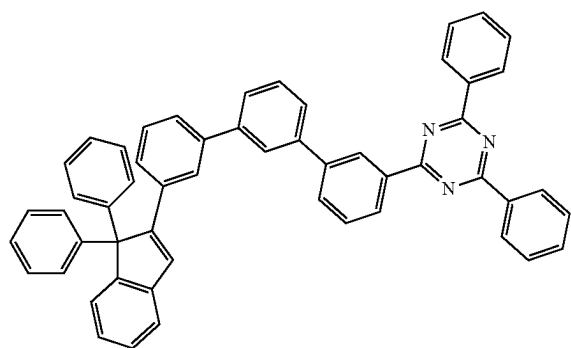
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R300



R297



The compound of Chemical Formula 1 of the present invention may be synthesized using general synthesis methods (refer to *Chem. Rev.*, 60:313 (1960); *J. Chem. SOC.* 4482 (1955); *Chem. Rev.* 95: 2457 (1995) or the like). Detailed synthesis processes of the compounds of the present invention will be specifically described in synthesis examples to be described later.

2. Organic Electroluminescent Device

Meanwhile, another aspect of the present invention relates to an organic electroluminescent device (organic EL device) including the compound represented by Chemical Formula 1 according to the present invention.

Specifically, the present invention relates to an organic electroluminescent device including an anode, a cathode, and one or more organic material layers provided between the anode and the cathode, and at least one of the one or more organic material layers includes the compound represented by Chemical Formula 1. Herein, the compound may be used either alone or as a mixture of two or more.

The one or more organic material layers may be any one or more of a hole injection layer, a hole transport layer, a light emitting layer, a light emitting auxiliary layer, a lifetime improving layer, an electron transport layer, an electron transport auxiliary layer and an electron injection layer, and at least one organic material layer among these may include the compound represented by Chemical Formula 1.

The structure of the organic electroluminescent device according to the present invention described above is not particularly limited, but, when referring to FIG. 1 as one example, includes an anode (10) and a cathode (20) facing each other, and an organic layer (30) located between the anode (10) and the cathode (20). Herein, the organic layer (30) may include a hole transport layer (31), a light emitting layer (32) and an electron transport layer (34). In addition, a hole transport auxiliary layer (33) may be included between the hole transport layer (31) and the light emitting layer (32), and an electron transport auxiliary layer (35) may be included between the electron transport layer (34) and the light emitting layer (32).

When referring to FIG. 2 as another example of the present invention, the organic layer (30) may further include a hole injection layer (37) between the hole transport layer (31) and the anode (10), and may further include an electron injection layer (36) between the electron transport layer (34) and the cathode (20).

The hole injection layer (37) laminated between the hole transport layer (31) and the anode (10) in the present invention is a layer having a function of, as well as improving interfacial properties between ITO used as the anode and an organic material used as the hole transport layer (31), smoothing the ITO surface by being coated on the top of the ITO of which surface is not smooth, and those commonly used in the art may be used without particular limit, and for example, amine compounds may be used. However, the hole injection layer is not limited thereto.

In addition, the electron injection layer (36) is a layer laminated on the top of the electron transport layer and having a function of facilitating electron injection from the cathode and eventually improving power efficiency, and is not particularly limited as long as it is commonly used in the art. For example, materials such as LiF, Liq, NaCl, CsF, Li₂O or BaO may be used.

Although not shown in the drawings in the present invention, a light emitting auxiliary layer may be further included between the hole transport auxiliary layer (33) and the light emitting layer (32). The light emitting auxiliary layer may perform a role of adjusting a thickness of the

organic layer (30) while performing a role of transporting holes to the light emitting layer (32). The light emitting auxiliary layer may include a hole transport material, and may be formed with the same material as the hole transport layer (31).

In addition, although not shown in the drawings in the present invention, a lifetime improving layer may be further included between the electron transport auxiliary layer (35) and the light emitting layer (32). Holes migrating to the light emitting layer (32) by getting on an ionization potential level in an organic light emitting device are not able to diffuse or migrate to the electron transport layer by being blocked by a high energy barrier of the lifetime improving layer, and consequently, the lifetime improving layer has a function of limiting the holes in the light emitting layer. Such a function of limiting the holes in the light emitting layer prevents the holes from diffusing to the electron transport layer migrating electrons by reduction, and therefore, suppresses a lifetime decrease phenomenon caused through an irreversible decomposition reaction by oxidation, and thereby contributes to improving a lifetime of the organic light emitting device.

The compound represented by Chemical Formula 1 in the present invention has an EWG bonding to a 5-membered aromatic ring or a 5-membered aromatic heteroring such as indole, indazole, indene, benzofuran, benzothiophene or triazole, and may be used as a host material since, by having a similar energy level with carbazole, the energy level may be adjusted to be higher than a dopant energy level. Particularly, benzofuran and benzothiophene moieties are electron abundant, and when used as an electron transport layer material of an organic electroluminescent device, mobility increases, and as a result, an increase in the luminous efficiency and a decrease in the driving voltage may be expected. In addition, the 5-membered aromatic ring or the 5-membered aromatic heteroring of the present invention has a smaller molecular weight compared to existing compounds, and therefore, may be deposited at a relatively lower deposition temperature when deposited compared to other materials, and favorable processability and enhanced thermal stability may be obtained.

Accordingly, the compound represented by Chemical Formula 1 of the present invention may be used as a material of any one of a hole injection layer, a hole transport layer, a light emitting layer, an electron transport layer and an electron injection layer, an organic material layer of an organic electroluminescent device, but may be preferably used as a material of any of a light emitting layer, an electron transport layer, and an electron transport auxiliary layer further laminated on the electron transport layer, and more preferably used as a material of an electron transport layer or an electron transport auxiliary layer.

In addition, when using the compound according to the present invention as a light emitting layer material, the compound represented by Chemical Formula 1 may be specifically used as a phosphorescent host, a fluorescent host or a dopant material of the light emitting layer, and may be preferably used as a phosphorescent host (blue, green and/or red phosphorescent host material).

In addition, the organic electroluminescent device in the present invention has, as described above, an anode, one or more organic material layers and a cathode consecutively laminated, and in addition thereto, may further include an insulating layer or an adhesive layer at an interface between the electrode and the organic material layer.

Except that at least one or more of the organic material layers (for example, electron transport auxiliary layer) are

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formed to include the compound represented by Chemical Formula 1, the organic electroluminescent device of the present invention may be manufactured by forming other organic material layers and electrodes using materials and methods known in the art.

The organic material layer may be formed using a vacuum deposition method or a solution coating method. Examples of the solution coating method may include spin coating, dip coating, doctor blading, inkjet printing, thermal transfer method or the like, but are not limited thereto.

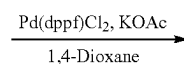
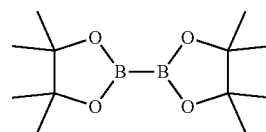
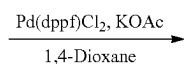
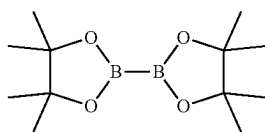
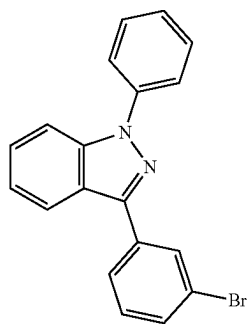
A substrate capable of being used in the present invention is not particularly limited, and silicon wafers, quartz, glass plates, metal plates, plastic films, sheets and the like may be used.

The anode material may be prepared using, for example, a conductor having high work function so as to have smooth hole injection, and examples thereof may include metals such as vanadium, chromium, copper, zinc or gold, or alloys thereof; metal oxides such as zinc oxide, indium oxide, indium tin oxide (ITO) or indium zinc oxide (IZO); combinations of metals and oxides such as ZnO:Al or SnO₂:Sb; conductive polymers such as polythiophene, poly(3-methylthiophene), poly[3,4-(ethylene-1,2-dioxy)thiophene] (PEDT), polypyrrole or polyaniline; carbon black, and the like, but are not limited thereto.

The cathode material may be prepared using, for example, a conductor having low work function so as to have smooth electron injection, and examples thereof may include metals such as magnesium, calcium, sodium, potassium, titanium, indium, yttrium, lithium, gadolinium, aluminum, silver, tin or lead, or alloys thereof; and multilayer-structured materials such as LiF/Al or LiO₂/Al, but are not limited thereto.

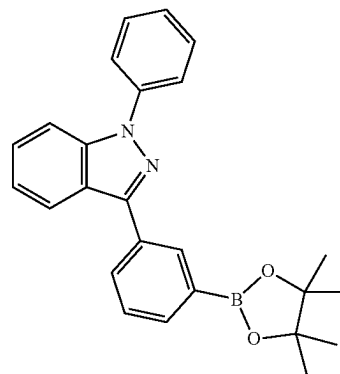
Hereinafter, the present invention will be described in detail with reference to examples as follows. However, the following examples are for illustrative purposes only, and the present invention is not limited to the following examples.

[Preparation Example 1] Synthesis of A1



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A1

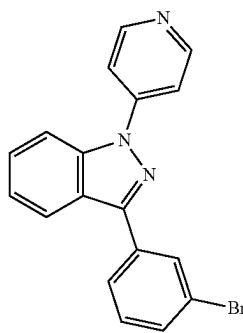
Under a nitrogen stream, 3-(3-bromophenyl)-1-phenyl-1H-indazole (8.5 g, 24.4 mmol), 4,4',4'',5,5',5''-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A1 (6.8 g, 17.1 mmol, yield 70%).

GC-Mass (theoretical value: 396.3 g/mol, measured value: 396 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.45~7.55 (m, 8H), 7.62 (s, 1H), 7.84~9.2 (m, 4H)

[Preparation Example 2] Synthesis of A2

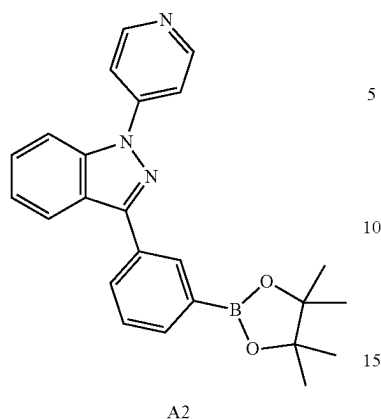


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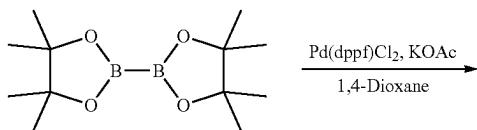
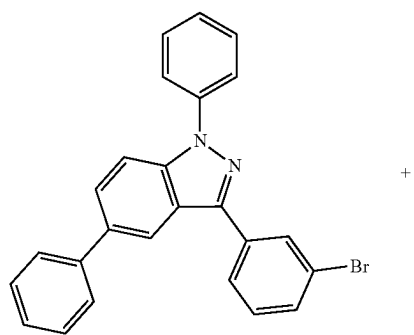
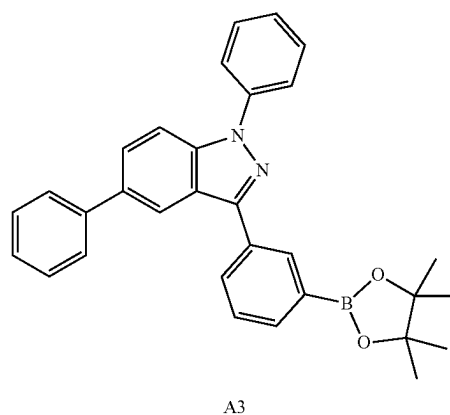
Under a nitrogen stream, 3-(3-bromophenyl)-1-(pyridin-4-yl)-1H-indazole (8.5 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A2 (7.3 g, 18.3 mmol, yield 75%).

GC-Mass (theoretical value: 397.3 g/mol, measured value: 397 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.45~7.55 (m, 5H), 7.62 (s, 1H), 7.84~9.2 (m, 4H), 8.11~12 (d, 2H)

[Preparation Example 3] Synthesis of A3

**110**
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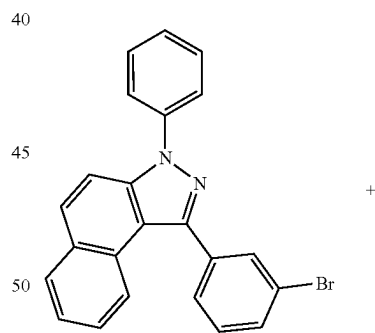
Under a nitrogen stream, 3-(3-bromophenyl)-1,5-diphenyl-1H-indazole (10.4 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A3 (8.6 g, 18.3 mmol, yield 75%).

GC-Mass (theoretical value: 472.4 g/mol, measured value: 472 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.44~7.59 (m, 14H), 7.62 (s, 1H), 8.23~25 (d, 2H)

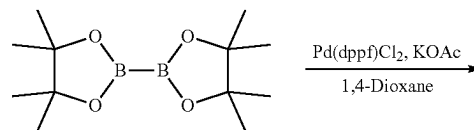
[Preparation Example 4] Synthesis of A4



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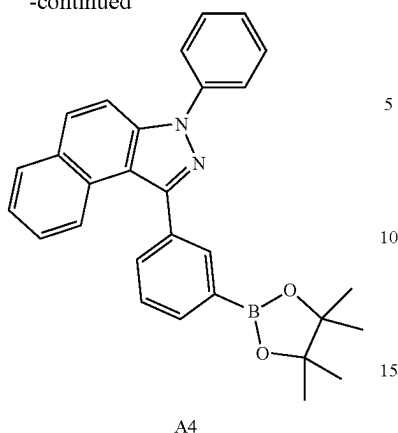
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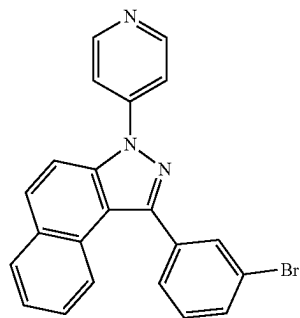
Under a nitrogen stream, 1-(3-bromophenyl)-3-phenyl-3H-benzo[e]indazole (9.7 g, 24.4 mmol), 4,4,4',4'',5,5,5'',5''')-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A4 (7.6 g, 17.1 mmol, yield 70%).

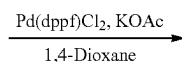
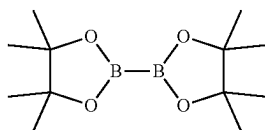
GC-Mass (theoretical value: 446.4 g/mol, measured value: 446 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.45~7.55 (m, 10H), 7.62 (s, 1H), 7.81~9.9 (m, 4H)

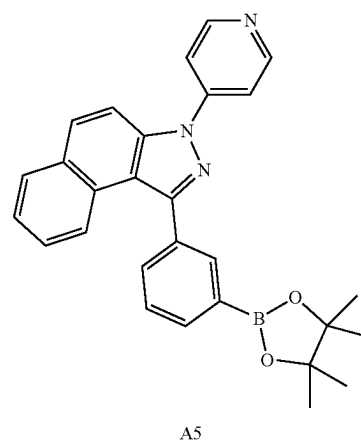
[Preparation Example 5] Synthesis of A5



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**112**

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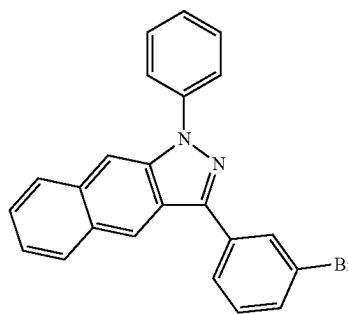
Under a nitrogen stream, 1-(3-bromophenyl)-3-(pyridin-4-yl)-3H-benzo[e]indazole (9.8 g, 24.4 mmol), 4,4,4',4'',5,5,5'',5''')-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A5 (7.6 g, 17.1 mmol, yield 70%).

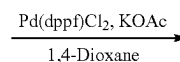
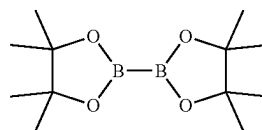
GC-Mass (theoretical value: 447.4 g/mol, measured value: 447 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.45~7.55 (m, 9H), 7.62 (s, 1H), 7.81~9.9 (m, 2H), 8.21~23 (d, 2H)

[Preparation Example 6] Synthesis of A6

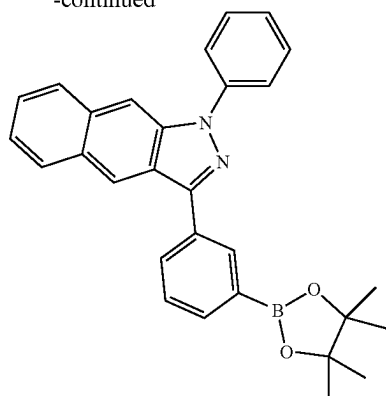


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A6

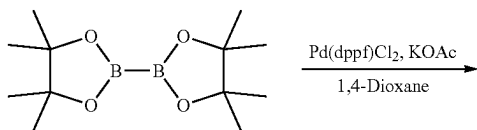
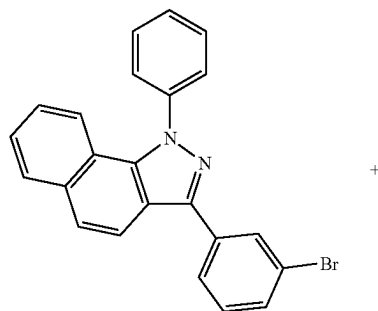
Under a nitrogen stream, 3-(3-bromophenyl)-1-phenyl-1H-benzo[f]indazole (9.7 g, 24.4 mmol), 4,4,4',4',5,5',5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A6 (7.6 g, 17.1 mmol, yield 70%).

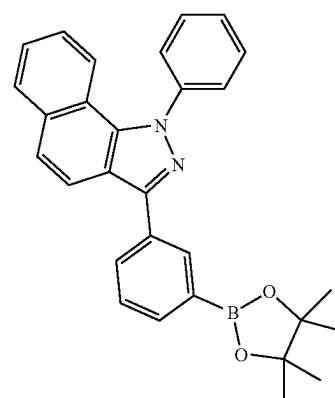
GC-Mass (theoretical value: 446.4 g/mol, measured value: 446 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.45~7.55 (m, 10H), 7.58 (s, 1H), 7.61 (s, 1H), 7.78 (s, 1H), 7.81~86 (m, 2H)

[Preparation Example 7] Synthesis of A7

**114**

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A7

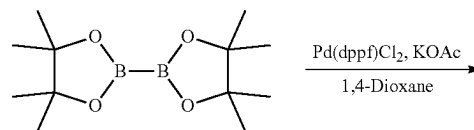
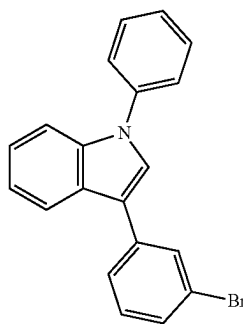
Under a nitrogen stream, 3-(3-bromophenyl)-1-phenyl-1H-benzo[g]indazole (9.7 g, 24.4 mmol), 4,4,4',4',5,5',5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

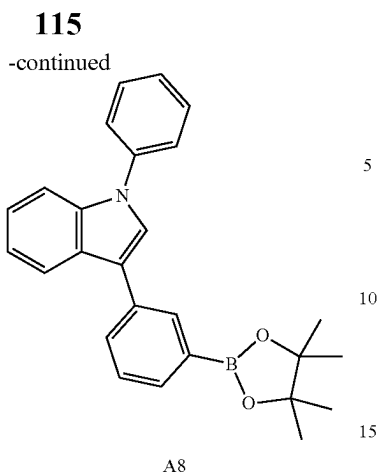
After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A7 (7.6 g, 17.1 mmol, yield 70%).

GC-Mass (theoretical value: 446.36 g/mol, measured value: 446 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.45~7.54 (m, 10H), 7.62 (s, 1H), 7.81~88 (m, 4H)

[Preparation Example 8] Synthesis of A8





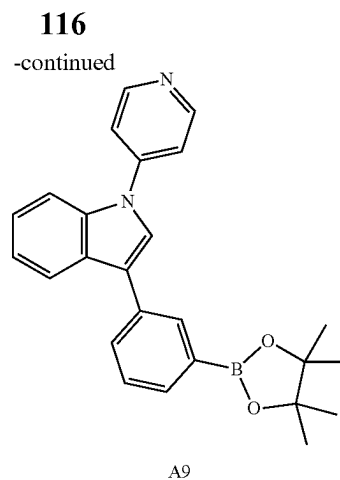
Under a nitrogen stream, 3-(3-bromophenyl)-1-phenyl-1H-indole (8.5 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A8 (6.7 g, 17.1 mmol, yield 70%).

GC-Mass (theoretical value: 395.3 g/mol, measured value: 395 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.43~7.55 (m, 9H), 7.62 (s, 1H), 7.84~9.2 (m, 4H)

[Preparation Example 9] Synthesis of A9



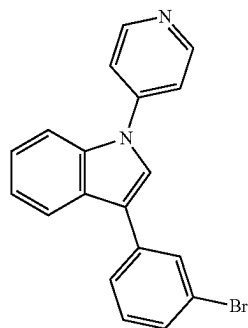
Under a nitrogen stream, 3-(3-bromophenyl)-1-(pyridin-4-yl)-1H-indole (8.5 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A9 (6.8 g, 17.1 mmol, yield 70%).

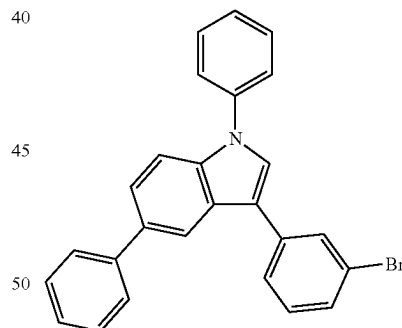
GC-Mass (theoretical value: 396.3 g/mol, measured value: 396 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.41~7.55 (m, 6H), 7.62 (s, 1H), 7.84~9.2 (m, 4H), 8.11~12 (d, 2H)

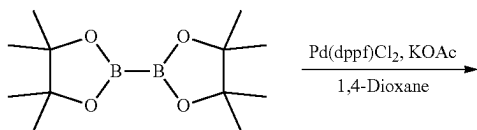
[Preparation Example 10] Synthesis of A10



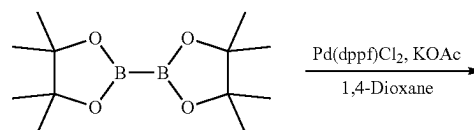
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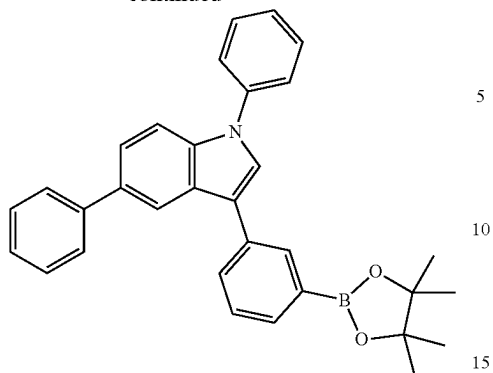
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A10

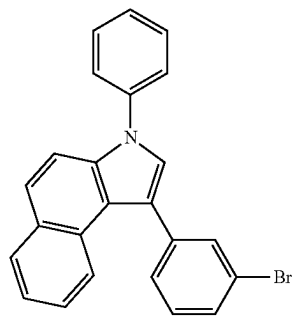
Under a nitrogen stream, 3-(3-bromophenyl)-1,5-diphenyl-1H-indole (10.3 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A10 (8.6 g, 18.3 mmol, yield 75%).

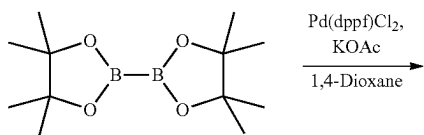
GC-Mass (theoretical value: 471.4 g/mol, measured value: 471 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.41~7.60 (m, 15H), 7.63 (s, 1H), 8.23~25 (d, 2H)

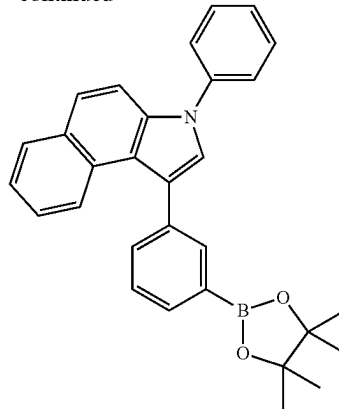
[Preparation Example 11] Synthesis of A11



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**118**

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A11

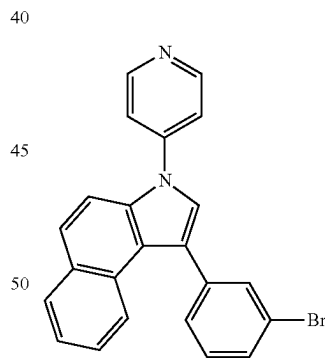
Under a nitrogen stream, 1-(3-bromophenyl)-3-phenyl-3H-benzo[e]indole (9.7 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A11 (7.6 g, 17.1 mmol, yield 70%).

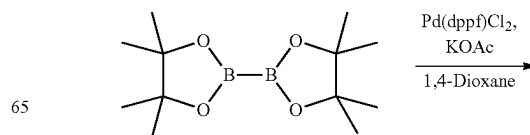
GC-Mass (theoretical value: 445.4 g/mol, measured value: 445 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.41~7.55 (m, 11H), 7.62 (s, 1H), 7.81~99 (m, 4H)

[Preparation Example 12] Synthesis of A12

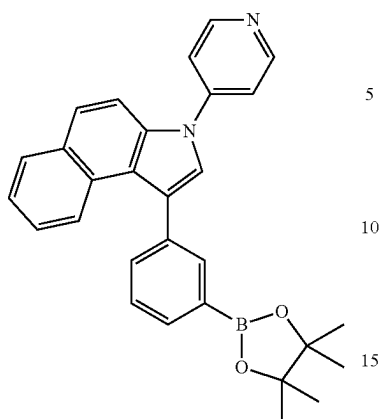


+



119

-continued



A12

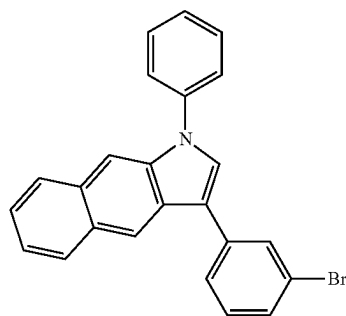
Under a nitrogen stream, 1-(3-bromophenyl)-3-(pyridin-4-yl)-3H-benzo[e]indazole (9.7 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A12 (7.6 g, 17.1 mmol, yield 70%).

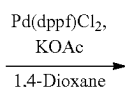
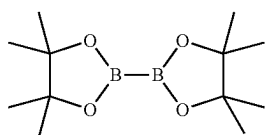
GC-Mass (theoretical value: 446.4 g/mol, measured value: 446 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.42~7.55 (m, 10H), 7.62 (s, 1H), 7.81~9.9 (m, 2H), 8.21~23 (d, 2H)

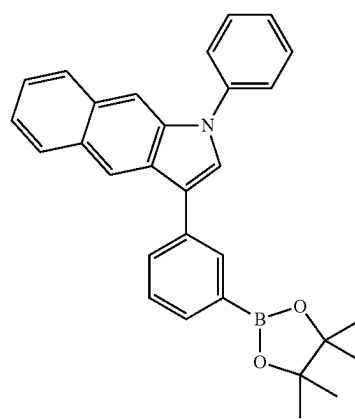
[Preparation Example 13] Synthesis of A13



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**120**

-continued



A13

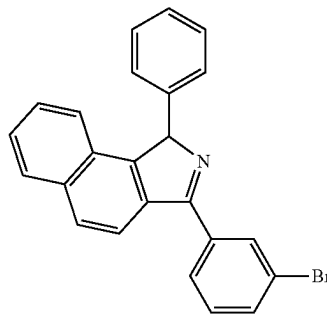
Under a nitrogen stream, 3-(3-bromophenyl)-1-phenyl-1H-benzo[f]indole (9.7 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A13 (7.6 g, 17.1 mmol, yield 70%).

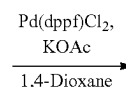
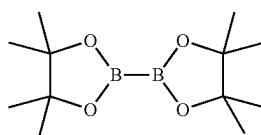
GC-Mass (theoretical value: 445.4 g/mol, measured value: 445 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.42~7.55 (m, 11H), 7.58 (s, 1H), 7.61 (s, 1H), 7.78 (s, 1H), 7.81~8.5 (m, 2H)

[Preparation Example 14] Synthesis of A14

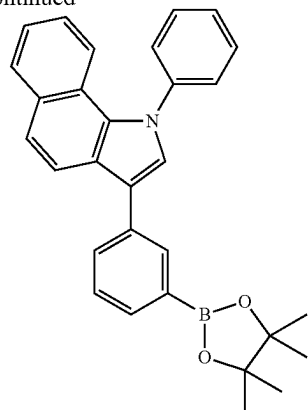


+



121

-continued



A14

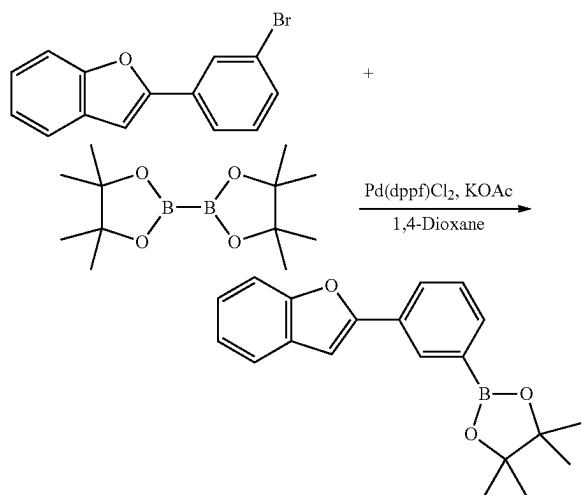
Under a nitrogen stream, 3-(3-bromophenyl)-1-phenyl-1H-benzo[g]indazole (9.7 g, 24.4 mmol), 4,4,4',4',5,5',5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A14 (7.6 g, 17.1 mmol, yield 70%).

GC-Mass (theoretical value: 445.37 g/mol, measured value: 445 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.42~7.54 (m, 11H), 7.62 (s, 1H), 7.82~8.9 (m, 4H)

[Preparation Example 15] Synthesis of A15



A15

Under a nitrogen stream, 2-(3-bromophenyl)benzofuran (6.7 g, 24.4 mmol), 4,4,4',4',5,5',5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄,

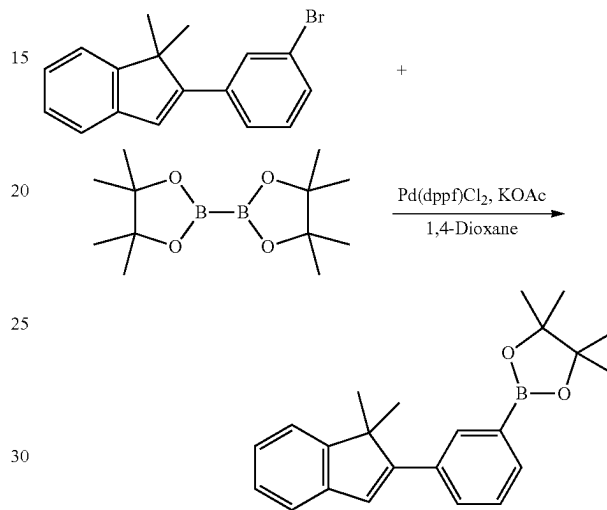
122

and the result was purified using column chromatography to obtain target Compound A15 (5.5 g, 17.1 mmol, yield 70%).

GC-Mass (theoretical value: 320.2 g/mol, measured value: 320 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.11 (s, 1H), 7.58 (s, 1H), 7.67~7.89 (m, 7H)

[Preparation Example 16] Synthesis of A16



A16

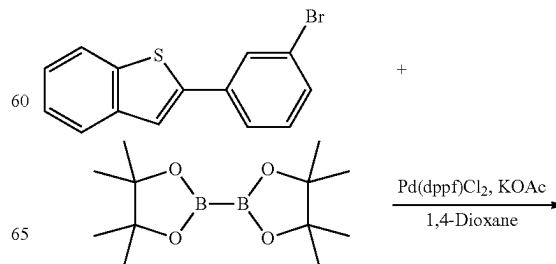
Under a nitrogen stream, 2-(3-bromophenyl)-1,1-dimethyl-1H-indene (7.3 g, 24.4 mmol), 4,4,4',4',5,5',5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ 0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A16 (5.9 g, 17.1 mmol, yield 70%).

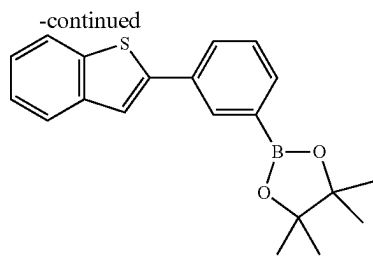
GC-Mass (theoretical value: 346.28 g/mol, measured value: 346 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 1.69 (s, 6H), 7.01 (s, 1H), 7.45 (s, 1H), 7.57~7.77 (m, 7H)

[Preparation Example 17] Synthesis of A17



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A17

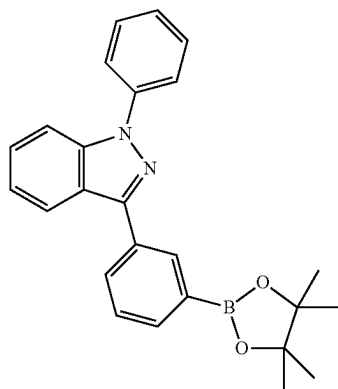
Under a nitrogen stream, 2-(3-bromophenyl)benzo[b]thiophene (7.0 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂ (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A17 (6.1 g, 18.3 mmol, yield 75%).

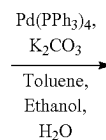
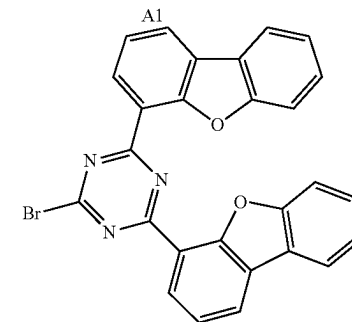
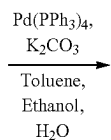
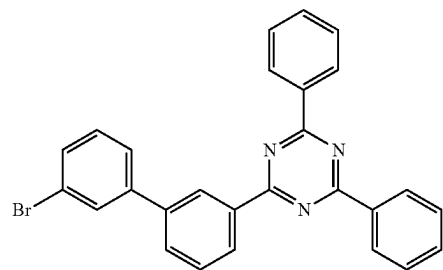
GC-Mass (theoretical value: 336.26 g/mol, measured value: 336 g/mol)

¹H-NMR: δ 1.25 (s, 12H), 7.51 (s, 1H), 7.59 (s, 1H), 7.72~7.92 (m, 7H)

[Synthesis Example 1] Synthesis of R1

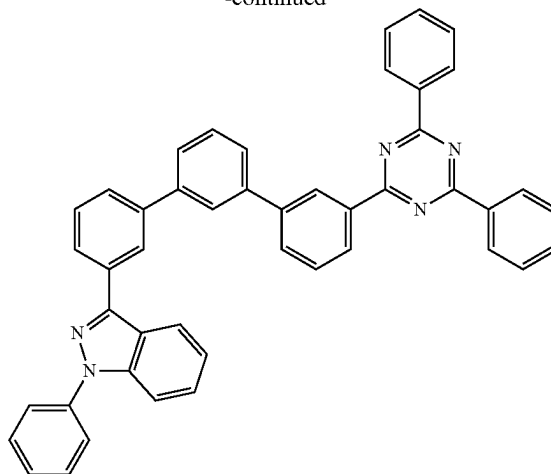


A1



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-continued



R1

Under a nitrogen stream, A1 (6.8 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

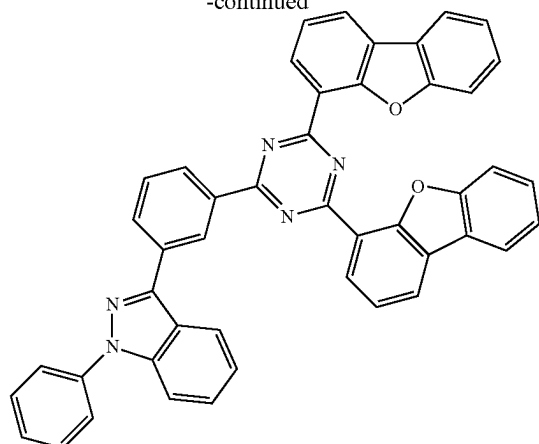
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R1 (8.4 g, 12.8 mmol, yield 75%).

GC-Mass (theoretical value: 653.8 g/mol, measured value: 653 g/mol)

[Synthesis Example 2] Synthesis of R8

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-continued



R8

5

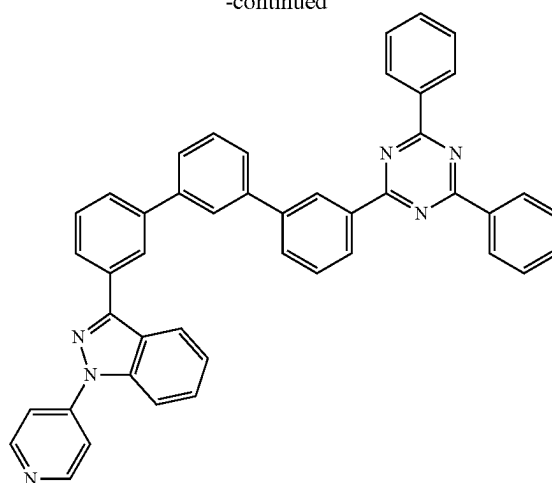
10

15

20

126

-continued



R21

25

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35

Under a nitrogen stream, A1 (6.8 g 17.1 mmol), 2-bromo-4,6-bis(dibenzo[b,d]furan-4-yl)-1,3,5-triazine (9.2 g 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R8 (9.0 g, 13.1 mmol, yield 77%).

GC-Mass (theoretical value: 681.8 g/mol, measured value: 682 g/mol)

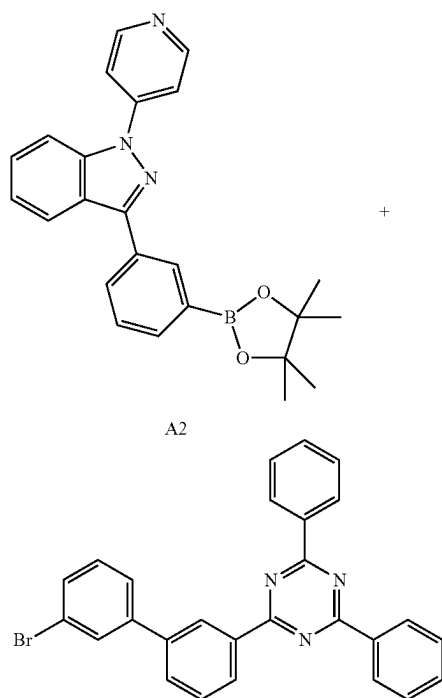
Under a nitrogen stream, A2 (6.8 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R21 (7.8 g, 11.9 mmol, yield 70%).

GC-Mass (theoretical value: 654.8 g/mol, measured value: 655 g/mol)

[Synthesis Example 3] Synthesis of R21

[Synthesis Example 4] Synthesis of R41



A2

40

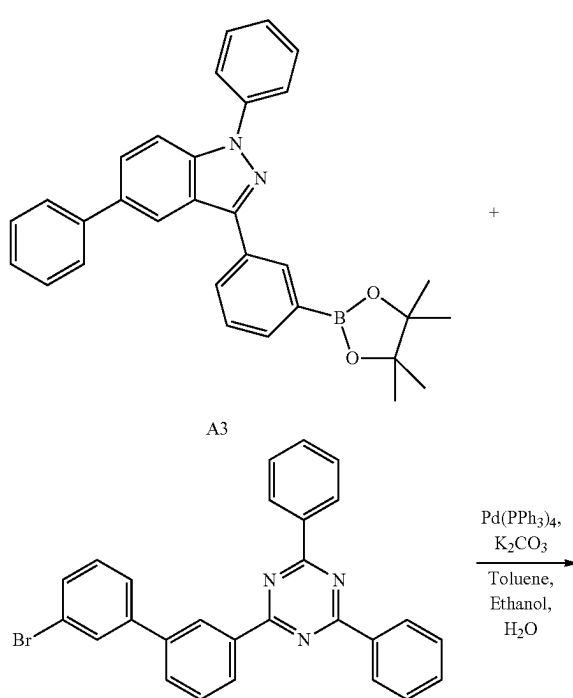
45

50

55

60

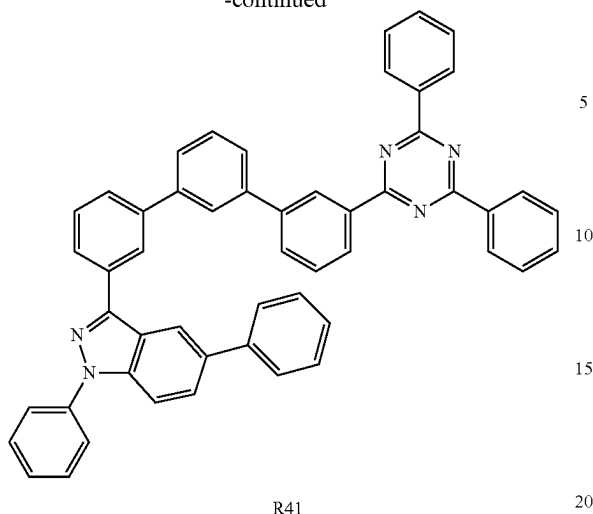
65



A3

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-continued

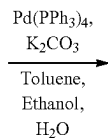
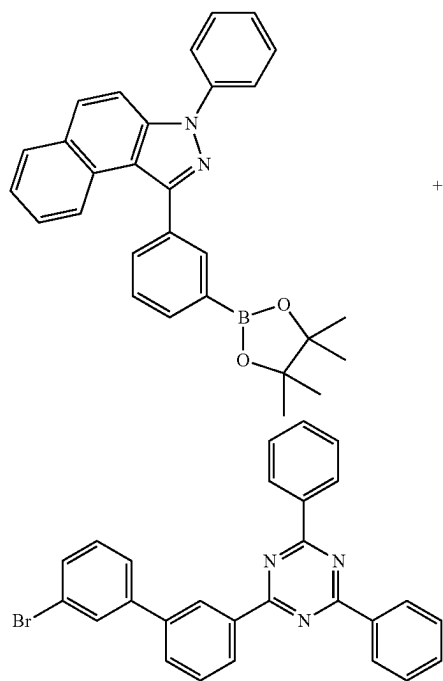


Under a nitrogen stream, A3 (8.6 g, 18.3 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (9.3 g, 20.1 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 73.1 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

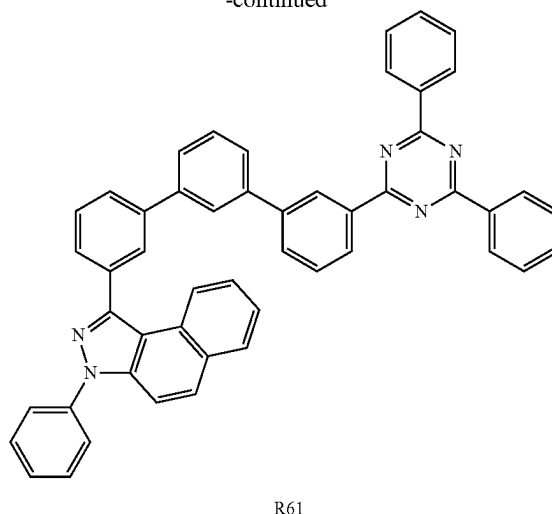
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R41 (10.0 g, 13.7 mmol, yield 75%).

GC-Mass (theoretical value: 729.9 g/mol, measured value: 730 g/mol)

[Synthesis Example 5] Synthesis of R61

**128**

-continued

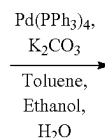
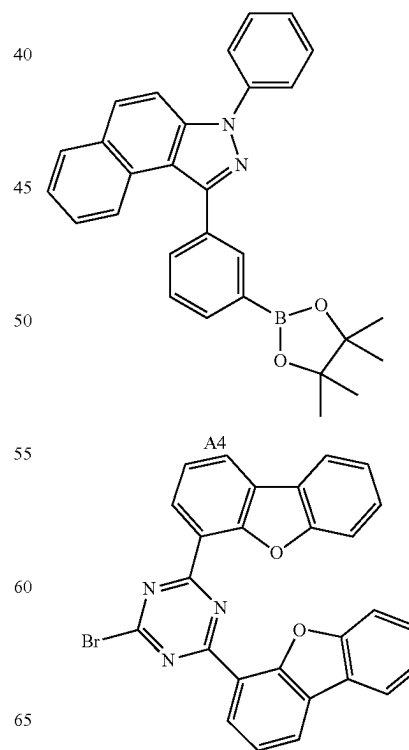


Under a nitrogen stream, A4 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 73.1 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R61 (9.0 g, 12.8 mmol, yield 75%).

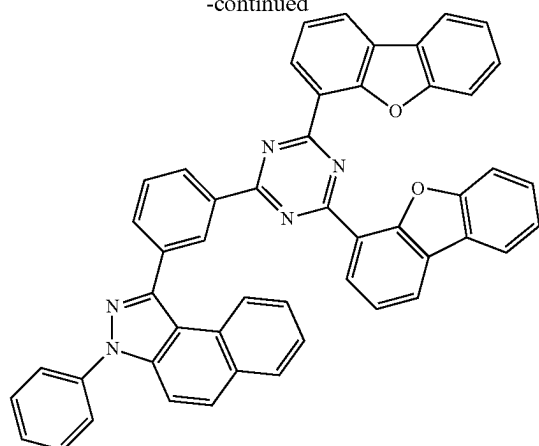
GC-Mass (theoretical value: 703.9 g/mol, measured value: 703 g/mol)

[Synthesis Example 6] Synthesis of R68



129

-continued



R68

Under a nitrogen stream, A4 (7.6 g, 17.1 mmol), 2-bromo-4,6-bis(dibenzo[b,d]furan-4-yl)-1,3,5-triazine (9.2 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

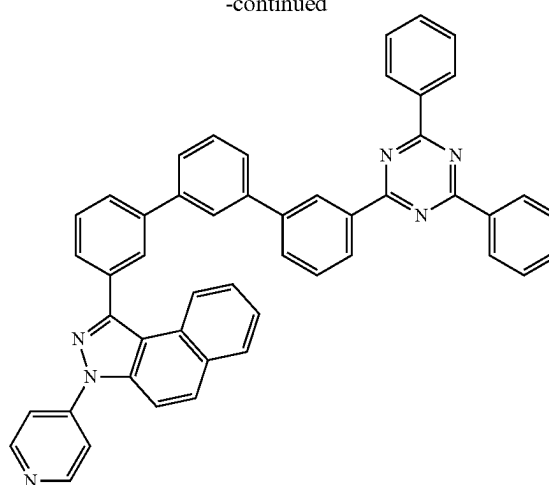
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R68 (9.4 g, 12.8 mmol, yield 75%).

GC-Mass (theoretical value: 731.8 g/mol, measured value: 731 g/mol)

[Synthesis Example 7] Synthesis of R81

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-continued



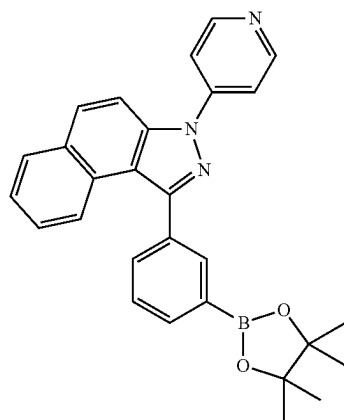
R81

Under a nitrogen stream, A5 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

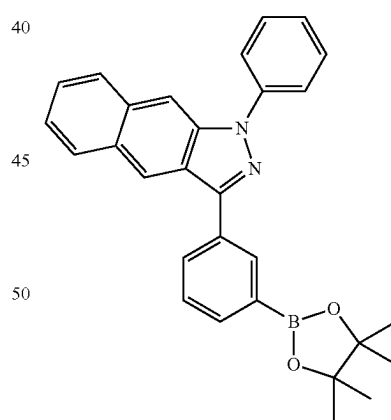
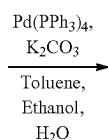
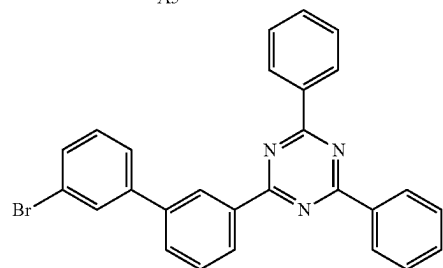
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R81 (8.9 g 12.6 mmol, yield 74%).

GC-Mass (theoretical value: 703.9 g/mol, measured value: 704 g/mol)

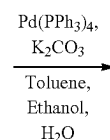
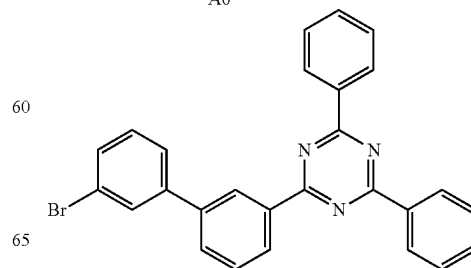
[Synthesis Example 8] Synthesis of R101



A5

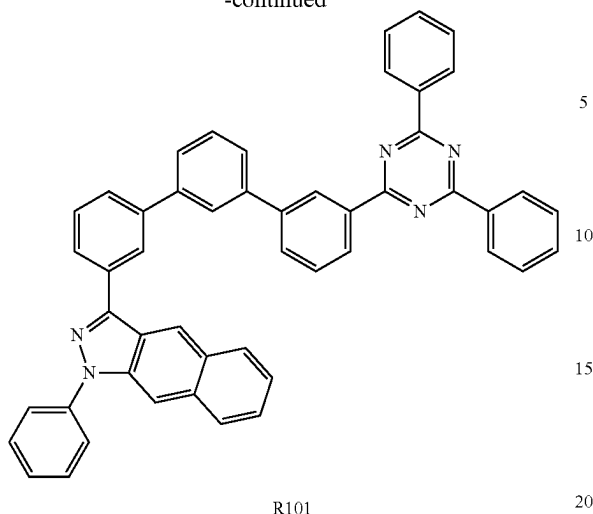


A6



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-continued



Under a nitrogen stream, A6 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

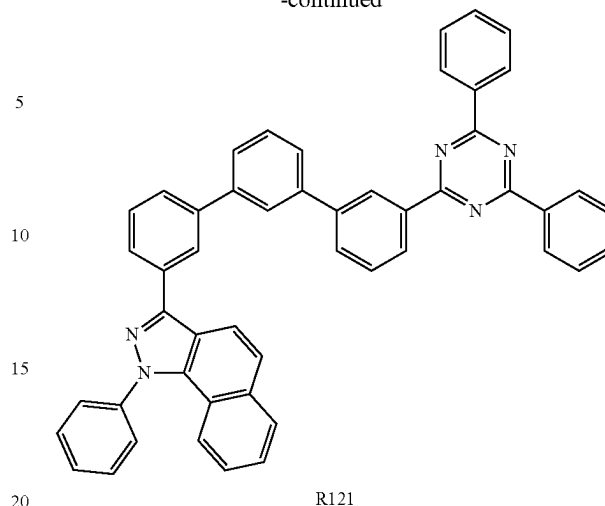
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R101 (9.0 g, 12.8 mmol, yield 75%).

GC-Mass (theoretical value: 703.9 g/mol, measured value: 703 g/mol)

[Synthesis Example 9] Synthesis of R121

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-continued

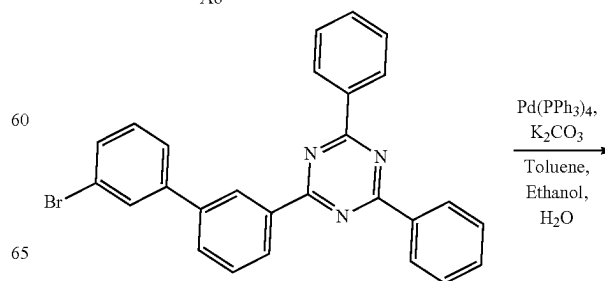
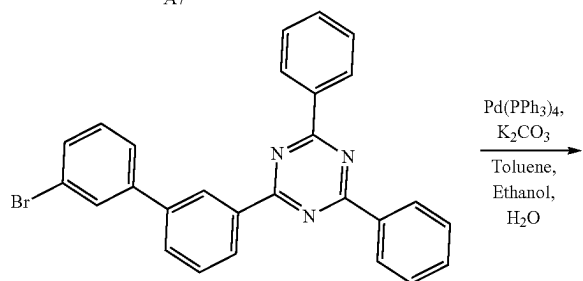
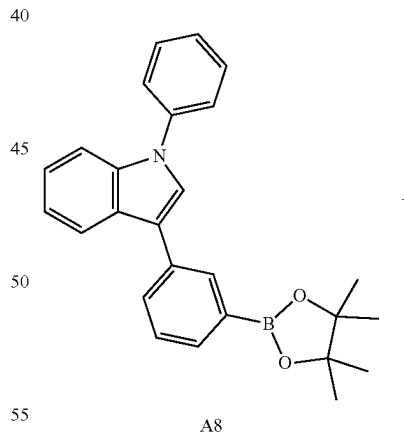
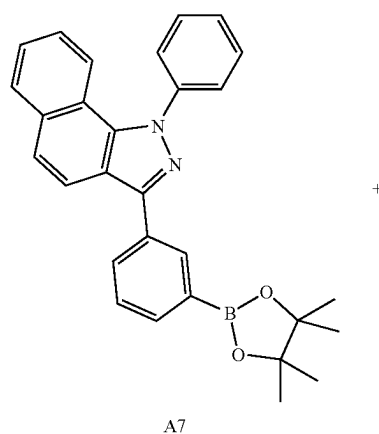


Under a nitrogen stream, A7 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R121 (8.9 g, 12.0 mmol, yield 74%).

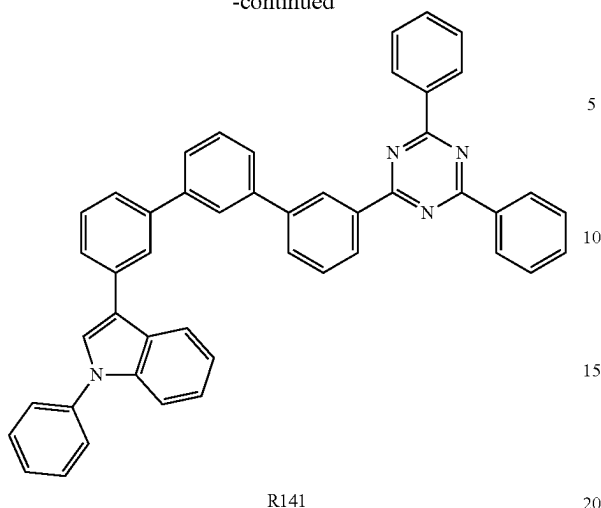
GC-Mass (theoretical value: 703.9 g/mol, measured value: 703 g/mol)

[Synthesis Example 10] Synthesis of R141



133

-continued

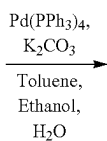
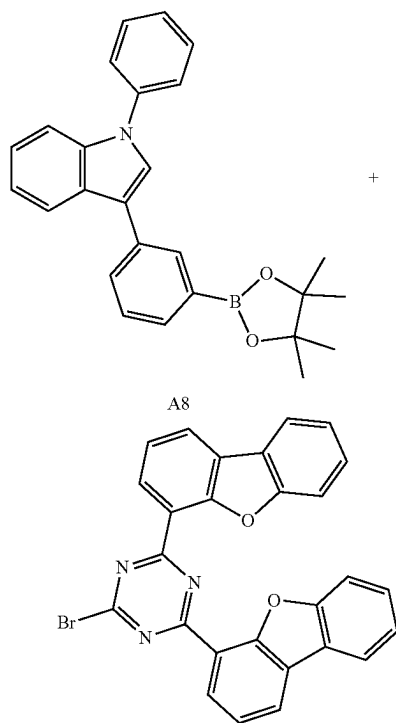


Under a nitrogen stream, A8 (6.7 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

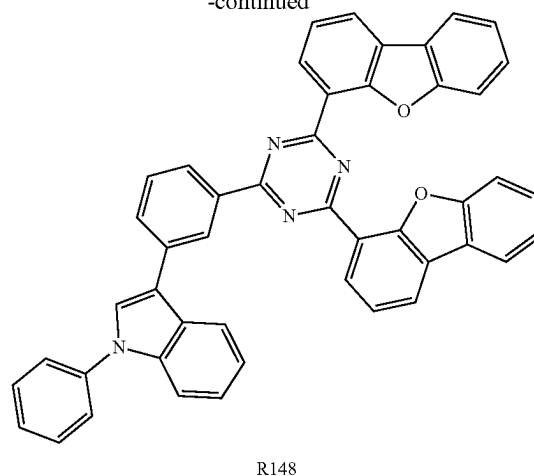
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R141 (8.3 g, 12.8 mmol, yield 75%).

GC-Mass (theoretical value: 652.8 g/mol, measured value: 653 g/mol)

[Synthesis Example 11] Synthesis of R148

**134**

-continued

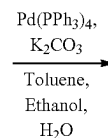
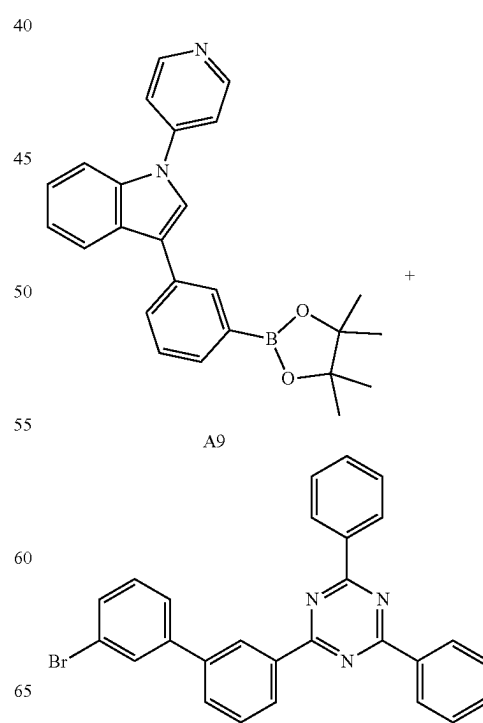


Under a nitrogen stream, A8 (6.7 g, 17.1 mmol), 2-bromo-4,6-bis(dibenzo[b,d]furan-4-yl)-1,3,5-triazine (9.2 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R148 (8.7 g, 12.8 mmol, yield 75%).

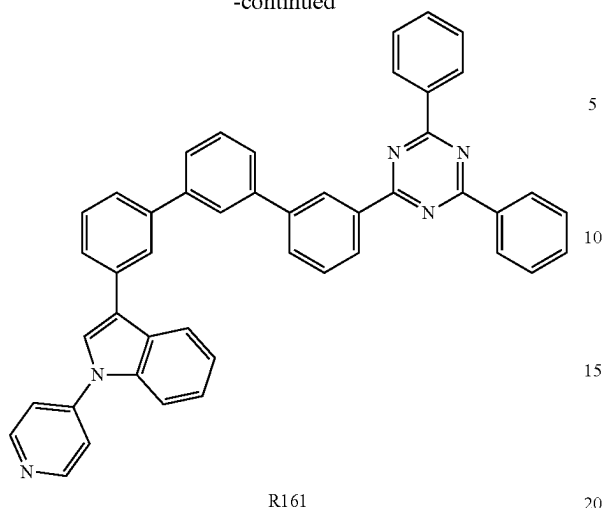
GC-Mass (theoretical value: 680.8 g/mol, measured value: 681 g/mol)

[Synthesis Example 12] Synthesis of R161



135

-continued

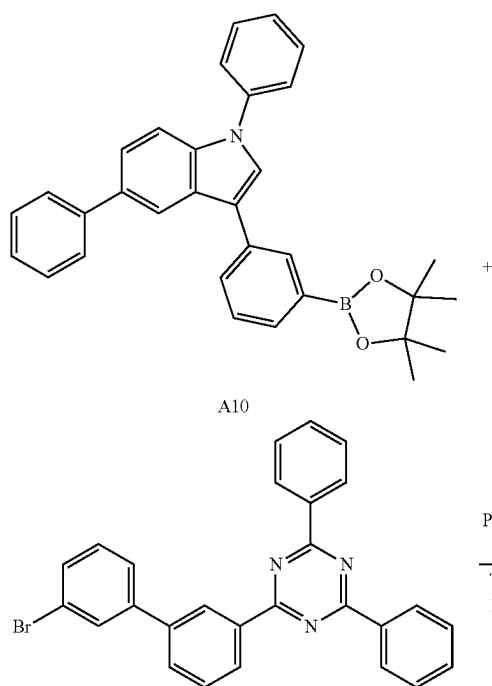


Under a nitrogen stream, A9 (6.8 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

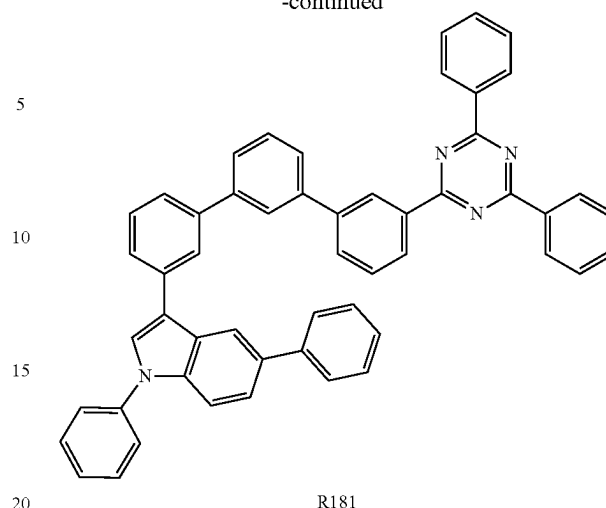
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R161 (7.9 g, 11.1 mmol, yield 71%).

GC-Mass (theoretical value: 653.8 g/mol, measured value: 654 g/mol)

[Synthesis Example 13] Synthesis of R181

**136**

-continued

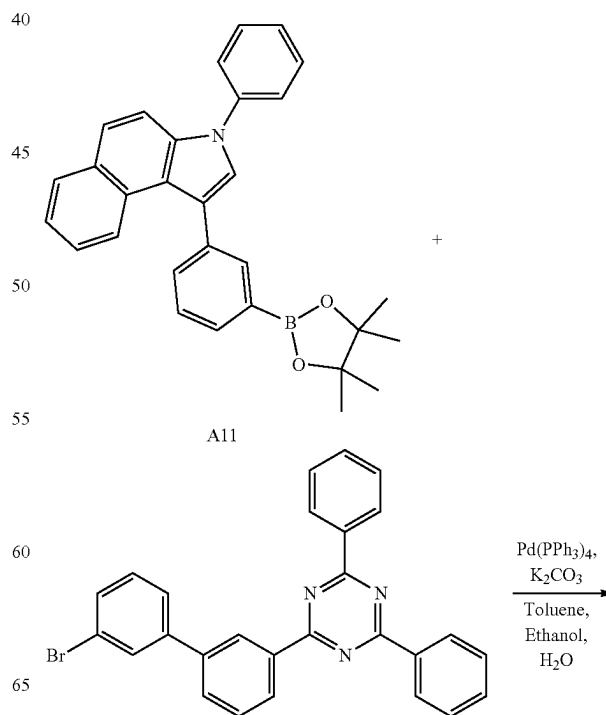


Under a nitrogen stream, A10 (8.6 g, 18.3 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (9.3 g, 20.1 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R181 (10.0 g, 13.7 mmol, yield 75%).

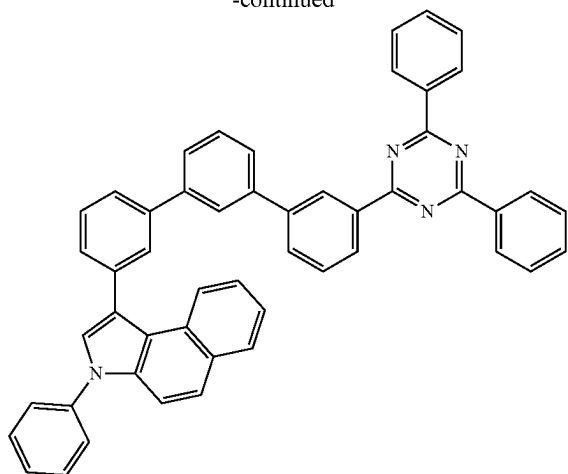
GC-Mass (theoretical value: 729.9 g/mol, measured value: 730 g/mol)

[Synthesis Example 14] Synthesis of R201



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-continued



R201

Under a nitrogen stream, A11 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

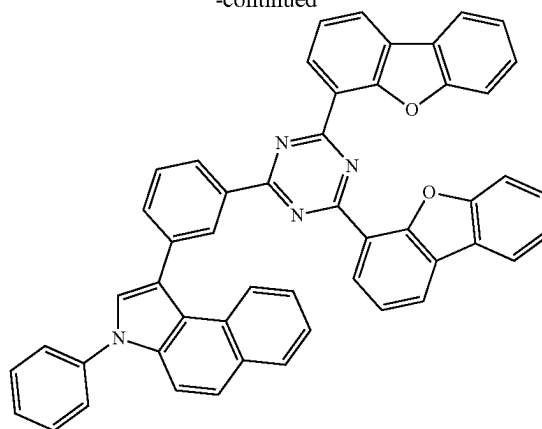
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R201 (9.0 g, 12.8 mmol, yield 75%).

GC-Mass (theoretical value: 703.9 g/mol, measured value: 703 g/mol)

[Synthesis Example 15] Synthesis of R208

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-continued



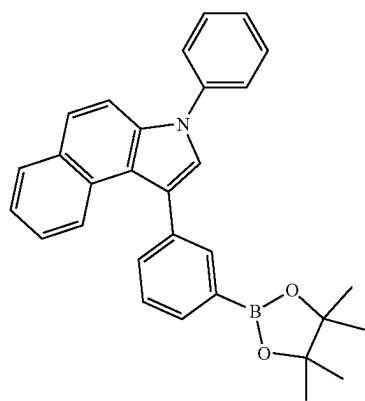
R208

Under a nitrogen stream, A11 (7.6 g, 17.1 mmol), 2-bromo-4,6-bis(dibenzo[b,d]furan-4-yl)-1,3,5-triazine (9.2 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

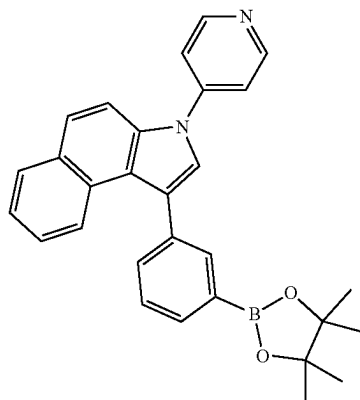
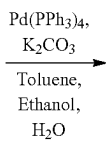
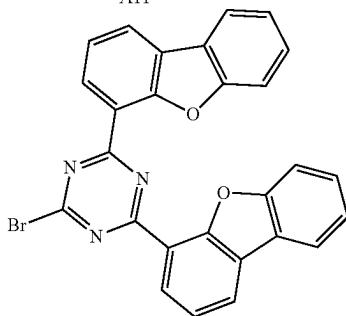
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R208 (9.4 g, 12.8 mmol, yield 75%).

GC-Mass (theoretical value: 731.8 g/mol, measured value: 731 g/mol)

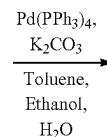
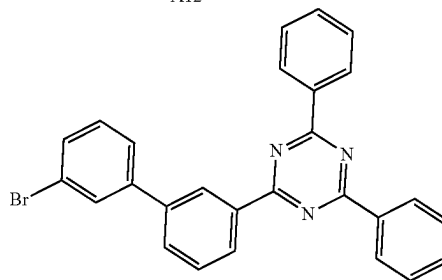
[Synthesis Example 16] Synthesis of R221



A11

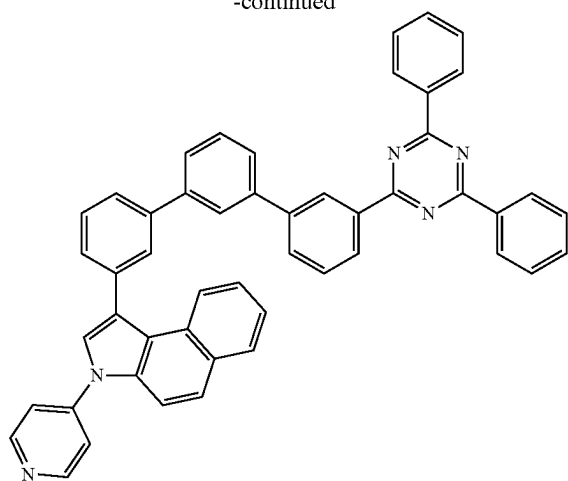


A12



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-continued



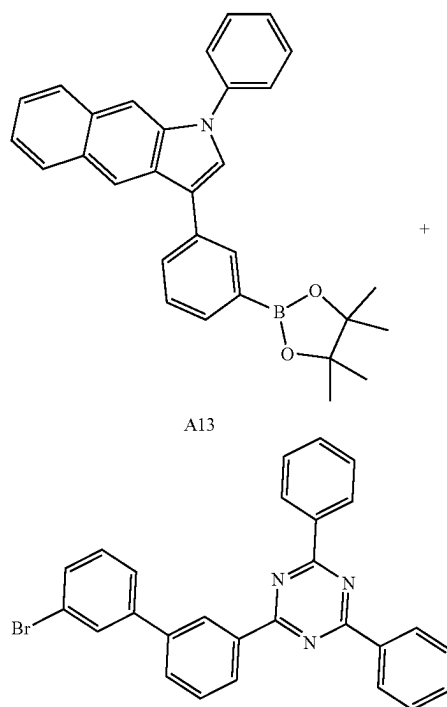
R221

Under a nitrogen stream, A12 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

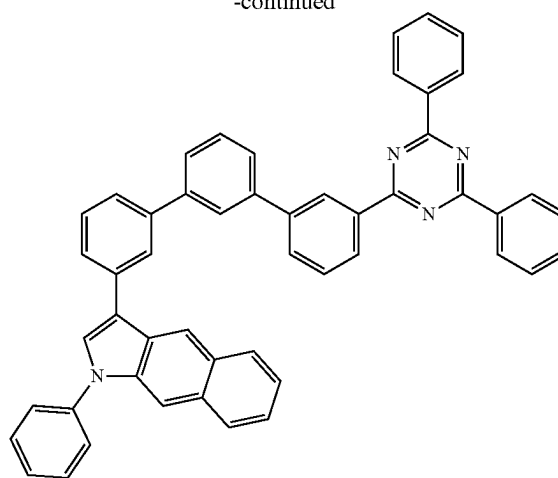
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R221 (8.9 g, 12.6 mmol, yield 74%).

GC-Mass (theoretical value: 703.9 g/mol, measured value: 704 g/mol)

[Synthesis Example 17] Synthesis of R241

**140**

-continued



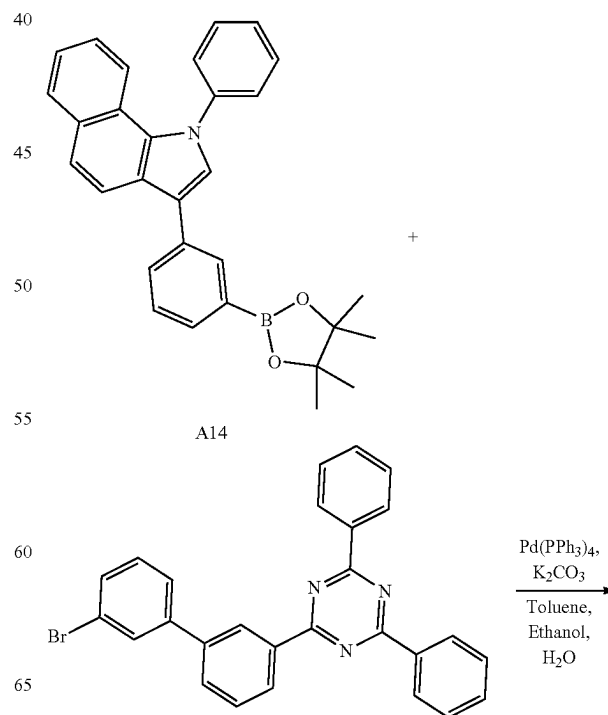
R241

Under a nitrogen stream, A13 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R241 (9.0 g, 12.8 mmol, yield 75%).

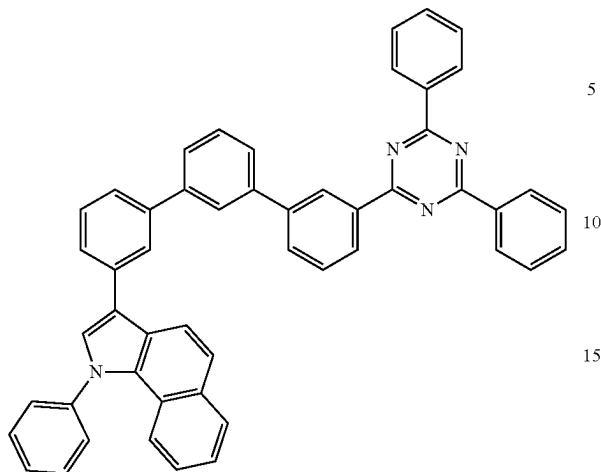
GC-Mass (theoretical value: 702.9 g/mol, measured value: 703 g/mol)

[Synthesis Example 18] Synthesis of R261



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-continued



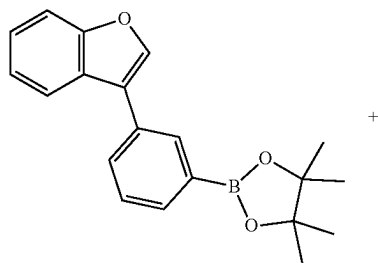
R261

Under a nitrogen stream, A14 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

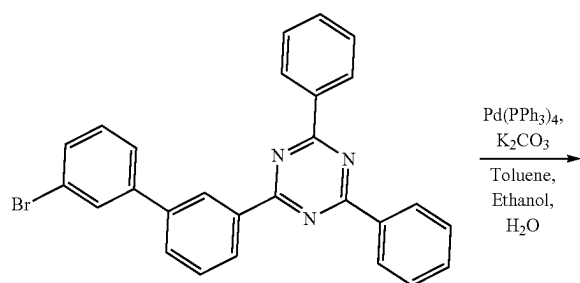
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R261 (8.9 g, 12.0 mmol, yield 74%).

GC-Mass (theoretical value: 702.9 g/mol, measured value: 703 g/mol)

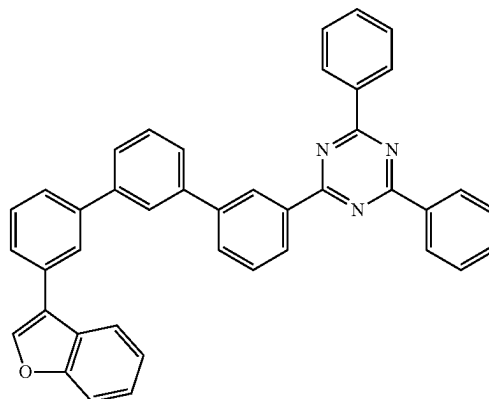
[Synthesis Example 19] Synthesis of R281



A15

**142**

-continued



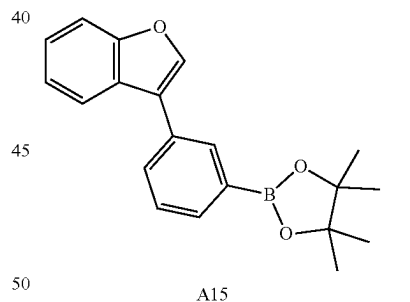
R281

Under a nitrogen stream, A15 (5.5 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

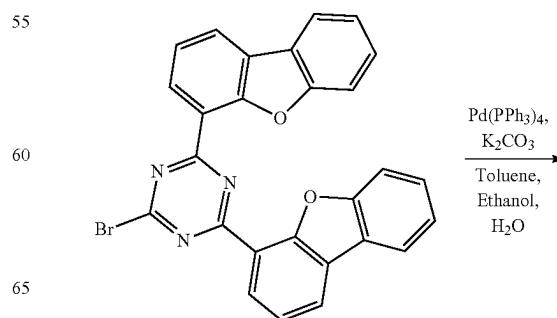
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R281 (9.9 g, 12.8 mmol, yield 75%).

GC-Mass (theoretical value: 577.7 g/mol, measured value: 578 g/mol)

[Synthesis Example 20] Synthesis of R285

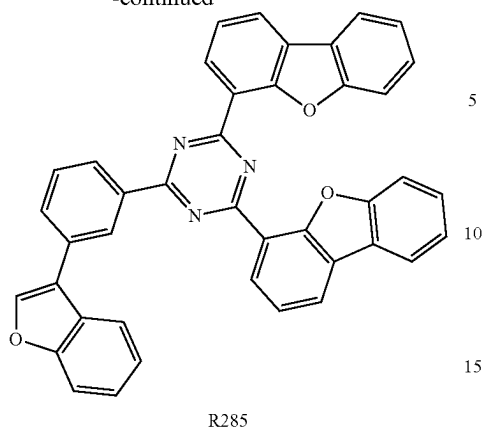


A15



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-continued

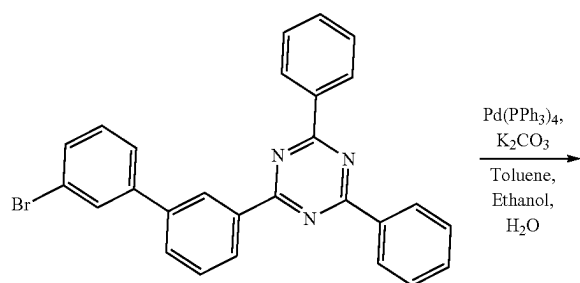
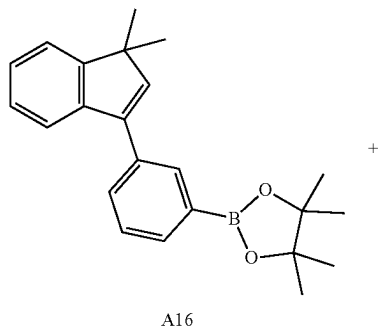


Under a nitrogen stream, A15 (5.5 g 17.1 mmol), 2-bromo-4,6-bis(dibenzo[b,d]furan-4-yl)-1,3,5-triazine (9.2 g 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

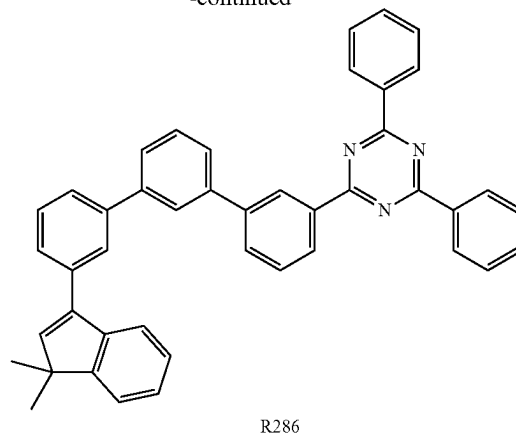
After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R285 (8.0 g, 13.1 mmol, yield 77%).

GC-Mass (theoretical value: 605.7 g/mol, measured value: 606 g/mol)

[Synthesis Example 21] Synthesis of R286

**144**

-continued

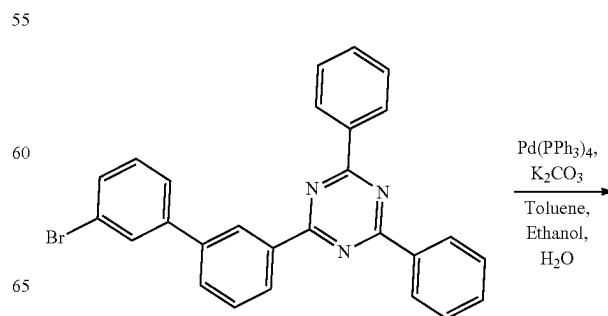
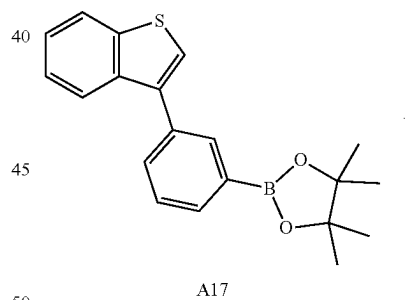


Under a nitrogen stream, A16 (5.9 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R286 (7.2 g, 11.9 mmol, yield 70%).

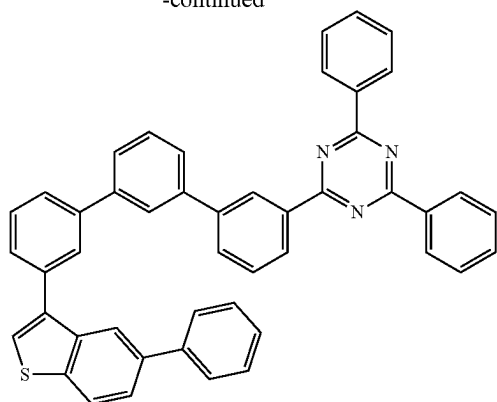
GC-Mass (theoretical value: 603.8 g/mol, measured value: 604 g/mol)

[Synthesis Example 22] Synthesis of R291



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-continued



R291

Under a nitrogen stream, A17 (6.1 g, 18.3 mmol), 2-(3'-
bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (9.3 g,
20.1 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbon-
ate (7.6 g, 73.1 mmol) and toluene/H₂O/ethanol (80 ml/40
ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

After the reaction was finished, the organic layer was
separated using methylene chloride, and water was removed
using MgSO₄. After the solvent of the organic layer was
removed, the result was purified using column chromatog-
raphy to obtain target Compound R291 (9.2 g, 13.7 mmol,
yield 75%).

GC-Mass (theoretical value: 669.9 g/mol, measured
value: 670 g/mol)

Examples 1 to 6 Manufacture of Green Organic Electroluminescent Device

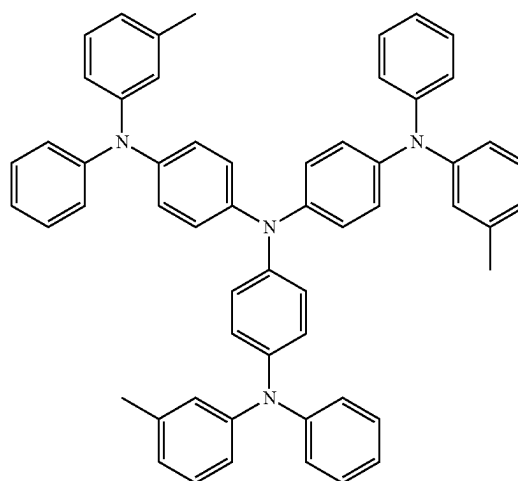
After high purity sublimation purifying the compounds
synthesized in the synthesis examples using commonly
known methods, green organic electroluminescent devices
were manufactured using the following procedure.

First, a glass substrate on which indium tin oxide (ITO)
was coated as a thin film to a thickness of 1500 Å was
ultrasonic cleaned using distilled water. After the cleaning
with distilled water was finished, the substrate was ultra-
sonic cleaned with solvents of isopropyl alcohol, acetone,
methanol and the like, dried, then transferred to a UV
OZONE washer (Power sonic 405, manufactured by
Hwashin Tech. Co., Ltd.), and then, after cleaning the
substrate for 5 minutes using UV, the substrate was trans-
ferred to a vacuum deposition apparatus.

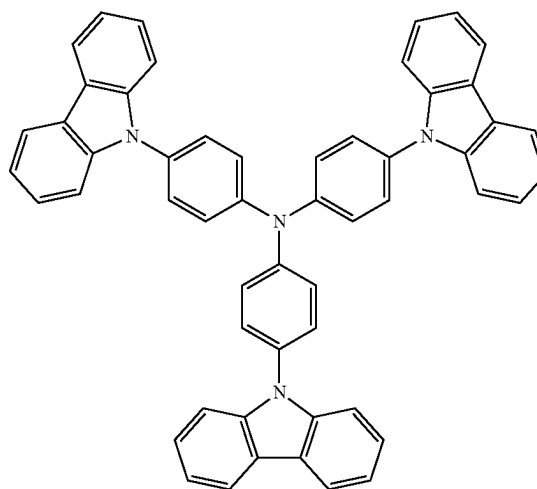
On the transparent ITO electrode prepared as above,
m-MTDATA (60 nm)/TCTA (80 nm)/each compound of R1,
R8, R141, R148, R201 and R208+10% Ir(ppy)₃ (300 nm)/
BCP (10 nm)/Alq₃ (30 nm)/LiF (1 nm)/Al (200 nm) were
laminated in this order to manufacture an organic electrolu-
minescent device.

Structures of m-MTDATA, TCTA, Ir(ppy)₃, CBP and
BCP are as follows.

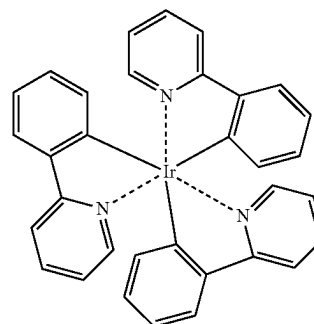
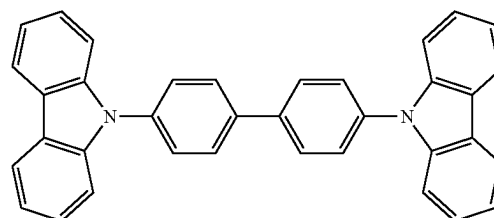
146



m-MTDATA



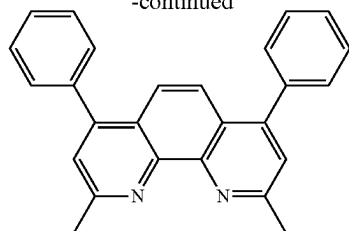
TCTA

Ir(ppy)₃

CBP

147

-continued



BCP

Comparative Example 1 Manufacture of Green Organic Electroluminescent Device

A green organic electroluminescent device was manufactured in the same manner as in Example 1 except that CBP was used instead of Compound R141 as the light emitting host material when forming the light emitting layer.

Evaluation Example 1

For each of the green organic electroluminescent devices manufactured in Examples 1 to 6 and Comparative Example 2, driving voltage, current efficiency and light emission peak at current density of (10) mA/cm² were measured, and the results are shown in the following Table 1.

TABLE 1

Sample	Host	Driving Voltage (V)	EL Peak (nm)	Current Efficiency (cd/A)
Example 1	R1	5.2	515	12.5
Example 2	R8	4.3	515	11.4
Example 3	R141	5.1	515	12.3
Example 4	R148	4.2	515	11.1
Example 5	R201	5.4	515	12.2
Example 6	R208	4.4	515	10.8
Comparative Example 1	CBP	7.1	516	7.2

As shown in Table 1, it was seen that using the compound (R1, R8, R141, R148, R201, R208) according to the present invention in a light emitting layer of the green organic EL device (Examples 1 to 6) resulted in more superior performance in terms of efficiency and driving voltage compared to the green organic EL device using existing CBP (Comparative Example 1).

Examples 7 to 28 Manufacture of Blue Organic Electroluminescent Device

After high purity sublimation purifying the compounds synthesized in the synthesis examples using commonly known methods, blue organic electroluminescent devices were manufactured using the following procedure.

First, a glass substrate on which indium tin oxide (ITO) was coated as a thin film to a thickness of 1500 Å was ultrasonic cleaned using distilled water. After the cleaning with distilled water was finished, the substrate was ultrasonic cleaned with solvents of isopropyl alcohol, acetone, methanol and the like, dried, then transferred to a UV OZONE washer (Power sonic 405, manufactured by Hwashin Tech. Co., Ltd), and then, after cleaning the substrate for 5 minutes using UV, the substrate was transferred to a vacuum deposition apparatus.

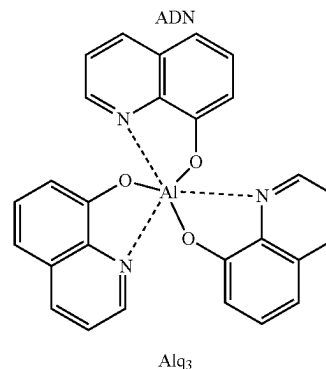
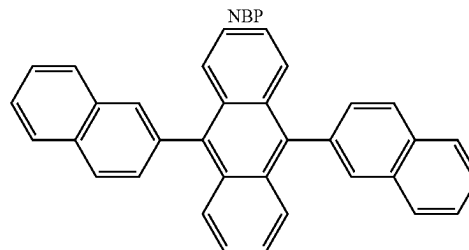
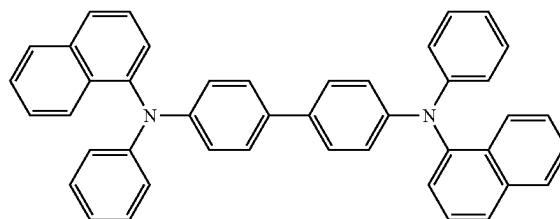
148

On the transparent ITO electrode prepared as above, DS-205 (Doosan Corporation Electro-Materials, 80 nm)/NPB (15 nm)/ADN+5% DS-405 (Doosan Corporation Electro-Materials, 30 nm)/each compound of R1, R8, R21, R41, R61, R68, R81, R101, R121, R141, R141, R148, R161, R181, R201, R208, R221, R241, R261, R281, R285, R286 and R291 (5 nm)/Alq₃ (25 nm)/LiF (1 nm)/Al (200 nm) were laminated in this order to manufacture an organic electroluminescent device.

Comparative Example 2 Manufacture of Blue Organic Electroluminescent Device

A blue organic electroluminescent device was manufactured in the same manner as in Example 6 except that R1 was not used as the electron transport auxiliary layer material, and Alq₃, the electron transport layer material, was deposited to 30 nm instead of 25 nm.

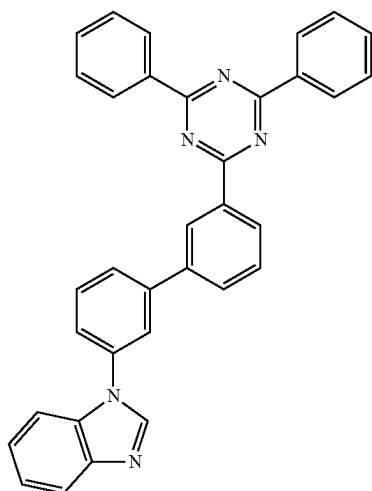
Structures of NPB, AND and Alq₃ used in Examples 6 to 28 and Comparative Example 2 are as follows.



Comparative Example 3 Manufacture of Blue Organic Electroluminescent Device

A blue organic electroluminescent device was manufactured in the same manner as in Example 6 except that A1 of the following structural formula was used instead of R1 as the electron transport auxiliary layer material.

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Evaluation Example 2

For each of the organic electroluminescent devices manufactured in Examples 7 to 28 and Comparative Examples 2 and 3, driving voltage, light emission wavelength and current efficiency at current density of 10 mA/cm² were measured, and the results are shown in the following Table 2.

TABLE 2

Sample	Electron Transport Auxiliary Layer	Driving Voltage (V)	Light Emission Peak (nm)	Current Efficiency (cd/A)
Example 7	R1	4.4	458	8.3
Example 8	R8	3.9	458	7.5
Example 9	R21	4.4	458	8.0
Example 10	R41	4.4	458	8.2
Example 11	R61	4.4	458	8.1
Example 12	R68	4.4	458	8.2
Example 13	R81	4.4	458	8.3
Example 14	R101	4.4	458	8.0
Example 15	R121	4.4	458	8.2
Example 16	R141	4.3	458	7.8
Example 17	R148	3.5	458	7.1
Example 18	R161	4.4	458	7.3
Example 19	R181	4.5	458	7.4
Example 20	R201	4.1	458	7.5
Example 21	R208	3.9	458	7.0
Example 22	R221	4.2	458	7.5
Example 23	R241	4.1	458	7.6
Example 24	R261	4.5	458	7.3
Example 25	R281	3.8	458	6.9
Example 26	R285	3.4	458	6.8
Example 27	R286	4.7	458	7.2
Example 28	R291	3.8	458	6.9
Comparative Example 2	Alq ₃	4.8	458	6.2
Comparative Example 3	A1	4.7	457	6.5

As shown in Table 2, it was seen that the blue organic electroluminescent devices using the compound of the present invention in an electron transport auxiliary layer (Examples 6 to 28) exhibited superior performance in terms of current efficiency, light emission peak and driving voltage compared to the blue organic electroluminescent devices without an electron transport auxiliary layer (Comparative Examples 2 and 3).

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Examples 29 to 32 Manufacture of Blue Organic Electroluminescent Device

After high purity sublimation purifying the compounds synthesized in the synthesis examples using commonly known methods, blue organic electroluminescent devices were manufactured using the following procedure.

First, a glass substrate on which indium tin oxide (ITO) was coated as a thin film to a thickness of 1500 Å was ultrasonic cleaned using distilled water. After the cleaning with distilled water was finished, the substrate was ultrasonic cleaned with solvents of isopropyl alcohol, acetone, methanol and the like, dried, then transferred to a UV OZONE washer (Power sonic 405, manufactured by Hwashin Tech. Co., Ltd.), and then, after cleaning the substrate for 5 minutes using UV, the substrate was transferred to a vacuum deposition apparatus.

On the transparent ITO electrode prepared as above, DS-205 (Doosan Corporation Electro-Materials, 80 nm)/NPB (15 nm)/ADN+5% DS-405 (Doosan Corporation Electro-Materials, 30 nm)/each compound of R1, R8, R21 and R41 (30 nm)/LiF (1 nm)/Al (200 nm) were laminated in this order to manufacture an organic electroluminescent device.

Comparative Example 4 Manufacture of Blue Organic Electroluminescent Device

A blue organic electroluminescent device was manufactured in the same manner as in Example 29 except that Alq₃ was used instead of R1 as the electron transport layer material.

Comparative Example 5 Manufacture of Blue Organic Electroluminescent Device

A blue organic electroluminescent device was manufactured in the same manner as in Example 29 except that R1 was not used as the electron transport layer material.

Evaluation Example 3

For each of the blue organic electroluminescent devices manufactured in Examples 29 to 32 and Comparative Examples 2 and 3, driving voltage, current efficiency and light emission wavelength at current density of 10 mA/cm² were measured, and the results are shown in the following Table 3.

TABLE 3

Sample	Electron Transport Layer	Driving Voltage (V)	Light Emission Peak (nm)	Current Efficiency (cd/A)
Example 29	R1	4.5	455	8.7
Example 30	R8	3.9	455	7.8
Example 31	R21	4.4	455	8.4
Example 32	R41	4.2	455	9.3
Comparative Example 4	Alq ₃	4.7	458	5.5
Comparative Example 5	—	4.8	460	6.2

As shown in Table 3, it was seen that the blue organic electroluminescent devices using the compound of the present invention in an electron transport layer (Examples 29 to 32) exhibited superior performance in terms of driving voltage, light emission peak and current efficiency compared to the blue organic electroluminescent device using existing

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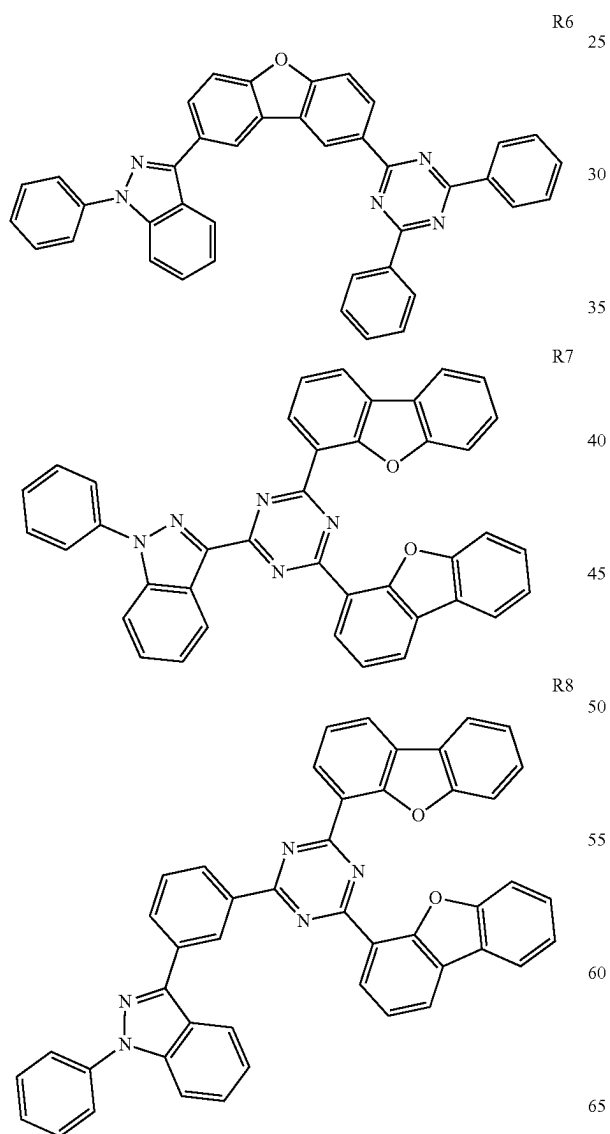
Alq₃ in an electron transport layer (Comparative Example 4), and the blue organic electroluminescent device without an electron transport layer (Comparative Example 5).

REFERENCE NUMERAL

- 10: Anode
- 20: Cathode
- 30: Organic Layer
- 31: Hole Transport Layer
- 32: Light Emitting Layer
- 33: Hole Transport Auxiliary Layer
- 34: Electron Transport Layer
- 35: Electron Transport Auxiliary Layer
- 36: Electron Injection Layer
- 37: Hole Injection Layer

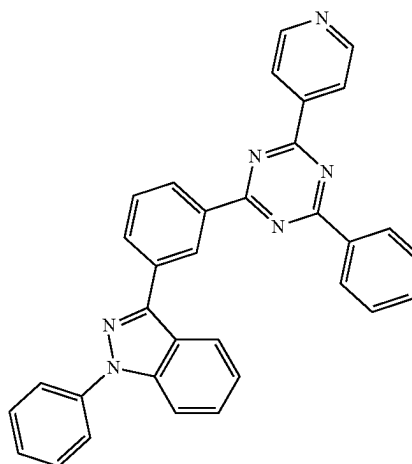
The invention claimed is:

1. A compound selected from the group consisting of the following compounds:

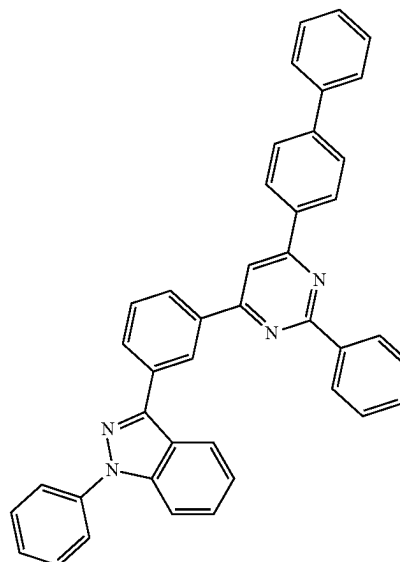
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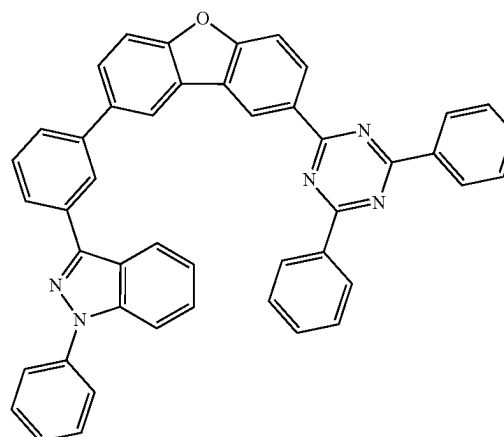
R9



R11

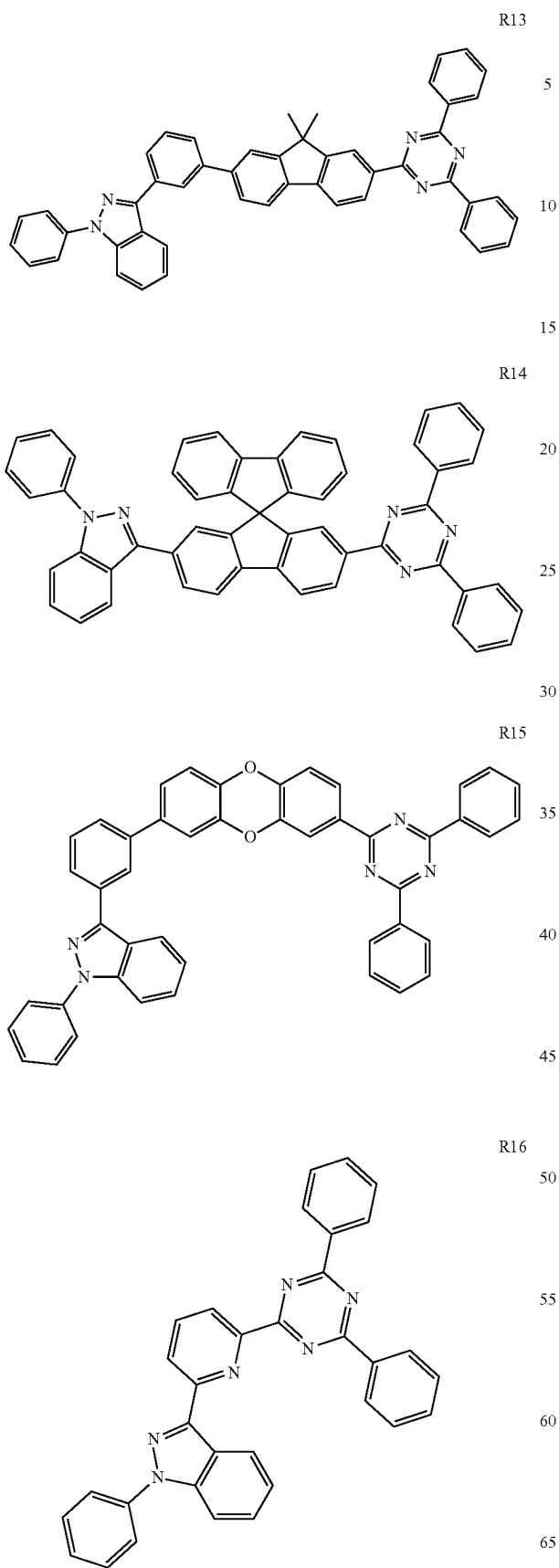


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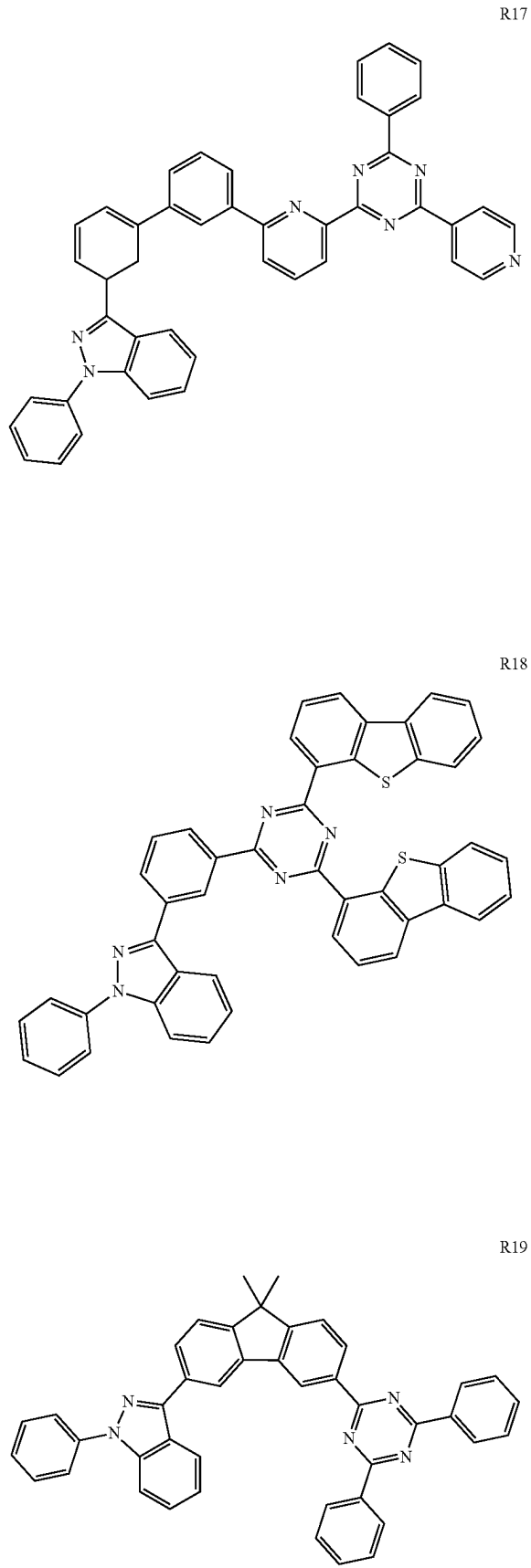


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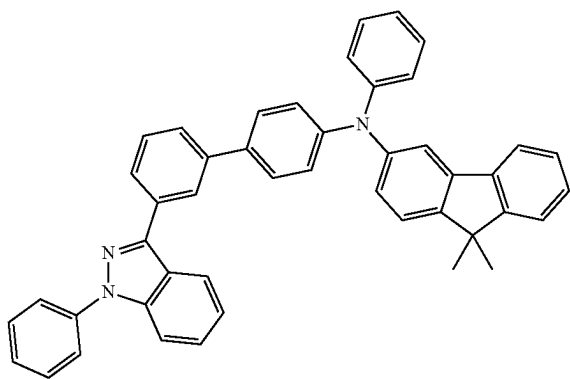
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R20



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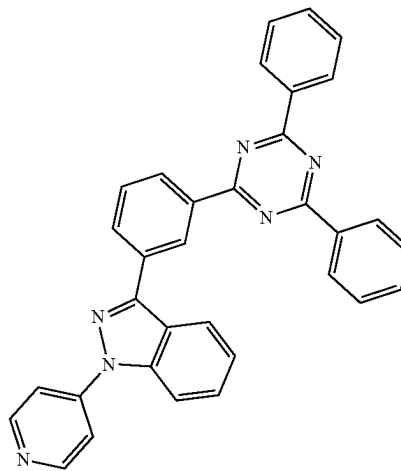
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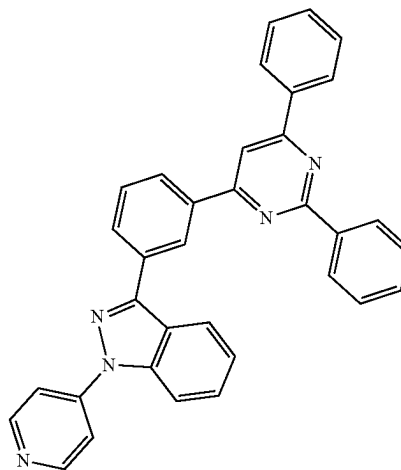
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R23



R24



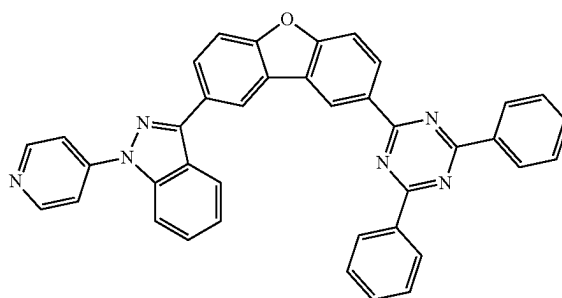
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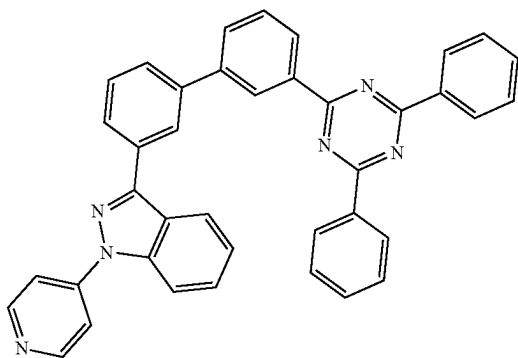
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R26



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R22

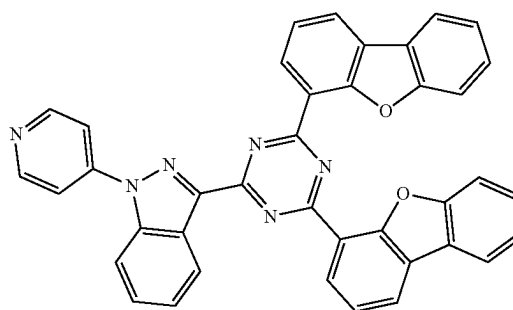


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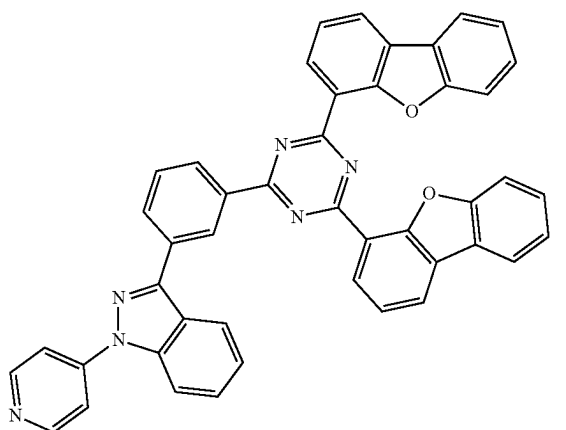
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R27



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R28

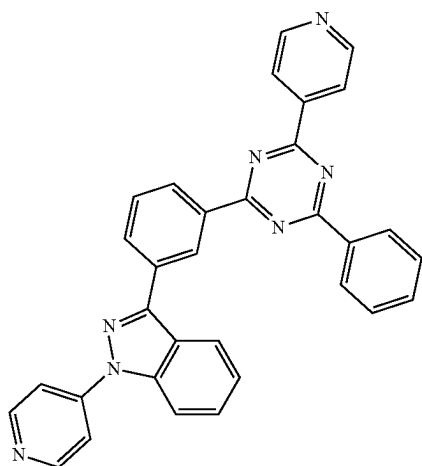
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R29

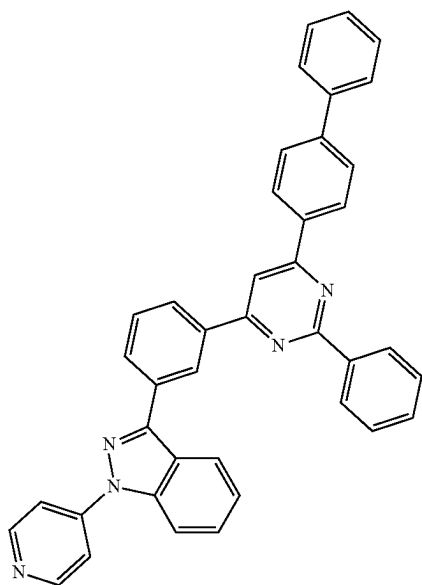


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R31



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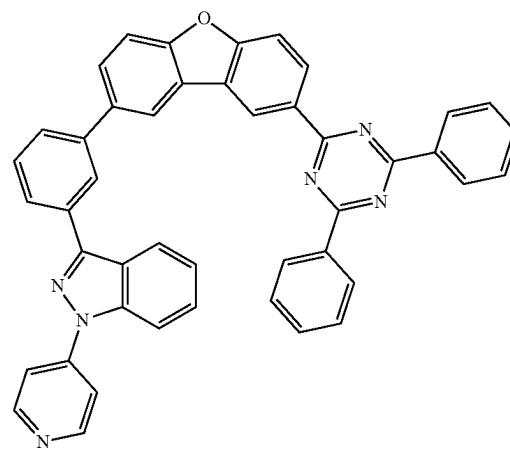
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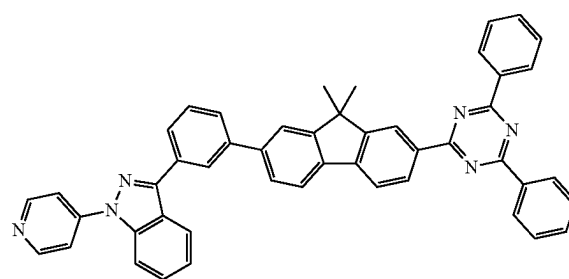
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R32



R33

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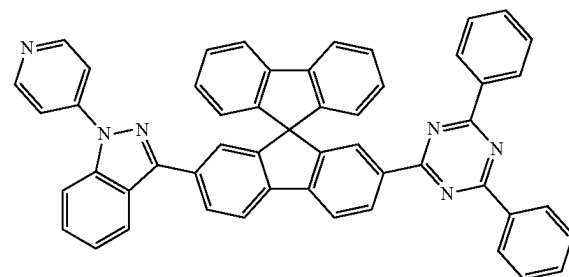
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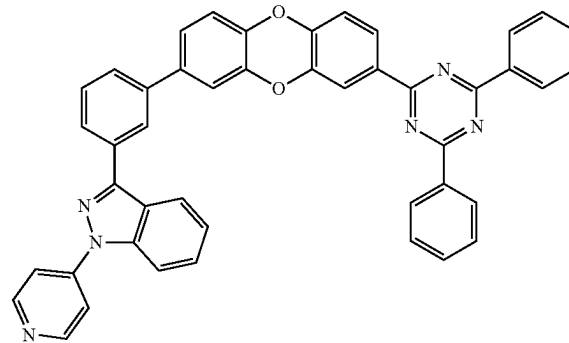
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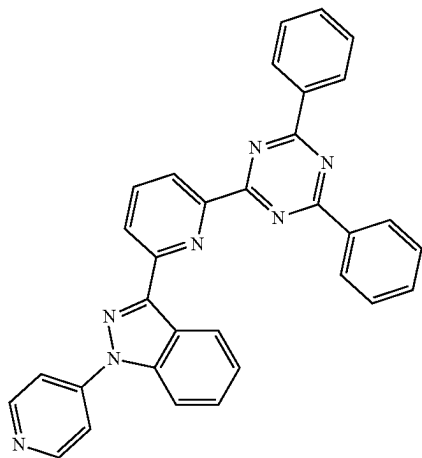


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R36

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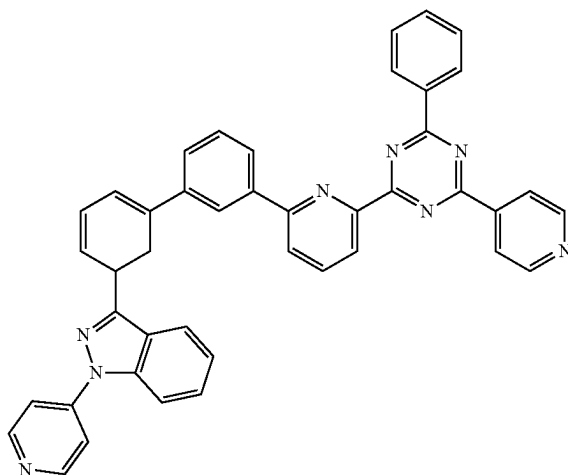
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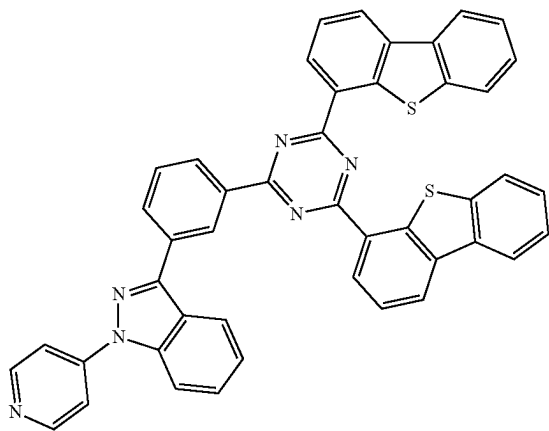
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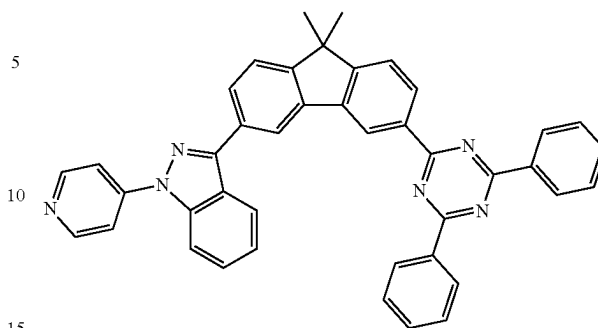
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R39



R40

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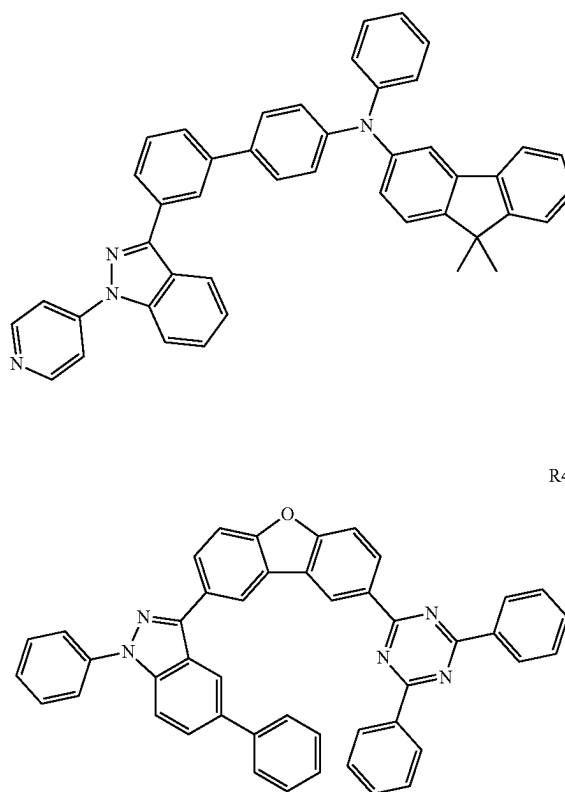
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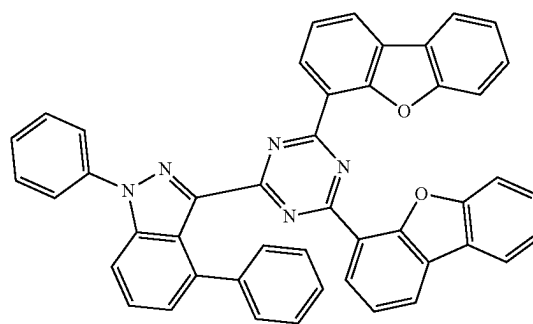
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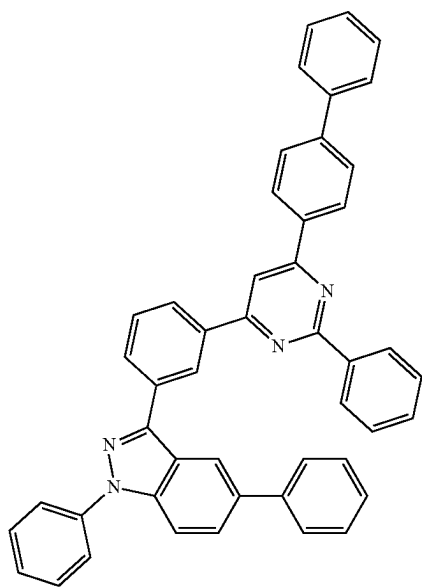
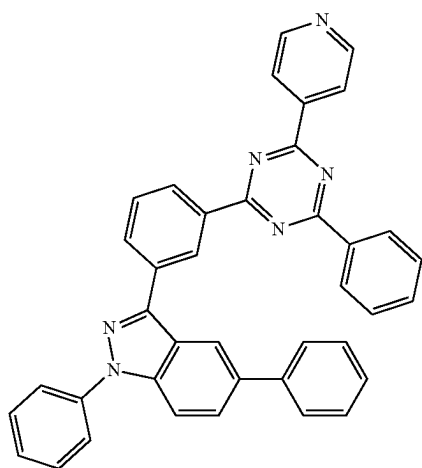
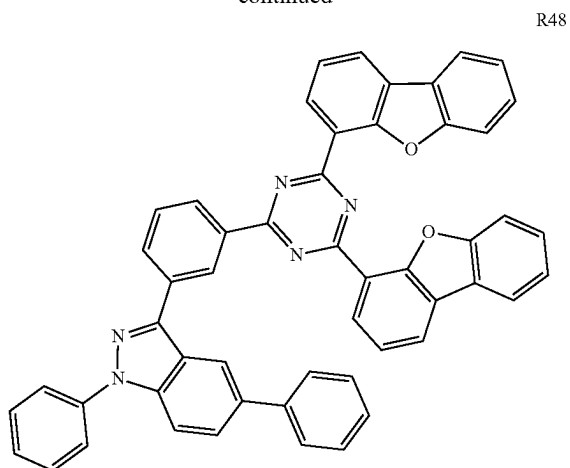
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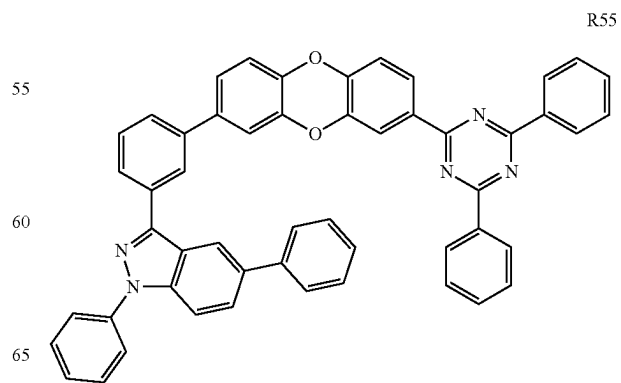
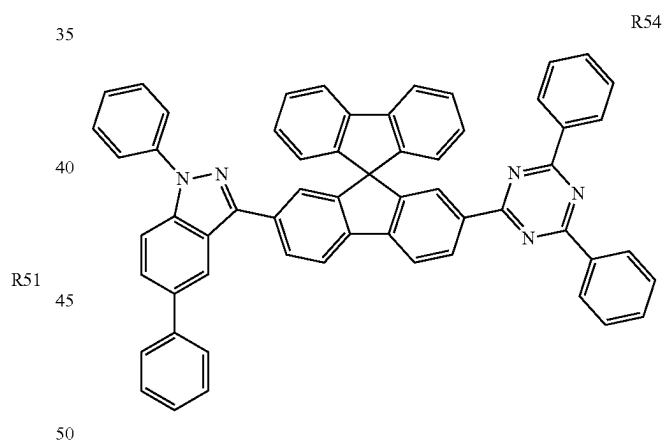
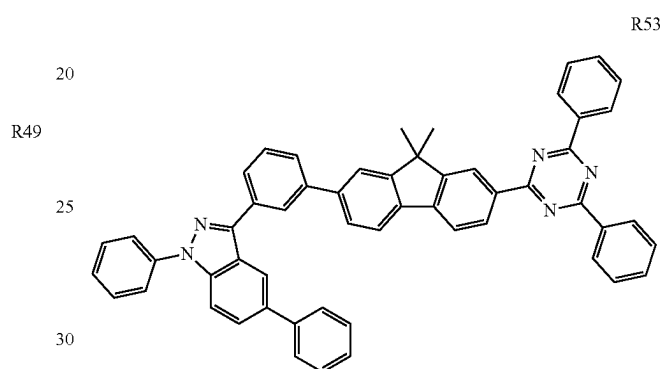
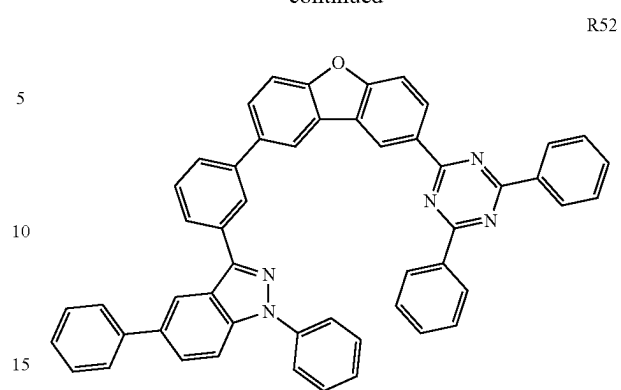


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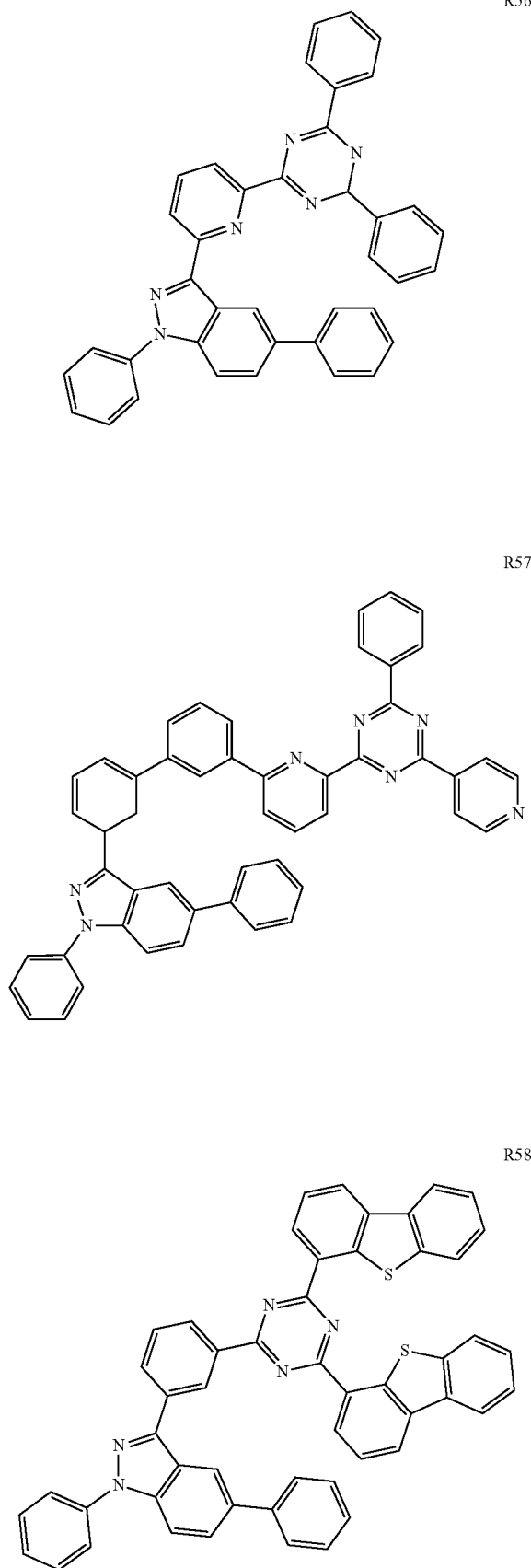
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163

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**164**

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R56

R59

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R57 25

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R58 50

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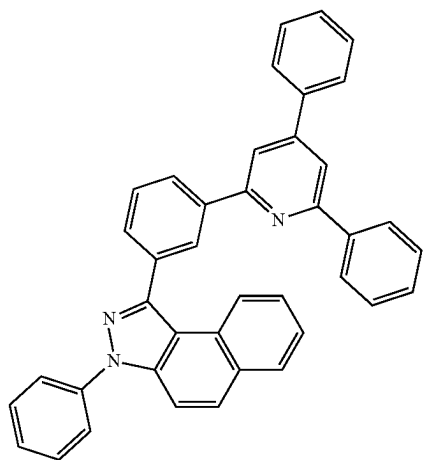
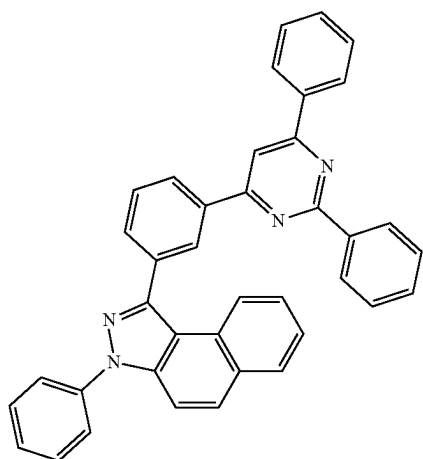
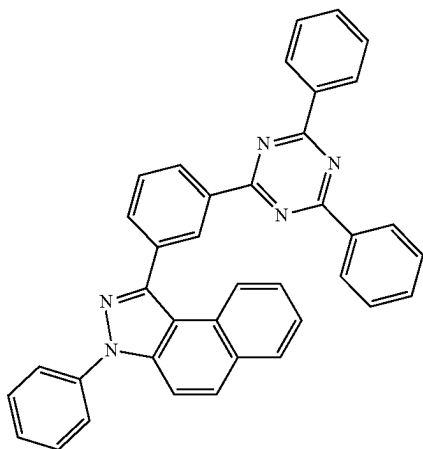
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R62

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R66

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R64

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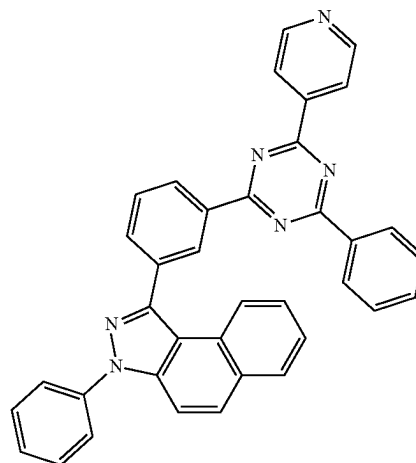
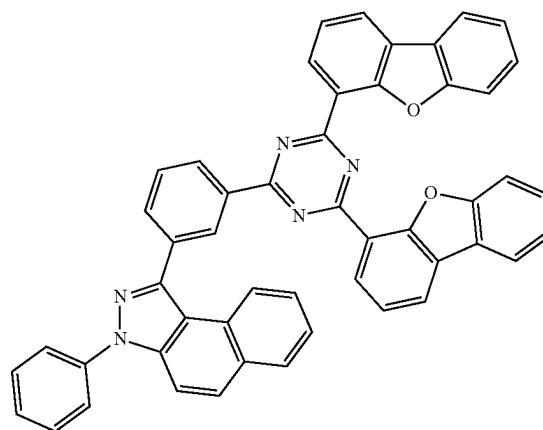
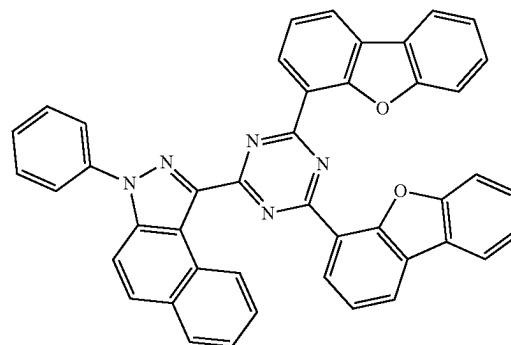
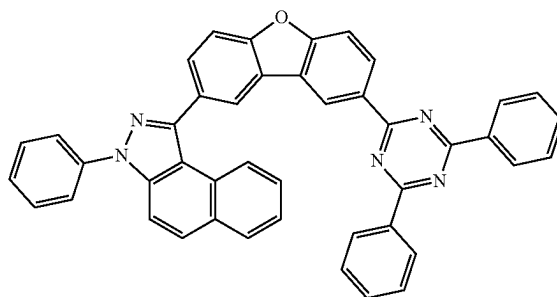
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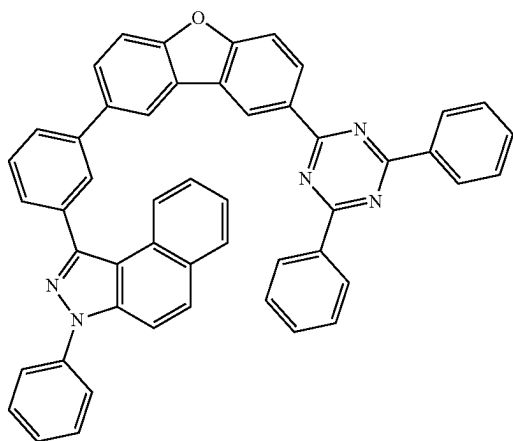
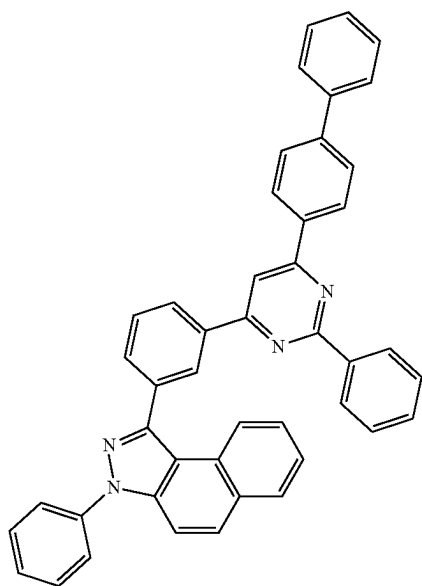
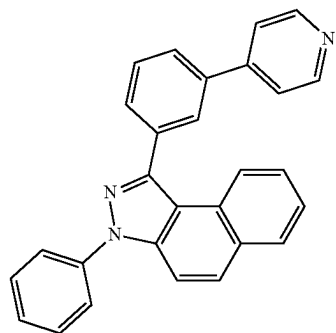
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R71

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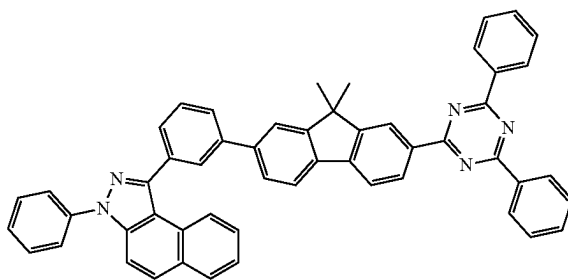
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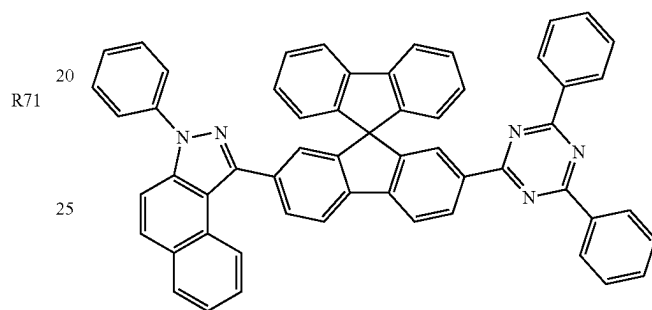
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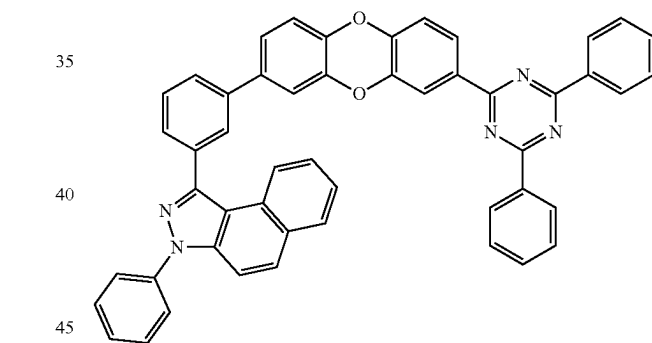
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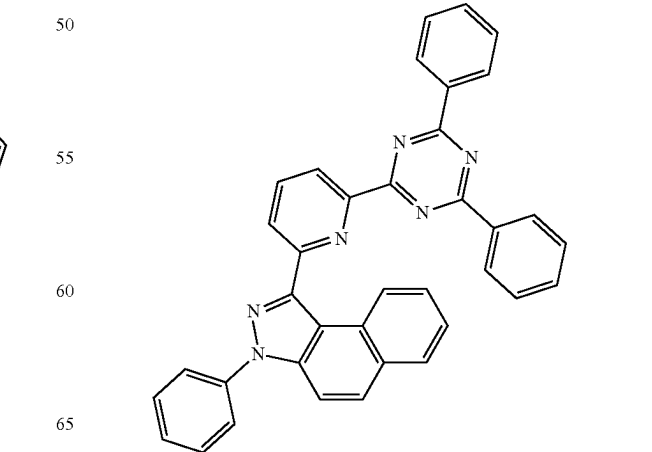
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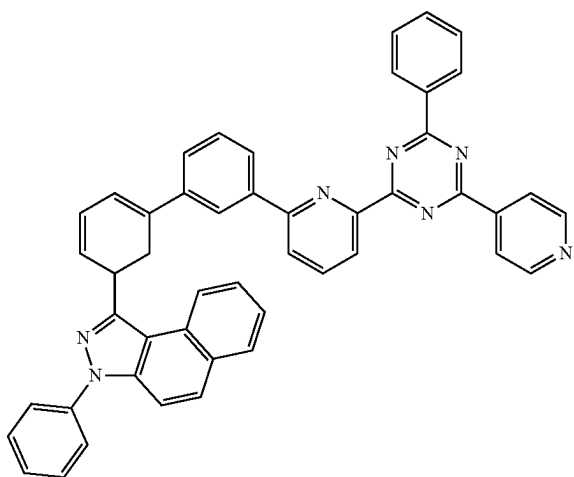
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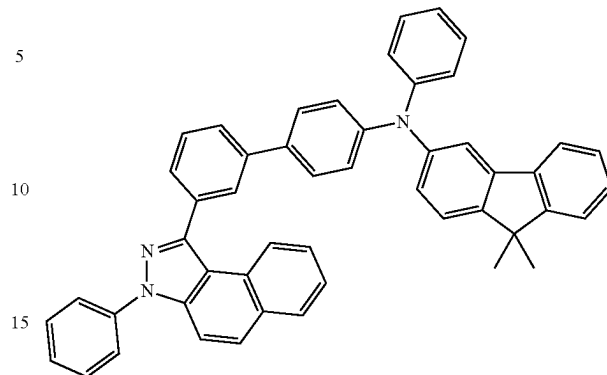
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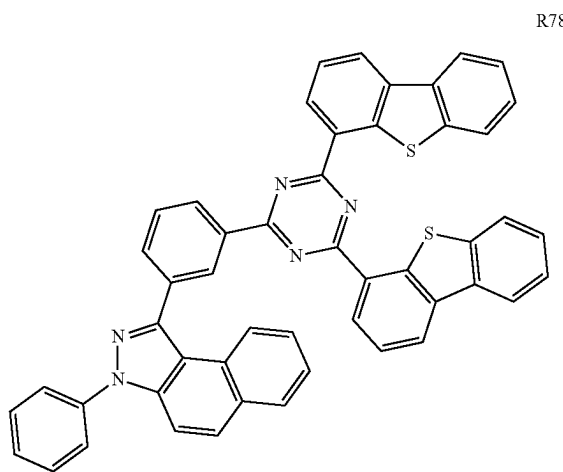
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R81



R78

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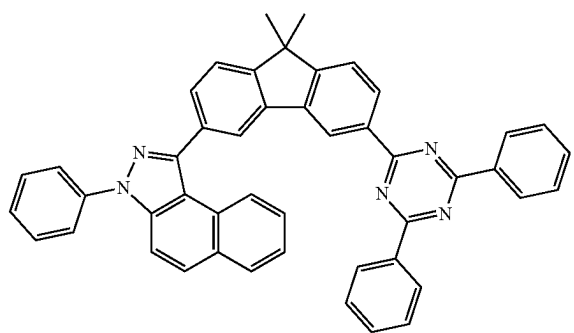
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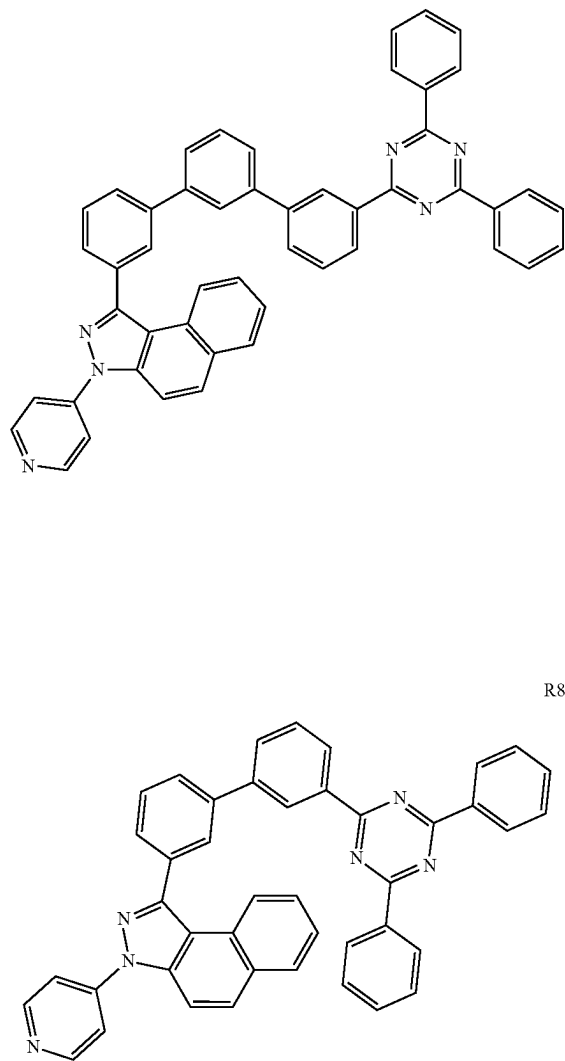
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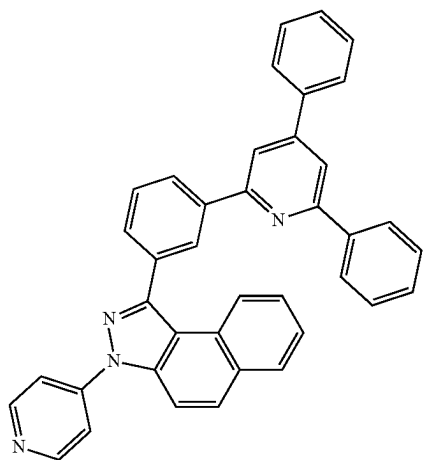
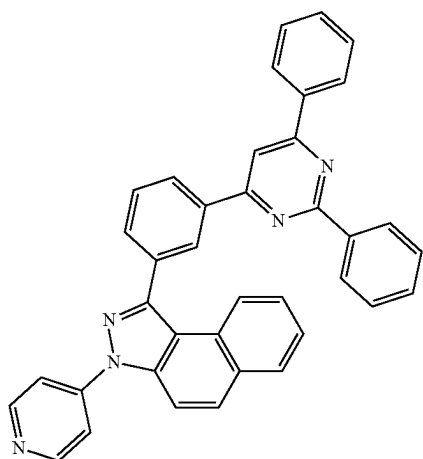
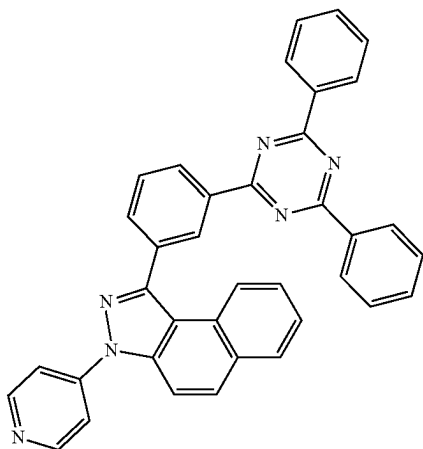
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R82

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**172**

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R83

R86

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R84

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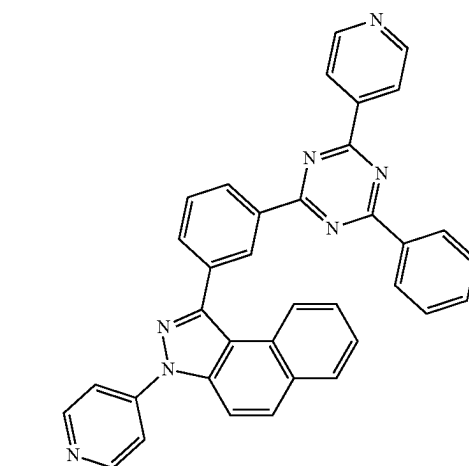
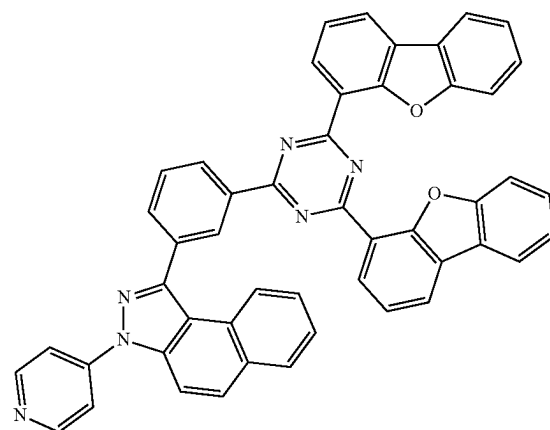
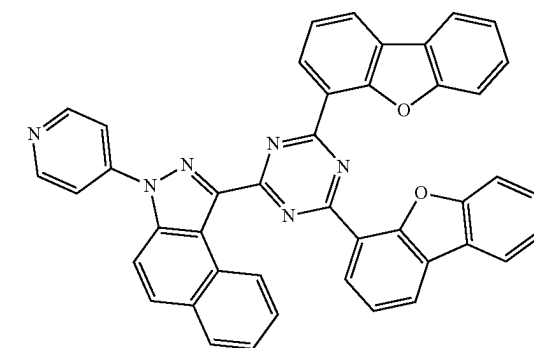
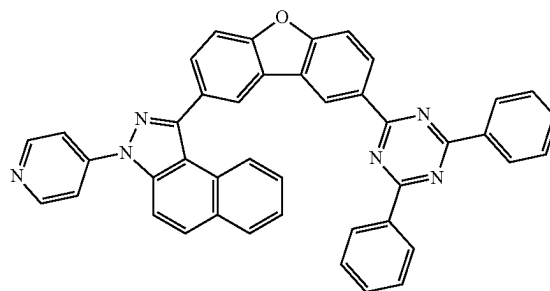
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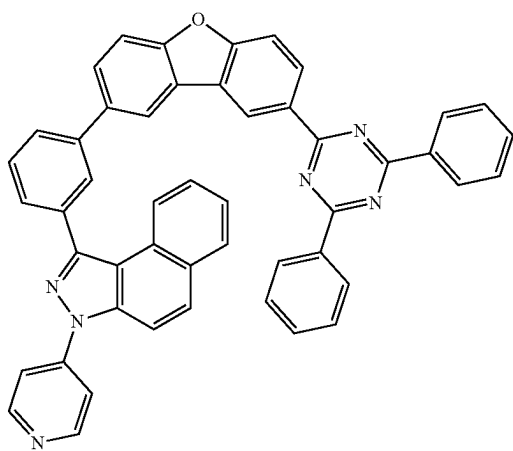
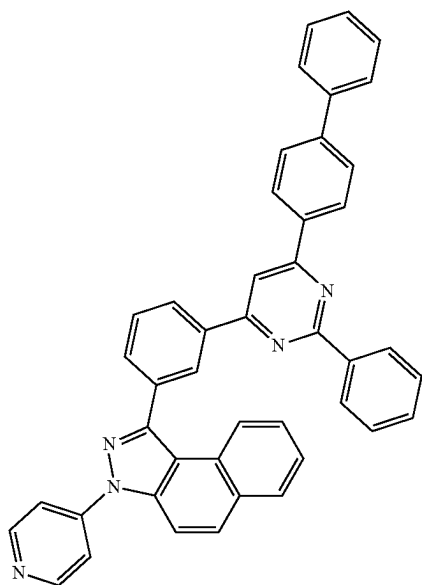
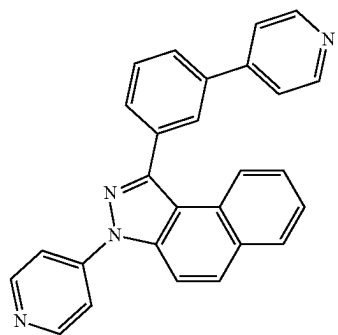
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R91

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R92

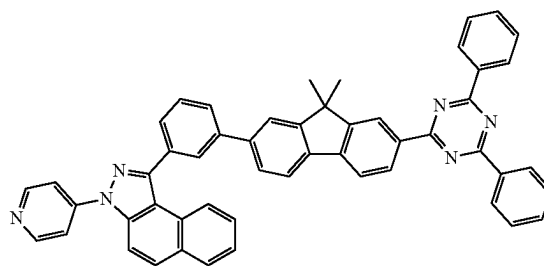
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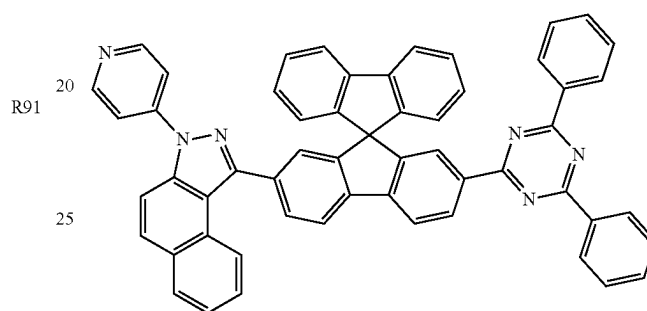
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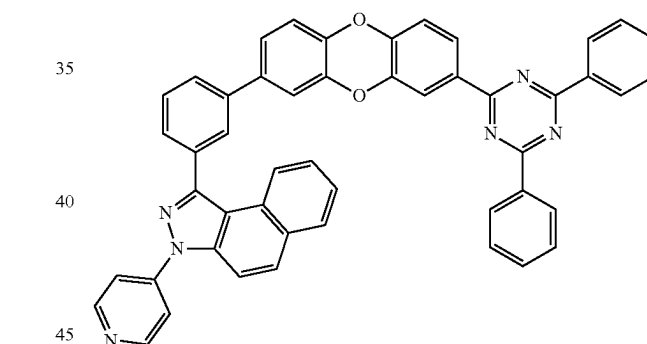
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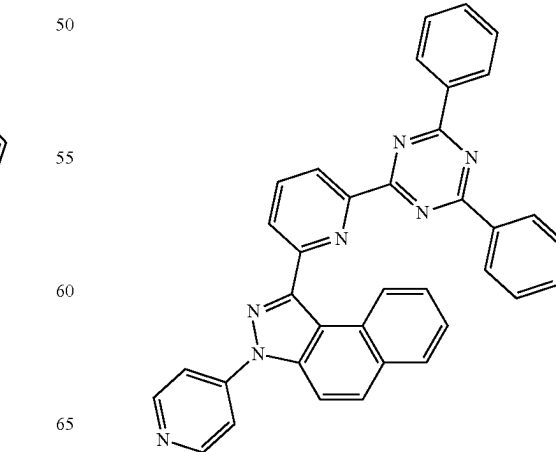
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R95



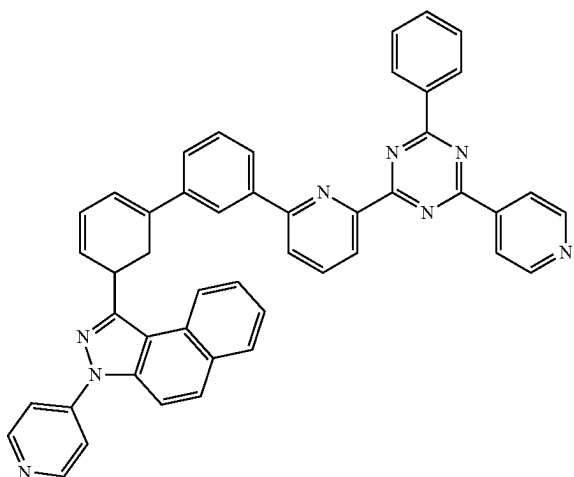
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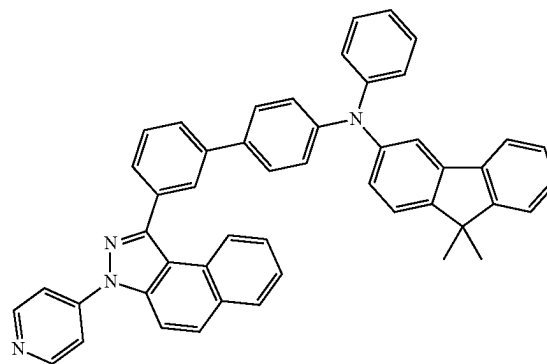
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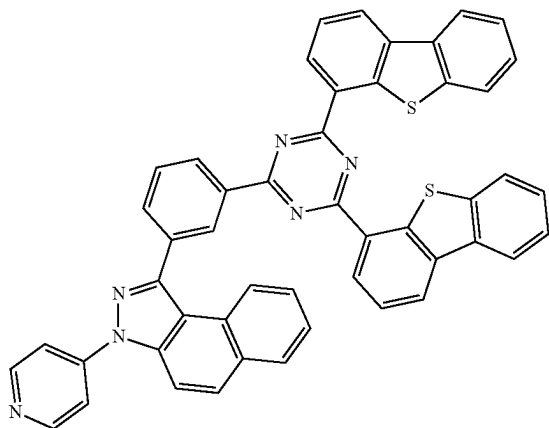
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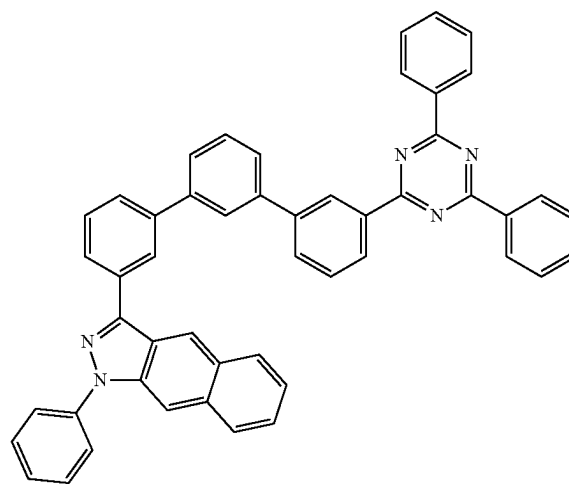
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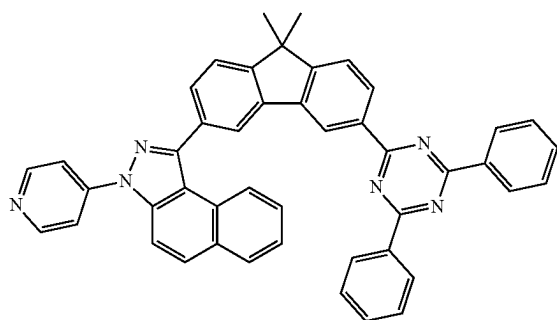
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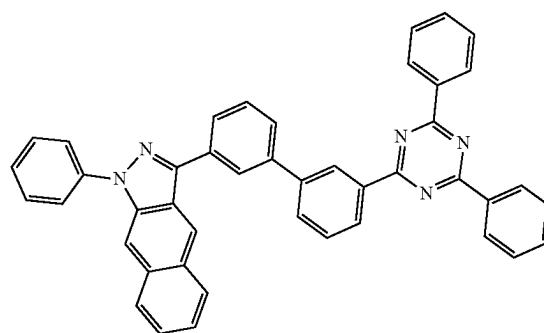
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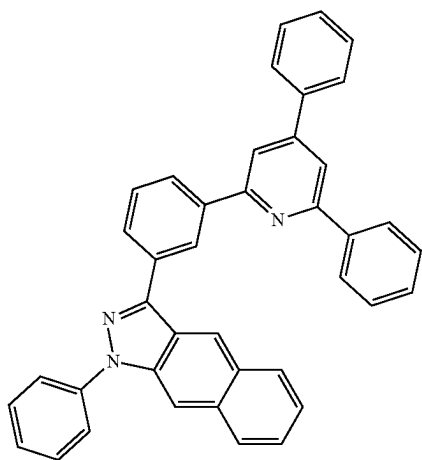
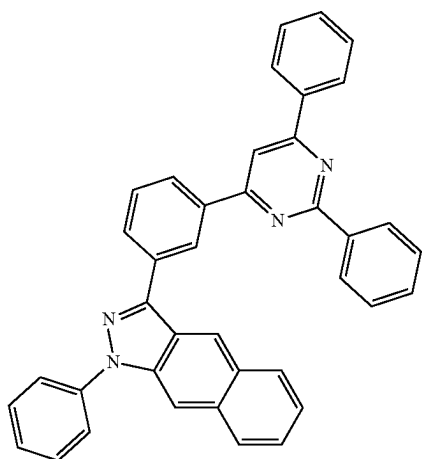
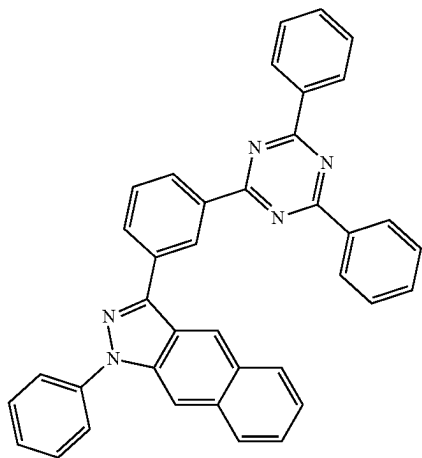


R102



177

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**178**

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R103

R106

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R104

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R105

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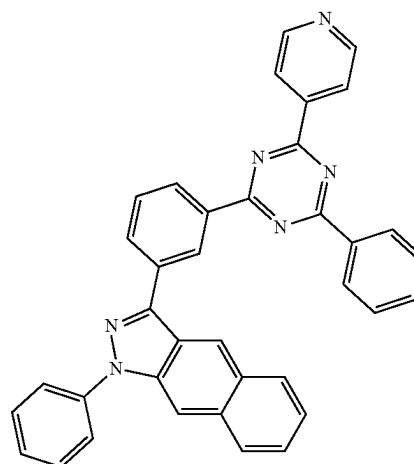
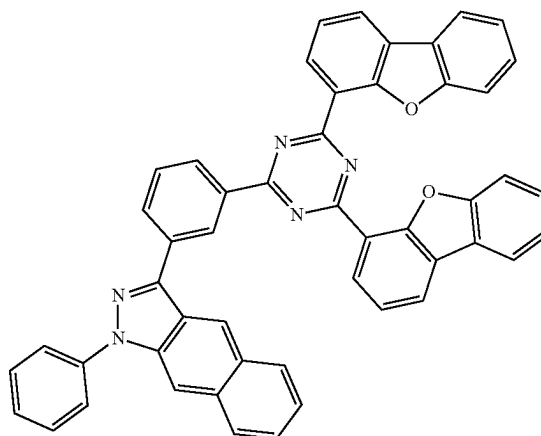
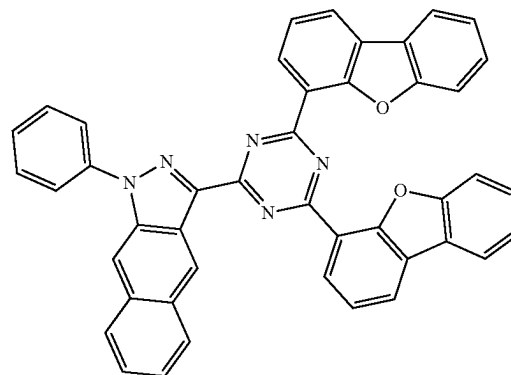
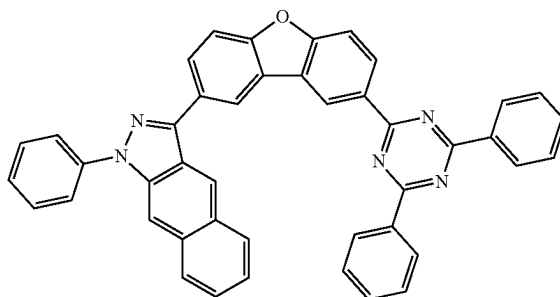
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R107

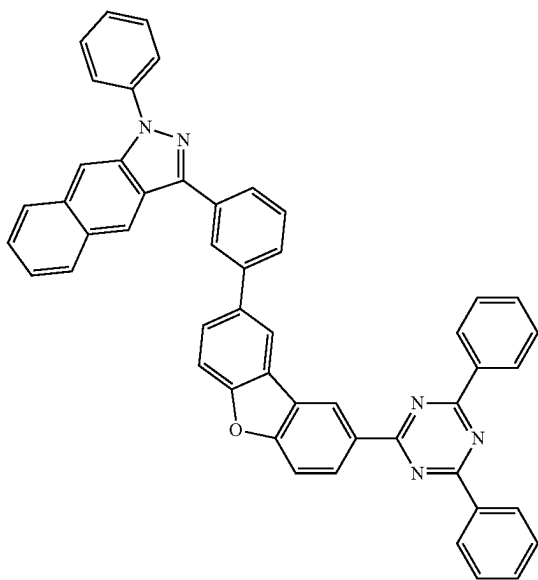
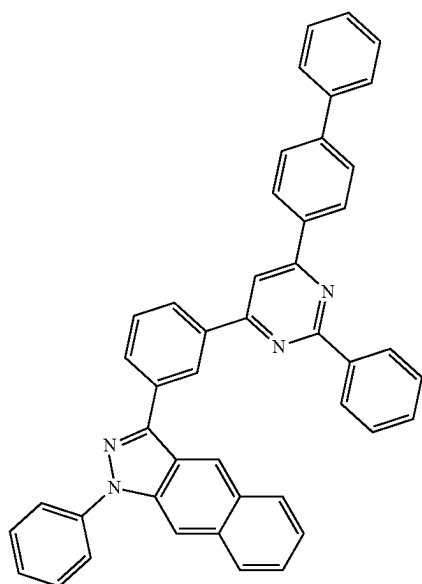
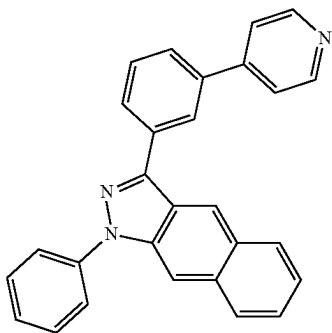
R108

R109



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**180**

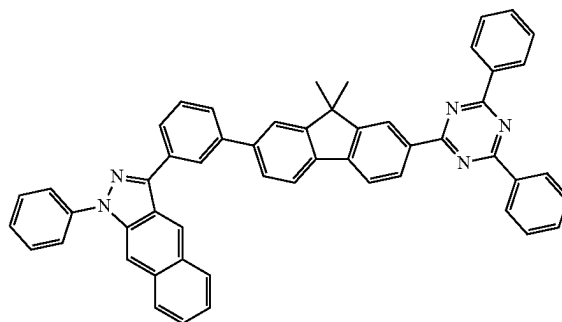
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R110

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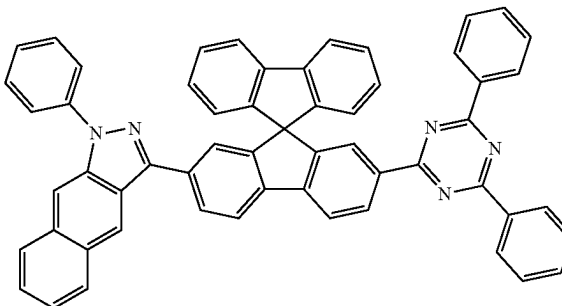
R113

R111

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R114

R112

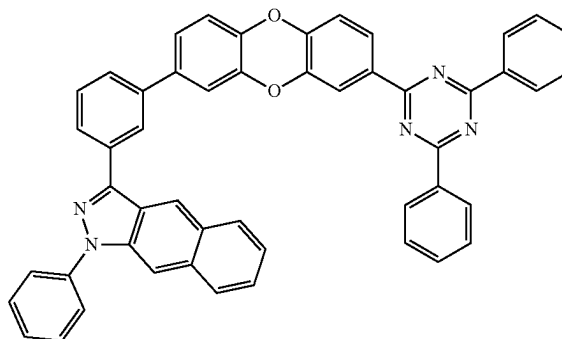
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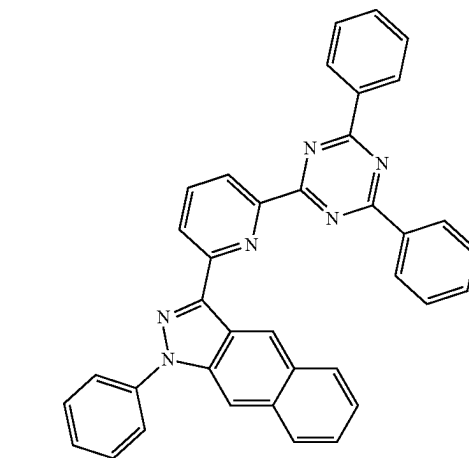
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R115

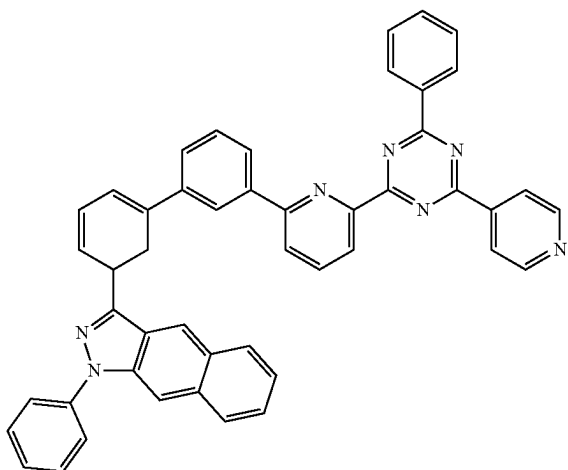
R116



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R117



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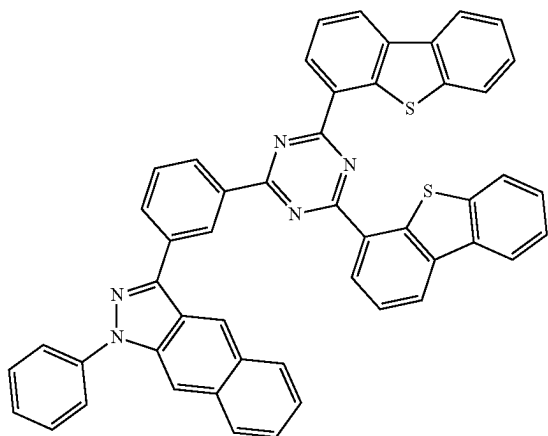
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R118



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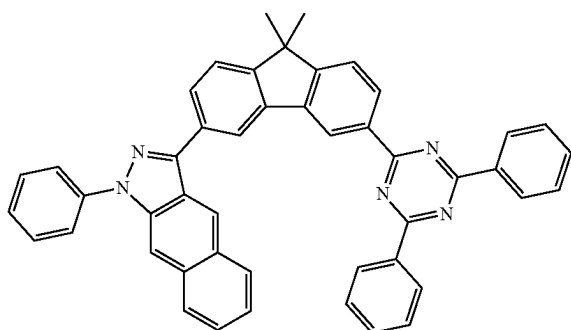
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R119



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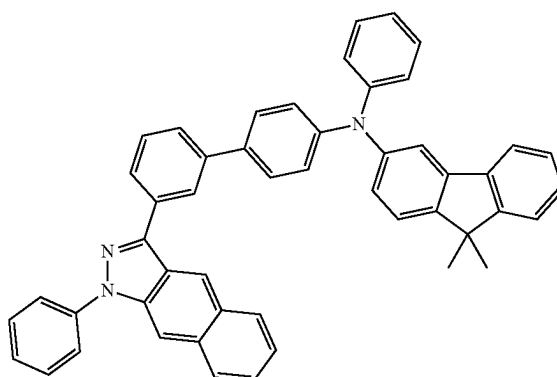
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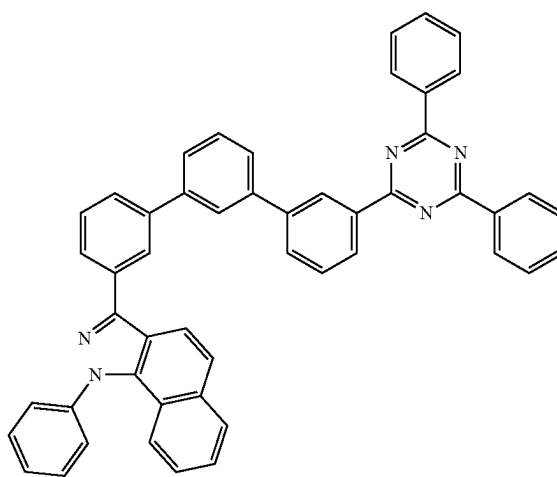
182

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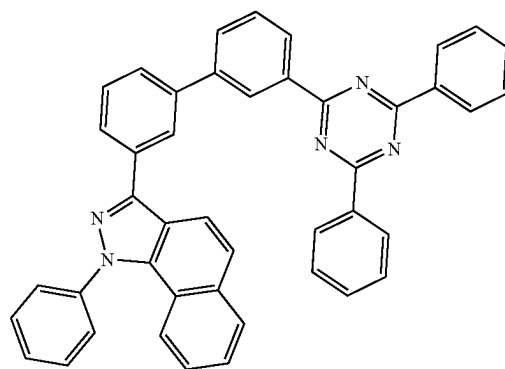
R120



R121

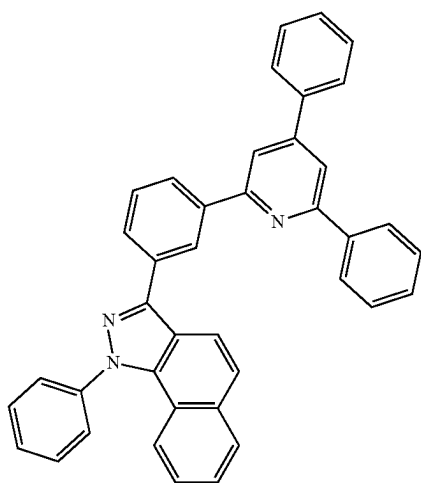
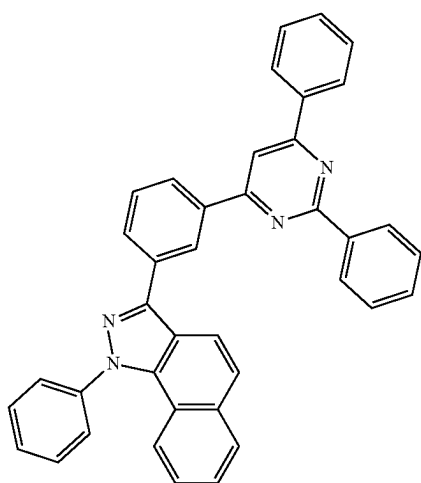
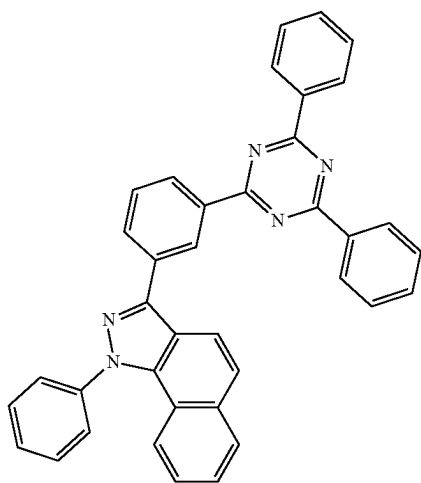


R122



183

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**184**

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R123

R126

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R124

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R125

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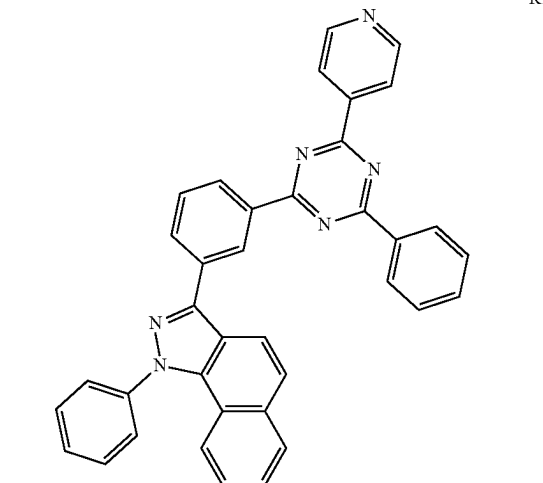
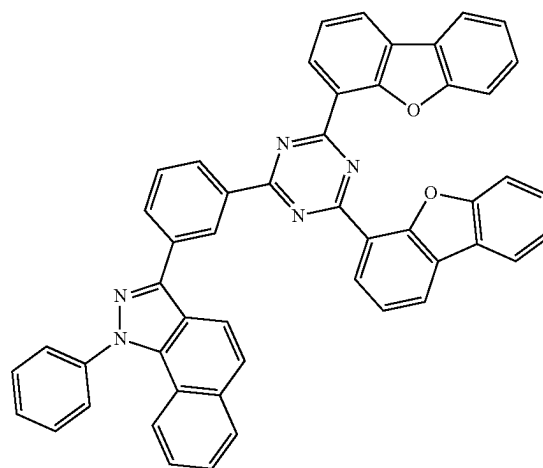
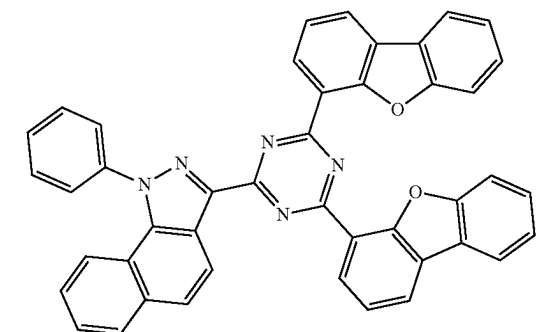
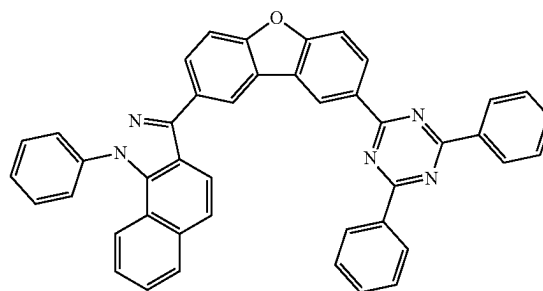
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R127

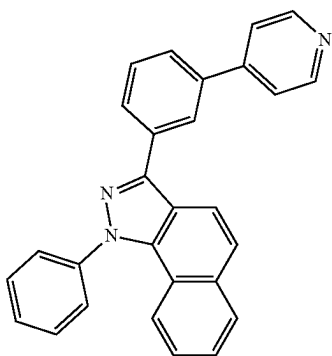
R128

R129



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**186**

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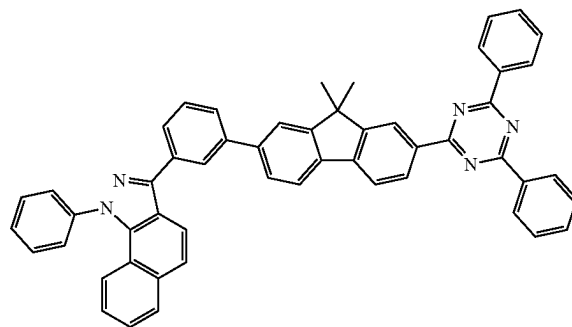
R130

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R133



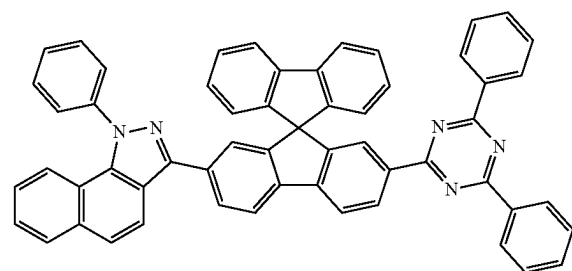
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R131

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R134

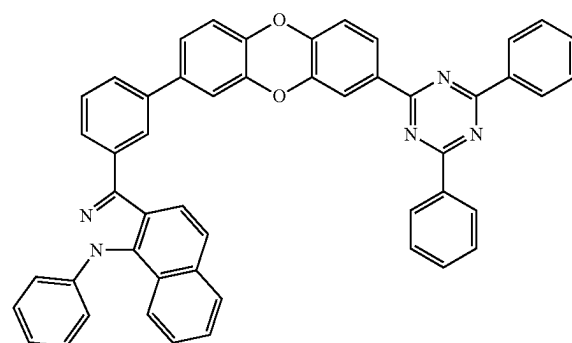


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R135



R132

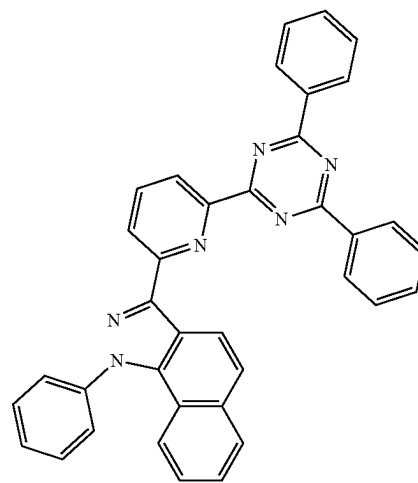
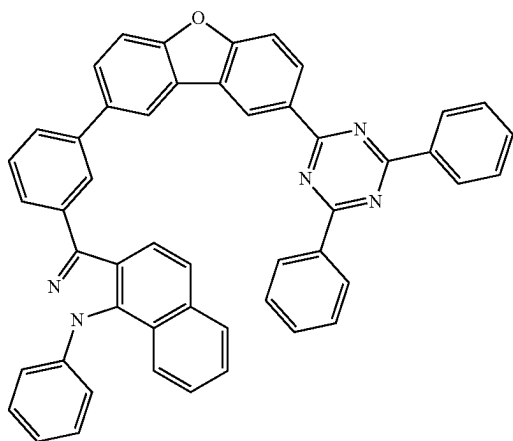
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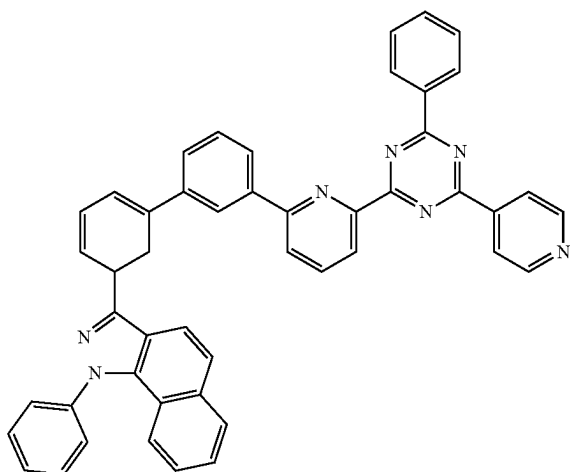
R136



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R137



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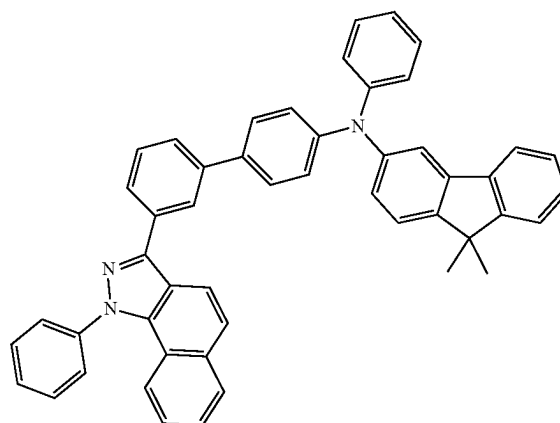
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R140



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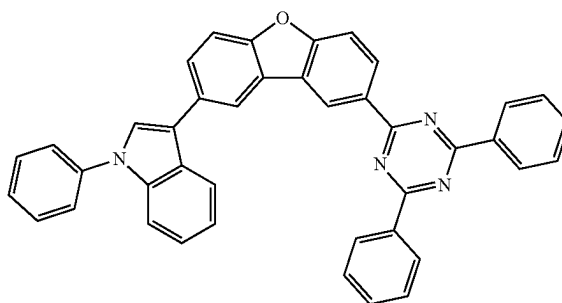
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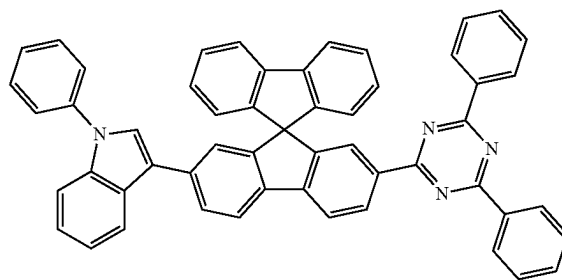
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R146

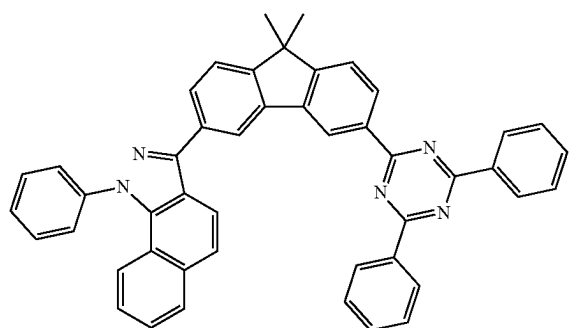


R154



R157

R139

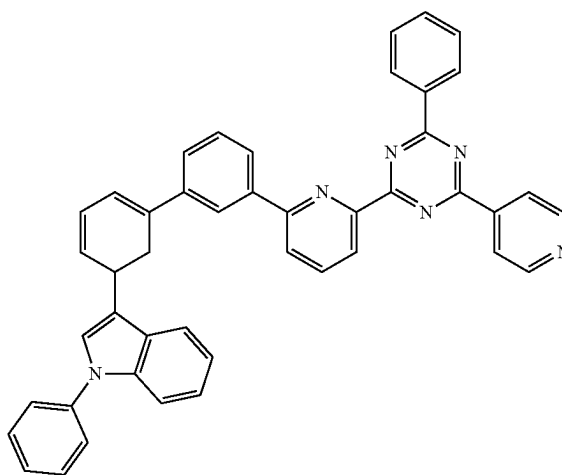


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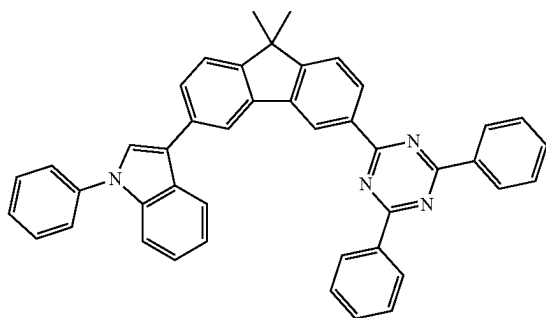
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R159



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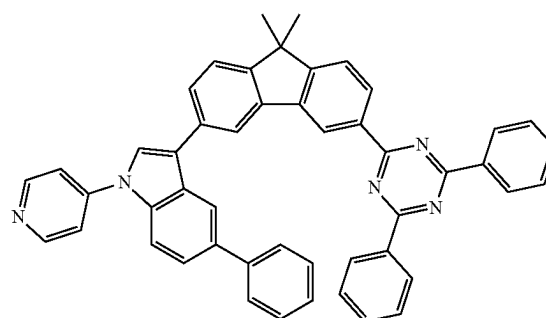
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R179



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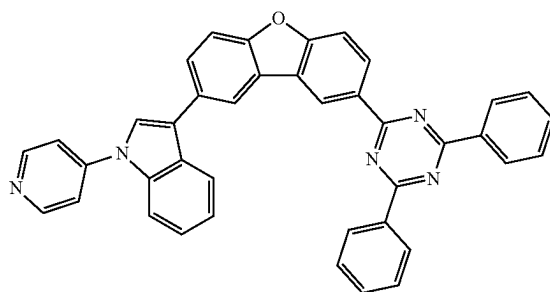
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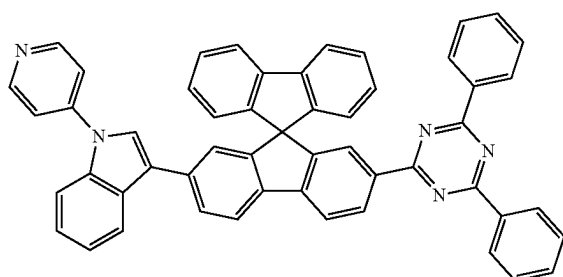
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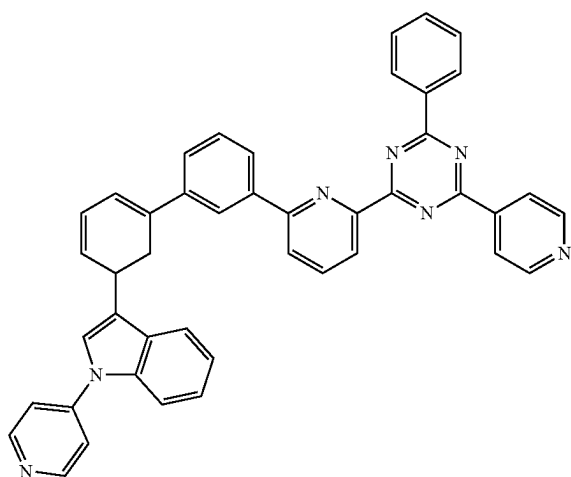
R166



R174



R177



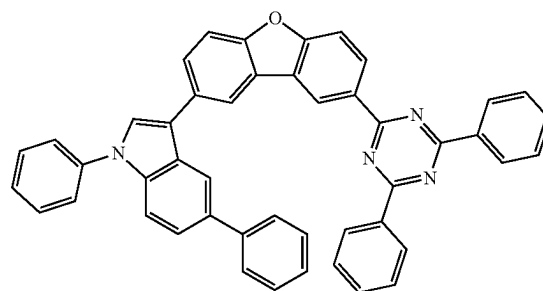
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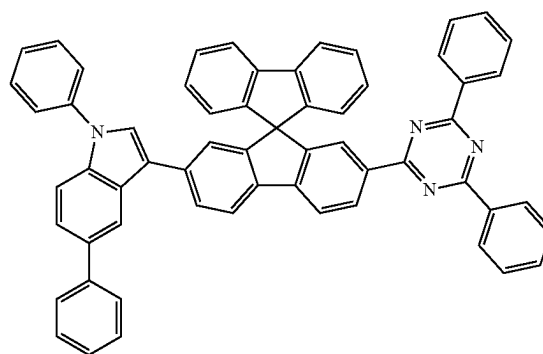
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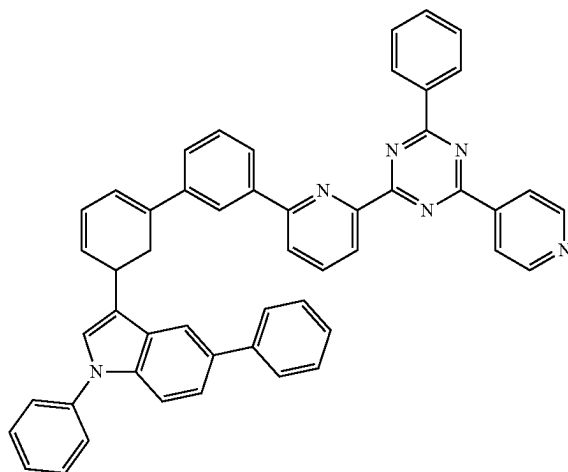
R186



R194



R197

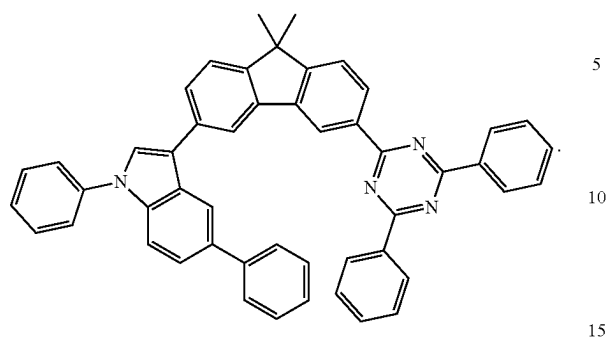


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R199



2. An organic electroluminescent device comprising:

an anode;

a cathode; and

one or more organic material layers provided between the anode and the cathode,

wherein at least one of the one or more organic material layers includes a compound of claim 1.

3. The organic electroluminescent device of claim 2, wherein the organic material layer includes one or more layers selected from the group consisting of a hole injection layer, a hole transport layer, a hole transport auxiliary layer, an electron transport layer, an electron transport auxiliary layer and a light emitting layer.

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